

Animal Welfare, Ethics and Jurisprudence VMD-511 (2+0)

"The greatness of a nation and its moral progress can be judged by the way its animals are treated"

By Mahatma Gandhi.

QUOTATIONS ABOUT ANIMAL RIGHTS

"We cannot have peace among men whose hearts find delight in killing any living creature." - Rachel Carson, American Author and Marine Biologist

"Every year tens of thousands of animals suffer and die in laboratory tests of cosmetics and household products, despite the fact that the test results do not help prevent or treat accidental or purposeful misuse of the products. Please join me in using your voice for those whose cries are forever sealed behind the laboratory doors." - Woody Harrelson, Actor

"The brute animals have all the same sensations of pain as human beings, and consequently endure as much pain when their body is hurt; but in their case the cruelty of torment is greater, because they have no mind to bear them up against their sufferings, and no hope to look forward to when enduring the last extreme pain." - Thomas Chalmers, Scottish Religious Leader

"True benevolence or compassion, extends itself through the whole of existence and sympathizes with the distress of every creature capable of sensation." - Joseph Addison, English Poet

"Life is life's greatest gift. Guard the life of another creature as you would your own because it is your own. On life's scale of values, the smallest is no less precious to the creature who owns it than the largest." - Lloyd Biggle Jr.

"A man is ethical only when life, as such, is sacred to him, that of plants and animals as well as that of his fellowman, and when he devotes himself helpfully to all life that is in need of help." - Dr. Albert Schweitzer, Alsatian Theologian, Musician, and Medical Missionary

"My doctrine is this: that if we see cruelty or wrong that we have the power to stop, and we do nothing, we make ourselves sharers in the guilt." - Anna Sewell, English Novelist

"Providing food and shelter is not proving love for your pet. Those too, but proper care and protection from harm make the truest sense of responsible pet ownership." - John D. Carraway,

DVM, enscribed hanging office plaque of his professional motto, FancyBoy and JoJo's Vet

"First they went after the Communists, and I did not stand up, because I was not a Communist. Then they went after the homosexuals and infirm, and I did not stand up, because I was neither. Then they went after the Jews, and I did not stand up, because I was not a Jew. Then they went after the Catholics, and I did not stand up, because I was Protestant. Finally, they went after me, and there was no one left to stand up for me." - Martin Neimoller, German Pastor who spent 8½ years in a Nazi Concentration Camp

"He who is cruel to animals becomes hard also in his dealings with men. We can judge the heart of a man by his treatment of animals." - Immanuel Kant, German Philosopher

"Animals are such agreeable friends, they ask no questions, they pass no criticisms." - George Eliot, Female English Novelist (Mary Ann Evans)

"We cannot glimpse the essential life of a caged animal, only the shadow of its former beauty." - Julia Allen Field (1937-)

"The awful wrongs and sufferings forced upon the innocent, helpless, faithful animal race, form the blackest chapter in the whole world's history." - Edward Freeman, English Freelance Historian and Journalist

"The assumption that animals are without rights, and the illusion that our treatment of them has no moral significance, is a positively outrageous example of Western crudity and barbarity. Universal compassion is the only guarantee of morality." - Arthur Schopenhauer, German Philosopher

"A man is truly ethical only when he obeys the compulsion to help all life which he is able to assist, and shrinks from injuring anything that lives." - Albert Schweitzer, Alsatian Theologian, Musician, and Medical Missionary

"Cherish the wonderful ways our animals are models of living life to the fullest, loving unconditionally, embracing the passion within, and being completely in the present moment." - Susan White

"I have always felt that the way we treat animals is a pretty good indicator of the compassion we are capable of for the human race." - Ali McGraw, American Actress

"The time will come when men such as I will look upon the murder of animals as they now look upon the murder of men." - Leonardo Da Vinci, Italian Painter, Sculptor, Architect, Musician, Engineer, and Scientist

"The greatness of a nation and its moral progress can be judged by the way its animals are treated." - Mahatma Gandhi, Indian Statesman and Philosopher

"Of all the creatures, man is the most detestable. Of the entire brood, he's the one that possesses malice. He is the only creature that inflicts pain for sport, knowing it to be pain. The fact that man knows right from wrong proves his intellectual superiority to the other creatures; but the fact that he

can do wrong proves his moral inferiority to any creature that cannot." - Mark Twain, American Novelist

"You have to love animals for what they are or leave them alone. The best thing you can do if you love them is leave them alone and see that other people do too." - Pat Derby (1942-)

"What is man without the beasts? If all the beasts were gone, men would die from great loneliness of spirit. For whatever happens to the beasts, soon happens to man. All things are connected." - Chief Seattle American Indian Duwamish Tribe

"True human goodness, in all its purity and freedom, can come to the fore only when its recipient has no power. Mankind's true moral test, its fundamental test (which lies deeply buried from view), consists of its attitude towards those who are at its mercy: animals. And in this respect mankind has suffered a fundamental debacle, a debacle so fundamental that all others stem from it." - Milan Kundera, Czech Novelist

"If only we can overcome cruelty, to human and animal, with love and compassion we shall stand at the threshold of a new era in human moral and spiritual evolution - and realize, at last, our most unique quality: humanity." - Jane Goodall, Author of Reason for Hope

"Until we have the courage to recognize cruelty for what it is...whether its victim is human or animal...we cannot expect things to be much better in this world...We cannot have peace among men whose hearts delight in killing any living creature. By every act that glorifies or even tolerates such moronic delight in killing we set back the progress of humanity." - Rachel Carson, American Biologist and Novelist

"We must fight against the spirit of unconscious cruelty with which we treat the animals. Animals suffer as much as we do. True humanity does not allow us to impose such sufferings on them. It is our duty to make the whole world recognize it. Until we extend our circle of compassion to all living things, humanity will not find peace." - Albert Schweitzer, Alsatian Theologian, Musician, and Medical Missionary

"The man who kills the animals today is the man who kills the people who get in his way tomorrow." - Dr. Diane Fossey, Woman in the Mists (Dr. Fossey was killed by someone who killed animals yesterday.)

"To sin by silence when they should protest makes cowards out of men." - Abraham Lincoln, Sixteenth President of the United States

"The worst sin toward our fellow creatures is not to hate them, but to be indifferent to them; that's the essence of humanity." - George Bernard Shaw, Irish Playwright and Critic

"Cruelty to animals is one of the most significant vices of a low and ignoble people. Wherever one notices them, they constitute a sign of ignorance and brutality which cannot be painted over even by all the evidence of wealth and luxury." - Alexander Von Humbolt, German Naturalist

"The quiet conscience is the invention of the devil. No one of us may permit any preventable pain to

be inflicted even though the responsibility for that pain is not ours. No one may shut his eyes and think that the pain which is therefore not visible, is nonexistent." - Albert Schweitzer, Alsatian Theologian, Musician, and Medical Missionary

"You cannot do a kindness too soon, because you never know how soon it will be too late." - Ralph Waldo Emerson, American Essayist and Philosopher

"It is as if life had said, 'I am going to send you into a world of cruelty. I shall make you sensitive to pain, fear, heat, cold, hunger and starvation. In this world of cruelty, I shall make you defenseless. In addition, I shall strike you dumb.' This is the kind of world that animals are born into." - Grace Johnson, Animal Activist

"All the beautiful sentiments in the world weigh less than a single lovely action." - James Russell Lowell, American Poet and Essayist

"Very little of the great cruelty shown by men can really be attributed to cruel instinct. Most of it comes from thoughtlessness or inherited habit. The roots of cruelty, therefore, are not so much strong as widespread. But the time will come when inhumanity protected by custom and thoughtlessness will succumb before humanity championed by thought. Let us work that this time may come." - Albert Schweitzer, Alsatian Theologian, Musician, and Medical Missionary

"To close your eyes will not ease another's pain." - Chinese Proverb

"The most solid comfort one can fall back upon is the thought that the business of one's life is to help in some small way to reduce the sum of ignorance, degradation, and misery on this earth." - George Eliot, Female English Novelist (Mary Ann Evans)

"These birds and animals and fish cannot speak, but they can suffer, and our God who created them, knows their sufferings, and will hold him who causes them to suffer unnecessarily to answer for it. It is a sin against their Creator." - George Q. Cannon, Animal Activist

"Cruelty to dumb animals is one of the distinguishing vices of the lowest and basest of the people - wherever it is found, it is a certain mark of ignorance and meanness." - Jones of Nayland

"It is totally unconscionable to subject defenseless animals to mutilation and death, just so a company can be the first to market a new shade of nail polish or a new, improved laundry detergent. It's cruel, it's brutal, it's inhumane, and most people don't want it." - Abigail 'Dear Abby' Van Buren Testifying before the House Judiciary Committee in support of the Consumer Products Safe Testing Act, March 1988

"The Animals of the planet are in desperate peril and they are fully aware of this. No less than human beings are doing in all parts of the world, they are seeking sanctuary." - Alice Walker, "Selected Writings," Living by the Word

"Love of animals is a universal impulse, a common ground on which all of us may meet. By loving and understanding animals, perhaps we humans shall come to understand each other." - Louis J.

Camuti, D.V.M., First American vet to devote his practice exclusively to cats.

"I abhor vivisection. It should at least be curbed. Better, it should be abolished. I know of no achievement through vivisection, no scientific discovery, that could not have been obtained without such barbarism and cruelty. The whole thing is evil." - Dr. Charles Mayo, Founder of the Mayo Clinic.

"All beings are fond of themselves, they like pleasure, they hate pain, they shun destruction, they want life and want to live long. To all life is dear; hence their life should be protected." ~ Mahavira

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MODULE-1: ANIMAL WELFARE

Learning outcome

In this module the student may get an introduction about the What is Animal Welfare, definition, status of animal welfare in India and animal welfare principles in AVMA.

ANIMAL WELFARE - INTRODUCTION

As per the Indian tradition and culture, animals always had respect and a special place in society. Hinduism, Buddhism and Jainism have always preached kindness and compassion to animals. Each Hindu God or Goddess is seen with an animal.

But today times have changed. With population explosion, urbanization and consumerism catching up, animals are easy prey for human greed. They are soft targets for any one trying to make a quick buck, and prime candidates for exploitation. Today even healthy, young animals are killed for leather, meat and tallow. Calves are kept hungry and emaciated, while the cow's milk is sold in the markets for human consumption. Chicken are kept in cramped batteries, either for the eggs or for slaughter. Slaughter animals are made to walk thousands of miles to slaughterhouses, or carted in trucks / tempos packed like sardines, with the result some of them die of suffocation before they reach their destination. Rhesus monkeys and Dancing Bears performing on the streets, with hordes of people watching them, is another eye sore in India. The fear of the baton falling on their backs is clearly visible in their eyes. Same is the case with thousands of animals performing in the circuses. Research on animals is another major issue that needs to be tackled. With tremendous pressure from animal welfare groups in the developed countries, companies are bringing their research work to India, where the laws are not strict and they can get away with just about anything.

In this situation, animal welfare in India takes on a whole new meaning. With so many animal issues and so many animals on the streets, the need of the day is to have several animal welfare societies in each community, whereas there is only one full fledged hospital in whole Mumbai city which is not at all sufficient to look after all the sick and injured animals. NGOs who are running the ABC centers also treat the sick, injured animals or house the abandon animals with very meager means and facilities. The Government of India is trying to encourage setting up of an SPCA in each District of the country, but there are few takers. IDA India also has a project on the agenda to build an animal hospital with latest technology and **equipment**.

The only ray of hope is the advent of television. Awareness about animal issues and environment is spreading **among** youngsters watching channels like Animal Planet, National Geographic and Discovery.

They are becoming aware of the place of animals in our lives and the need to give them the due respect. IDA India hopes to tap the potential of these youngsters for welfare of animals, and has formed the 'Compassionate Children's Club'. IDA India also regularly organized the animal welfare and awareness programs in schools, collages, and socials groups which is playing very effective role in bringing wider vision for animal issues among the adult and children.

ANIMAL WELFARE

- Animal welfare means how an animal is coping with the conditions in which it lives. An animal is in a good state of welfare if (as indicated by scientific evidence) it is healthy, comfortable, well nourished, safe, able to express innate behaviour, and if it is not suffering from unpleasant states such as pain, fear, and distress. Good animal welfare requires disease prevention and veterinary treatment, appropriate shelter, management, nutrition, humane handling and humane slaughter. Animal welfare refers to the state of the animal; the treatment that an animal receives is covered by other terms such as animal care, animal husbandry, and humane treatment. Protecting an animal's welfare means providing for its physical and mental needs.
- Ensuring animal welfare is a human responsibility that includes consideration for all aspects of animal well-being, including proper housing, management, nutrition, disease prevention and treatment, responsible care, humane handling, and, when necessary, humane euthanasia.
- There are numerous perspectives on animal welfare that are influenced by a person's values and experiences. There are also various means of measuring animal welfare, including (but not limited to) health, productivity, behavior, and physiological responses.

ANIMAL CRUELTY LAWS

- As civilization has advanced, there was a need to augment in ascendancy and regulations and set of laws of daily life. These governance and regulations mount from general repugnance over activities all over the world. One such fast growing activity is animal cruelty. Animal cruelty implies cruel gratuitous treatment of animals. Such behavior usually has a single point program - to subject animals and sometimes pets to unnecessary harm and pain. One major type of animal cruelty is torture

ANIMAL CRUELTY REPORT

- In a current report, People for Ethical Treatment of Animals (PETA) have released discomforting facts regarding the circus lives of animals. As per their reports, many famous circuses do not put into practice what they advocate and are active in animal cruelty on a customary basis. No foundation or entity takes care of the training sessions of the circuses. Tight collars, lashing, whipping, beatings and persecution are the order of the day for the animals at the circus. PETA also mentions that the actions of the animals during the circus shows **are** not even close to what they would call natural, in the sense that tigers don't jump around and then lay down in agreement, elephants don't stroll trunk to tail, and female elephants don't mount each other. All **these are** possible only by severe training, which may be translated into severe cruelty.

CIRCUS ANIMAL CRUELTY

- The animals in circuses are also customarily and hazardously exposed to multiple diseases. In some cases, they were not given essential medical first aid. Tuberculosis is seen as a main disease for the elephants in circuses. Some case reveal that the elephants were not treated medically for injuries inflicted during thrashing and other natural grounds.
- Animal brutality is a vague term and the terminology flickers from region to region, state to state and country to country. The classification of animal cruelty may be everything between wide-ranging and non-existent. While animal cruelty amongst people may either be existent.

ANIMAL CRUELTY STATISTICS

- Supremacy lies not in treading over the weaker species, but in being empathetic and compassionate enough to value their being. It doesn't mean that all of us will be an animal lover overnight - However the point is that if you don't like them, don't go out of your way to harm them either. Nature made all of us for a special reason and human being was not put at the head of the food chain to torment the other beings. We don't understand it now, but life has a law: What goes around, comes around. Let us not get so blinded and absorbed in our ravenousness that we forget how to be human beings.

DUTIES OF AWO(ANIMAL WELFARE OFFICERS)

1. An AWO (Animal Welfare Officer) should have a thorough knowledge of all the Animal Welfare Laws and especially the following laws:
 - The Prevention of Cruelty to Animals Act, 1960,
 - The Wild Life (Protection) Act, 1972 as amended in 2006.
 - Breeding of and Experiments on Animals (Control and Supervision) Amendment Rules, 2005
 - The Animal Birth Control (Dogs) Rules, 2001
 - The Performing Animals (Registration) Rules, 2001
 - The Performing Animals Rules, 1973
 - The Prevention of Cruelty to Animals (Licensing of Farriers) Rules,
 - The Prevention of Cruelty to Draught and Pack Animals Rules, 1965, amended 1968
 - The Transport of Animals (Amendment) Rules, 2001
2. The AWO is not an enforcement person. The holding of AWO Card does not entitle to take direct law enforcement action such as raids, checking of premises, stopping of vehicle on roads etc. You are an assisting Hony. Animal Welfare Officer, an assisting person & entitled to lodge a police complaint in the event of any cruelty being seen or reported to you. The AWO in case wish to inspect any illicit slaughter house/premises etc. he will be authorized specifically by AWBI on request from him. In the event of any cruelty being seen or reported you are requested to take the assistance of Police/Authorities concerned in addition to informing to the Board over telephone 044-24454958/24454935/e-mail:awbi@md3.vsnl.net.in/fax 044-24454330.
3. The Honorary Animal Welfare Officer is expected to prevent illicit slaughter and if any report of illicit slaughter is made, the case must be immediately reported to the concerned authorities and care should be taken to make sure that the illegal slaughtering is immediately stopped
4. Cruelties to animals during transportation are often severe and the AWO should immediately report incidents of cruelty to the authorities and ensure that strict legal action is enforced on the offenders.

5th point may be deleted as it is repeated

6. During natural calamities, the AWO must network with like-minded people in the neighborhood and organize for (pl delete for) immediate rescue operations and emergency relief for the helpless animals.
7. An important duty of an AWO is to make sure that animals like bears and monkeys are not used as performing animals in the neighborhood and not subjected to cruelty and suffering
8. Preventing the suffering that bullocks, horses, donkeys and camels undergo when used for draught energy, through sensitization and education of the animal owners is a key responsibility that AWOs must fulfill with complete dedication.
9. The AWO must do his or her best to prevent the misuse and abuse of oxytocin in milking animals by educating the dairy farmers
10. Participation in coordinating or organizing training programs for villagers on setting up self sustaining gaushalas is a task that AWOs must execute sincerely.
11. The AWO is expected to organize first aid, veterinary care, food and shelter for sick and destitute animals
12. Attention **towards** the proper care and management of birds and fish maintained in aviaries and aquaria and prevention of **cruelty** is another important duty that AWOs must vigilantly carry out.
13. The AWO is expected to be thorough with the PCA act, rules and regulations & he/she should be able to conduct training programs on legislation issues in animal welfare for police personnel and lawyers.
14. It is expected that AWOs organize mobile clinics and veterinary camps for stray animals and for animals belonging to poor people, especially in areas where veterinary services are not available
15. The AWO is expected to sensitize the community and take appropriate remedial measures to prevent cruelty to animals from happening – eg. animal sacrifice, Jallikattu, bull fights, cock fights and ram fights.
16. Establishing good rapport with the press and electronic media to create extensive public awareness on animal welfare issues is another key role that an AWO should carry out diligently.
17. The use of animals by the entertainment industry should be vigilantly monitored by the AWO by attending the film shoots and being physically present at religious and social events where animals are brought together to prevent cruelty to animals.
18. **AWOs** are expected to keep close liaison with law enforcement agencies such as Police, Forest, Customs & Excise, etc. for effective enforcement of animal welfare laws and work in close co-ordination with Members of Animal Welfare Board of **India**, General Public and Animal Welfare Organisations of the area to promote animal welfare in the country.

19. AWO is not entitled to collect any fine or accept cash from offenders. Such action will **amount** to bribery. Any AWO found to collect fine or money will be penalized through legal action and FIR will be lodged in addition to cancellation of AWO ID Card

AVMA ANIMAL WELFARE PRINCIPLES

The AVMA, as a medical authority for the health and welfare of animals in USA, offers the following eight integrated principles for developing and evaluating animal welfare policies, resolutions, and actions.

- The responsible use of animals for human purposes, such as companionship, food, fiber, recreation, work, education, exhibition, and research conducted for the benefit of both humans and animals, is consistent with the Veterinarian's Oath.
- Decisions regarding animal care, use, and welfare shall be made by balancing scientific knowledge and professional judgment with consideration of ethical and societal values.
- Animals must be provided water, food, proper handling, health care, and an environment appropriate to their care and use, with thoughtful consideration for their species-typical biology and behavior.
- Animals should be cared for in ways that minimize fear, pain, stress, and suffering.
- Procedures related to animal housing, management, care, and use should be continuously evaluated, and when indicated, refined or replaced.
- Conservation and management of animal populations should be humane, socially responsible, and scientifically prudent.
- Animals shall be treated with respect and dignity throughout their lives and, when necessary, provided a humane death.
- The veterinary profession shall continually strive to improve animal health and welfare through scientific research, education, collaboration, advocacy, and the development of legislation and regulations.

• MODULE-2: ROLE OF VETERINARIANS IN ANIMAL WELFARE

Learning outcome

This module elaborates on Animal Welfare and Veterinarian's Role in Animal Welfare, Management of Laboratory Animal Production and Maintenance Colonies Resource and Veterinarians and Institutional Animal Ethics Committees Resource in brief.

ROLE OF VETERINARIANS IN ANIMAL WELFARE

- For many veterinarians, animal welfare is a matter of animal health: As long as an animal is healthy, it is also happy. For the past forty years, however, behaviour biologists have shown repeatedly that, whereas poor health contributes to poor welfare, the opposite is not necessarily the case. There is more to good welfare than good health.
- Numerous studies have shown that, despite many generations of selection for various traits, the behavioural repertoire of domestic animals is almost identical to that of their wild ancestors. A broiler chicken has a similar demand for dust bathing as a jungle fowl. A thoroughbred racehorse has the same demand for social contact as the Przewalski horse. These are just two of many other examples. Consequently, to ensure a certain level of welfare in our domestic animals, it is important that they are able to perform their species-specific behaviour beyond eating behaviour, drinking behaviour, eliminative behaviour and resting behaviour.
- And that's where the controversy begins Which elements of the animal's ethogram are essential and which ones are not? Must a prey animal show flight behaviour regularly to be happy? Must a well-fed animal show foraging behaviour to be content? Is social contact through bars enough contact?
- The controversy consists of, on the one side, practitioners or producers that need to house and manage animals in such a way that their production is productive and, on the other side, animal protection people or animal rights people who are against any "exploitation" of the domestic animals.
- As with most issues, obviously the optimum lies somewhere in between these two extremes. We all have to compromise and so must our domestic animals. In return for regular food, water, shelter and medical care, the animals must give up some of their freedom. But how much they must give up, how restrictively we can keep them with good conscience, is a matter of dispute.
- The veterinary profession plays a key role in the controversy in that it possesses a detailed knowledge of both camps. Practicing veterinarians and veterinary inspectors are in regular contact with the animals, either in the clinic or during visits to the farms, either to treat sick animals, to institute preventive measures, or to control that legislation is being followed. At the same time veterinarians possess (or should possess) the scientific background that enable them to judge the welfare of their patients and clients.
- Public concern for the welfare of domestic animals is likely to increase in the future. At the present time behaviour biologists still struggle with the difficult task of "measuring happiness" in the animals. Similarly, programs for monitoring welfare in praxis are being developed in many countries. An important question in these developments is how the veterinary profession can best prepare itself for future challenges, both in the veterinary curriculum and in terms of continued education.

As public concern for animal welfare grew, people began asking awkward questions regarding, for example, the production of foiegras, the forced molting of laying hens, and the slaughter of horses for human consumption overseas.

- The public views veterinarians favorably, regarding them as experts in animal care and welfare. But the veterinary profession has had a love-hate relationship with animal advocates.
- The profession's economics and changing demographics—in 1960, less than 2% of veterinarians were female; today 45% are female—will have an impact on the way the profession deals with animal welfare and animal rights. Public opinion polls find that women are about 15 percentage points more positive toward animal welfare issues than are men. One could argue that the increased numbers of female veterinarians is already having an impact, even though there are still relatively few women in leadership positions in the profession. In addition, most pet caregivers who take their animals to veterinarians will favor those veterinarians who emphasize animal welfare.
- Because their animals are now considered members of the household, people are requesting advanced veterinary services—and paying more for them. The more veterinary hospitals respond to this change in pet caregiver attitudes by, for example, focusing more on animal welfare and working cooperatively with local humane societies, the more their clients are likely to trust them and accept the new economic realities of pet medicine.
- Moreover, as people have come to value their pets more highly, many veterinarians have responded by offering—and charging for—medical and surgical services that meet their clients' expanding demands for advanced healthcare.

INTRODUCTION

- Veterinarians play a central role in biomedical (including veterinary), wildlife and farm animal research. The comprehensive nature of veterinary training equips veterinary graduates to handle a diverse range of professional responsibilities relevant to the use of animals for scientific purposes and animal welfare. In the Indian context, veterinarians in scientific institutions may be divided into two main categories: those involved in the production, medicine and surgery of animals used for scientific purposes; and those involved on Institutional Animal Ethics Committees, and, of course, there are many veterinarians whose duties involve elements of both categories.
- The broad range of duties in the first category includes management of laboratory animal production and maintenance colonies, operation of preventative medicine programs, clinical medicine, provision of research support, maintenance of quarantine facilities, operation of in-house pathology programs, experimental surgery, and involvement in animal house design. For those veterinarians involved with Institutional Animal Ethics Committees their role is to promote animal welfare and regulatory compliance duties include review of proposals to use animals for scientific purposes, the minimisation of pain and distress, the monitoring of animals in research and teaching, the administration of animal ethics committees, provision of advice to the institution, provision of advice to researchers, provision of consultation and advice concerning compliance with relevant legislation and the Code of Practice for the Care and Use of Animals for Scientific Purposes, and training of research and technical personnel.
- Veterinarians also play a major role in the development of institutional and regulatory policy, codes of practice, regulation and monitoring.
- Although principally outside the scope of this text, qualification as a veterinarian also provides a strong base for veterinary graduates to embark upon a scientific research career in such diverse fields as immunology, molecular biology, physiology, biochemistry, virology, pathology, bacteriology, parasitology, animal management, wildlife research, agricultural animal research, and exotic disease research, and indeed there are veterinary graduates currently working as investigators in all these fields.

**MANAGEMENT OF LABORATORY ANIMAL PRODUCTION AND
MAINTENANCE COLONIES**

- A veterinarian responsible for the management of laboratory animals uses the strong scientific knowledge provided by a veterinary degree but must acquire a number of special skills and a variety of experience in order to fulfil the duties involved in this role. Although well equipped in the general principles of preventative and clinical medicine, surgery, genetics, the scientific process, the principles of animal management, pathology, quarantine, and nutrition, the laboratory animal veterinarian needs to apply this knowledge to a range of less familiar species. The focus of undergraduate training is on the domestic and farm animal species with little attention given to mice, rats, guinea pigs, and rabbits, let alone fish, amphibia, native animals and reptiles. The laboratory animal veterinarian therefore embarks early on a steep learning curve concerning anatomy, physiology and medicine of rodents, rabbits and other unusual species. The latter includes diagnosis and treatment of diseases one has not encountered before. In some situations, more familiar species, in which he or she has received considerable training, may be encountered, but the circumstances of their housing may be vastly different in the research environment. Many additional responsibilities such as personnel management of a team of animal technicians, financial management and environmental control are part of the role of these veterinarians. It is apparent that, in addition to the experience and training of a veterinarian, it is necessary to acquire some of the skills of the production engineer and human resources manager.
- The ultimate aim of the laboratory animal veterinarian is to provide to the researcher, in a timely and efficient manner, an experimental animal in a state appropriate to the intended research in terms of biological characteristics, genetic constitution and microbiological and general health status. In recent years genetic definition, by targeting specific genes, particularly in laboratory mice, has allowed investigators to more accurately define the biological roles of genes and the genetic components of disease processes. As the effects of genetic manipulation cannot be fully known in advance, this has placed more demands on veterinarians involved with the production and monitoring of these animals.
- Monitoring aspects will be discussed in the next section, but from a production viewpoint these animals may require special care, they may be more susceptible to disease and reproductive efficiency may be impaired. The veterinarian with experimental surgery commitments performs surgery to assist investigators in their research and this may extend to development of surgical techniques for specific protocols as well as training and providing advice in surgical techniques to investigators and animal technicians in some procedures. It may be seen that the laboratory animal veterinarian, as defined here, is something of a jack of all trades. Indeed they are masters of most of the component disciplines. It might be seen to be a section of the profession that has many challenges. Laboratory animal veterinarians will tell you that this is indeed true!

VETERINARIANS AND INSTITUTIONAL ANIMAL ETHICS COMMITTEES

- The role of veterinarians within the Animal Ethics Committee's sphere of activities and institutional animal welfare extends well beyond the formal ethics meetings.
- Training of investigators and technical staff is an important function carried out by veterinarians in many institutions using animals. This training usually includes formal instruction on compliance with the Code and relevant legislation, and extends to practical workshops in animal handling techniques, routine research procedures, anaesthetic methods, euthanasia techniques and general or specific surgical skills. This may be done in conjunction with training provided by research group leaders. The ongoing monitoring of animals and animal facilities is another essential role.
- The veterinarian is the ideal person to monitor pain and distress and instigate alleviation measures when necessary. As the Animal Ethics Committee is required to provide comment on the building or modification of animal facilities the veterinarian also has a central role to play in assisting the committee to determine the appropriateness of the housing environment.
- The concept of genetic modification of animals, while enabling scientists to concentrate on specific genes essential to the disease process, has brought with it a whole new set of potential welfare issues to those entrusted with the monitoring of research using animals. Veterinarians, conversant with this discipline acting in conjunction with animal technical staff and investigators, are well-positioned to assess the welfare and genetic stability of newly created genetically modified animals. They are also best able to institute special care when necessary and to implement measures where increased susceptibility to disease is involved.
- Given the sometimes sensitive nature of particular research protocols, and the unique role of veterinarians within institutions and their Animal Ethics Committees, veterinarians are often required to act as de facto "information or publicity officers" with potential for interaction with the media and the general community in public forums. The veterinarian's role may also include that of an independent complaints officer for concerns and issues raised by staff, students and members of the general community relating to the care and use of animals for research or teaching purposes.
- The veterinarian who is a member of an Institutional Animal Ethics Committee has, by nature of his or her training and experience, a unique and significant role in being able to provide beneficial advice to the Animal Ethics Committee, the investigators and/or teachers, and the institute itself whilst facilitating the accumulation of scientific knowledge in the most humane manner.

MODULE-3: ANIMAL WELFARE ORGANIZATIONS

Learning outcome

This module focusses on Animal Welfare Organizations present in India like Animal Welfare Division of MoEF, GoI and Animal Welfare Board of India (AWBI). It also focusses on funds, functions and the power of the Board (AWBI) to make regulations, Policies, Acts and Rules of AWBI. The recognised Animal Welfare Organizations by AWBI will also be discussed.

MINISTRY OF ENVIRONMENT AND FORESTS - ANIMAL WELFARE DIVISION

Under the Ministry of Environment and Forests, GoI, the Animal Welfare Division is entrusted with the implementation of the provisions of the Prevention of Cruelty to Animals Act, 1960 (59 of 1960). Plan Schemes are in operation for implementation of the statutory obligations under this Act. Two statutory organizations viz. Animal Welfare Board of India (AWBI) and committee for the Purpose of Supervision and Control of Experiments on Animals (CPCSEA) have also been set up under this Act. The mandate of the Animal Welfare Division is to prevent the infliction of unnecessary pain or suffering on animals.

- Under the scheme for shelter houses for looking after the animals, grant is provided to establish and maintain shelter houses for distressed animals in the country. Primarily, Non-Governmental Organisations (NGOs) and Society for Prevention of Cruelty to Animals (SPCAs) are given grants of upto Rs.22.50 lakhs for construction of boundary walls, shelters, water tank, drains, in-house dispensary, medical equipment, contingencies etc. In the current financial year, 13 shelter houses have been constructed up to November 2007.
- In the current financial year, under the Scheme for Birth Control and Immunization of Stray Dogs, 33,222 animal birth control operations have been conducted upto November 2007.
- Under the Scheme for Provision of Ambulance Services to Animals in Distress, 22 ambulances have been given to Animal Welfare Organisations (AWOs) upto November, 2007.
- The Animal Welfare Board of India (AWBI) has granted recognition to 55 newly established Animal Welfare organizations (AWOs) during the year.

ANIMAL WELFARE ORGANIZATIONS

- [The Animal Welfare Board of India](#)
- [Ministry of Environment & Forests](#)
- [RSPCA](#)
- [WSPA](#)
- [Blue Cross of India](#)
- [Blue Cross of Hyderabad](#)
- [People For Animals](#)
- [Wildlife Institute of India](#)
- [Visakha SPCA](#)
- [IPAN](#)
- [APOWA](#)
- [Bombay SPCA](#)
- [Help In Suffering](#)
- [CUPA](#)
- [Friendicoes Seca](#)
- [Madras SPCA](#)
- [In Defense of Animals](#)
- [Animal Rights International](#)
- [Pet Animal Welfare Society](#)
- [Calcutta SPCA](#)
- [Animal Rights Fund](#)

ANIMAL WELFARE BOARD OF INDIA

Establishment of Animal Welfare Board of India

- For the promotion of animal welfare generally, and for the purpose of protecting animals from being subjected to unnecessary pain or suffering, in particular, there shall be established by the Central Government, as soon as may be after the commencement of this Act, a Board to be called the * * (Animal Welfare Board of India).
 - Subs. Act 26 of 1982; S.3, for the words "Animal Welfare Board".
 - Sub-ibid, S.4 for the words "Animal Welfare Board".
- The Board shall be a body corporate having perpetual succession and a common seal with power, subject to the provisions of this Act, to acquire, hold and dispose of property and may by its name sue and be sued.
- Established in 1962 under Section 4 of the Prevention of Cruelty to Animals Act, 1960 (No. 59 of 1960), the Animal Welfare Board of India was started under the stewardship of Late Smt. Rukmini Devi Arundale, well known humanitarian. From ensuring that animal welfare laws in the country are diligently followed, to provide grants to Animal Welfare Organizations and advising the Government of India on animal welfare issues, the Board has been the face of the animal welfare movement in the country for the last 47 years.
- The Board consists of 28 Members. The term of office of Members is for a period of 3 years.

CONSTITUTION OF THE BOARD

1. The Board shall consist of the following persons, namely

- a. the Inspector General of Forests, Government of India, ex-officio;
- b. the Animal Husbandry Commissioner to the Government of India, ex-officio;

* (ba) two persons to represent respectively the Ministers of the Central Government dealing with Home Affairs and Education, to be appointed by the Central Government;

* inserted by Act 26 of 1982; S. 5 (a) (i)

(bb) one person to represent the Indian Board for Wildlife, to be appointed by the Central Government;

(bc) three persons who, in the option of the Central Government are or have been actively engaged in animal welfare work and are well-known humanitarians, to be nominated by the Central Government;

- c. one person to represent such association of veterinary practitioners as in the opinion of the Central Government ought to be represented on the Board, to be elected by that association in the prescribed manner;
- d. two persons to represent practitioners of modern and indigenous systems of medicine, to be nominated by the Central Government;
- e. * * one person to represent each of such two municipal corporations as in the opinion of the Central Government ought to be represented on the Board, to be elected by each of the said corporations in the prescribed manner;

* * Subs. - ibid S. 5 (a) (ii) for the original clause.

- f. one person to represent each of such three organizations actively interested in animal welfare as in the opinion of the Central Government ought to be represented on the Board, to be chosen by each of the said organizations in the prescribed manner;
- g. one person to represent of such three societies dealing with prevention of cruelty to animals as in the opinion of the Central Government ought to be represented on the Board, to be chosen in the prescribed manner;
- h. three persons to be nominated by the Central Government
- i. six Members of Parliament, four to be elected by the House of the People (Lok Sabha) and two by the Council of States (Rajya Sabha)

2. Any of the persons referred to in clause 9 a or clause (b) or *** clause (ba) or clause (bb) of sub-

section (1) may be depute any other person to attend any of the meetings of the Board.

*** Subs. by Act 26 of 1982: S. 5 (b),for the word,brackets and letter "Clause(b)

TERMS OF OFFICE AND CONDITIONS OF SERVICE OF MEMBERS OF THE BOARD

- **6. (1)The term for which the Board may be reconstituted under section 5 A shall be three years from the date of the reconstitution and the Chairman and other Members of the Board as so reconstituted shall hold office till the expiry of the term for which the Board has been so reconstituted.
- 2. Notwithstanding anything contained in sub-section (1):
 - (a) the term of office of an ex-officio Member shall continue so long as he holds the office by virtue of which he is such a Member;
 - (b) the term of office of a Member elected or chosen under clause (c), clause (e), clause (g), clause [h] clause (i) of section 5 to represent anybody of persons shall come to an end as soon as he ceases to be a Member of the body which elected him or in respect of which he was chosen;
 - (c) the term of office of a Member appointed, nominated, elected or chosen to fill a casual vacancy shall continue for the remainder of the term of office of the Member in whose place he is appointed, nominated, elected or chosen;
 - (d) the Central Government may, at any time, remove for reasons to be recorded in writing a member from office after giving him a reasonable opportunity of showing cause against the proposed removal and any vacancy caused by such removal shall be treated as casual vacancy for the purpose of clause (c).
- 3. The members of the Board shall receive such allowance, if any, as the Board may, subject to the previous approval of the Central Government, provided by regulations made in this behalf;
- 4. No act done or proceeding taken by the Board shall be questioned on the ground merely of the existence of any vacancy in, or defect in the constitution of the Board and in particular, and without prejudice to the generally of the foregoing, during the period intervening between the expiry of the term for which the Board has been reconstituted under section 5 A and its further reconstitution under that section, the ex-officio members of the Board shall discharge all the powers and function of the Board

SECRETARY AND OTHER EMPLOYEES OF THE BOARD

- 7. (1) The Central Government shall appoint *xxxxx the Secretary of the Board

- (2) Subject to such rules as may be made by the Central Government in this behalf, the Board may appoint such number of other officers and employees as may be necessary for the exercise of its powers and the discharge of its functions and may determine the terms and conditions of service of such officers and other employees by regulations made by it with the previous approval of the Central Government.

• FUNDS OF THE BOARD

8. The funds of the Board shall consist of grants made to it from time to time by the Government and of contributions, subscriptions, bequests, gifts and the like made to it by any local authority or by any other person.

FUNCTIONS

- To keep the law in force in India for the Prevention of Cruelty to Animals under constant study and to advise the government on the amendments to be undertaken in any such law from time to time.
- To advise the Central Government on the making of rules under the Act with a view to preventing unnecessary pain or suffering to animals generally, and more particularly when they are being transported from one place to another or when they are used as performing animals or when they are kept in captivity or confinement.
- To advise the Government or any local authority or other person on improvements in the design of vehicles so as to lessen the burden on draught animals.
- To take all such steps as the Board may think fit for amelioration of animals by encouraging, or providing for the construction of sheds, water troughs and the like and by providing for veterinary assistance to animals.
- To advise the Government or any local authority or other person in the design of slaughter houses or the maintenance of slaughter houses or in connection with slaughter of animals so that unnecessary pain or suffering, whether physical or mental, is eliminated in the pre-slaughter stages as far as possible, and animals are killed, wherever necessary, in as humane a manner as possible.
- To take all such steps as the Board may think *fit* to ensure that unwanted animals are destroyed by local authorities, whenever it is necessary to do so, either instantaneously or after being rendered insensible to pain or suffering.
- To encourage by the grant of financial assistance or otherwise, the formation or establishment of Pinjarapoles, rescue homes, animals shelters, sanctuaries and the like, where animals and birds may find a shelter when they have become old and useless or when they need protection.
- To co-operate with, and co-ordinate the work of associations or bodies established for the purpose of preventing unnecessary pain or suffering to animals or for the protection of animals and birds.
- To give financial assistance and other assistance to Animal Welfare Organisations functioning in any local area or to encourage the formation of Animal Welfare Organisations in any local area which shall work under the general supervision and guidance of the Board.
- To advise the Government on matters relating to the medical care and attention which may be provided in animal hospitals, and to give financial and other assistance to animal hospitals

whenever the Board think it is necessary to do so.

- To impart education in relation to the human treatment of animals and to encourage the formation of public opinion against the infliction of unnecessary pain or suffering to animals and for the promotion of animal welfare by means of lectures books, posters, cinematographic exhibitions and the like.
- To advise the Government on any matter connected with animal welfare or the Prevention of infliction of unnecessary pain or suffering on animals.

POWER OF THE BOARD TO MAKE REGULATIONS

10. The Board may, subject to the previous approval of the Central Government, make such regulations as it may think fit for the administration of its affairs and for carrying out its functions.

POLICY OF THE ANIMAL WELFARE BOARD OF INDIA

The Constitution of India Provides Animal Protection by Article 51 Ensuring "Compassion to all Living Creatures". Further, in pursuance of the recommendations of Datter Singh's Committee for providing prohibition of slaughter of cows and its progeny, the constitution in its Directive Principle of state policy provided protection to cow and its progeny by prohibiting slaughter of all milch animals, which especially identified cows and calf. In the above backdrop and constitutional mandate, the Govt. Of India by a Legislative Enactment by Parliament established the Animal Welfare Board of India under the provisions of the prevention of Cruelty to Animals Act, 1960.

The Animal Welfare Board of India has been constituted vide chapter II, Sections 4 to 10 of the Prevention of Cruelty to Animals Act, 1960. Section 9 lays down the functions of the board, while section 10 gives the Powers to make regulations, subject to the previous approval of the central government, to make regulations for carrying on its affairs and functions. The primary sources of funding of the AWBI are grants from the Government of India. However contributions, subscriptions, bequests and gifts will also be encouraged.

The Objective of the Board is to prevent any action resulting in the infliction of pain or cruelty and misuse of animals in the country. The Board should advise the government with regard to the development of instruments of law that will fulfill this objective. The board is unequivocal about its commitment to vegetarianism on ethical, moral and medical grounds. The Board in pursuance of the above constitutional and statutory mandate and directive have already resolved and demanded from the govt. of India, prohibition of meat export. In addition, the board is taking several other programmes for the saving animal life, very extreme pain of killing and slaughter like dog killing and cow slaughter etc. The Cattle population in India being the highest as $\frac{1}{4}$ of the world population about - 26 crores, the increasing emphasis is about its protection from slaughter in addition to the saving of dog population from mass killing by municipal corporations. Statement of Policy of the board (GUIDELINES OF THE FUNCTIONING OF THE BOARD) To achieve the above policy the

AWBI shall

A. In order to institutionalize the animal welfare movement in the country, the board will take action:

To establish State Animal Welfare Boards, to define their duties and to ensure that necessary provision of funds are made for the functioning of the board by the state.

To establish District Animal Welfare Boards which will monitor the implementation of PCA Act and the rules made there under, and to take action for the welfare of animals in the district.

- To make a database of NGOs in each district and to see that they liaise with and are helped by the State and District Boards and the local administration in their work.
- To establish one SPCA in each taluka, which will have a hospital/shelter, attached to it on land given by the local administration and which will have inspectors employed by it to check the abuse/misuse of animals.
- To derecognise SPCAs that are nonworking or misusing their mandate.
- To encourage the formation of animal welfare organizations and to give financial and other assistance.
- To monitor the activities of such NGOs.

B. In order to increase the number of groups working in the field of animal welfare and generate awareness on issues the Board shall: Make official master trainers/representatives of the Board whose duties will include monitoring all animal welfare activities and organizations working in the district, which is allotted to them and to conduct regular programmes of training and awareness generation for potential NGOs. They will also be put in charge of inspections in the districts. To conduct awareness programmes or help NGOs conduct awareness programmes through various media including pamphlets, TV and radio, workshops for specific issues.

- To conduct training/education programmes of specific target groups such as journalists, police, teachers and government officials.
- To conduct training Programmes of people already in the animal welfare movement in order to increase their effectiveness.

C. In order to strengthen the laws and implement the ones already in place, the board shall: Maintain a library of all existing laws which would include Constitutional Law, Central and state Acts, Rules, Regulations and bye-laws, Notification and Govt. Orders with judicial decisions upholding their implementation for animal protection, prohibition of slaughter, prohibition of animal sacrifices and prohibition and prosecution for them and other offences causing cruelty to animals. The Board will also suggest enactment of new laws or amendment in them or make within its powers such rules, regulations or issue administrative directions for enforcing and implementing them. Make a legal wing with a panel of lawyer's city wise that can be called to contest cases.

- Be proactive in initiation of cases involving issues of animal welfare.
- Issue a monthly newsletter on legal issues.
- Make suitable provision to assist NGOS with their legal cases.

D. In order to ensure that a network of shelters/hospitals is spread throughout India, the Board

shall: Help with financial aid in the running of such shelters. Assist in the augmentation of shelters already in existence and in providing for the construction of water troughs etc.

- The financial grants would be as far as possible in consonance with the board's decisions fixing the parameters and criteria for them in respect of the various NGOS functioning and requirements, capacity and working conditions and standards. The present criteria as fixed by the Board would continue till the Board further amends them.
- Recommend the funding of institution to the parent ministry for the construction of shelters/hospitals by established groups.

E. In order to make sure that the organizations that have been provided with funds from the Ministry or the board have utilized the funds prudently and honestly, the board will build up an effective system of inspections. This will be done by: Investigating all first time NGOS before any funds are released to them. Nominating Investigators who will be paid for their inspection.

- Blacklisting Fraudulent or incompetent organizations and informing the State and District authorities of their blacklisting.
- Moving for recovery of funds given to them.

F. The AWBI shall arrange for the transport or reimburse the costs incurred for the following: Where animals have been misused/abused or likely to be slaughtered and have to be transported to safe rescue centers and shelters and maintained as in the case of - Circuses Animals rescued from laboratories requiring rehabilitation Animals rescued from trains, trucks and other automobiles/vehicles Animals rescued from airports, from where they are being/illegally smuggled out/in The Board would be provided additional budget for the activities by the Ministry or the Govt. at the initiation of the ministry.

G. The board shall take all possible action to: Enforce the Animal Birth Control programme throughout India and will take every step necessary to see that the killing of dogs is also stopped by municipalities and replaced with vaccination and sterilization programmes Take every step to ensure that the sacrifice of animals for religious proposes is stopped.

- Take every step to ensure that the dissection of animals in schools, which has now been stopped by CBSE and ICSE, is also stopped by State Education Boards and is replaced by other models or systems of education.
- Regulate the use of animals in films and television by strictly enforcing the Performing Animal (Amendment) Rules, 2000 and regulating the granting of permission for the use of animals in the same, ensure presence and functioning of the appointees on the Censor Boards and by sending inspectors to the shooting of films and other locations where animals are being used for the purposes of entertainment. All producers and other people connected with the making of films be informed of the rules from time to time and it should be ensured that they have obtained the necessary approvals from the Board. The AWBI prohibits traveling menageries and exhibitions of animals.
- Regulate any form of the use of animals in sports through racing, rides. Bullock cart races and fights are forbidden under law and action be taken against any individual, organization or State Government using them. The Board also to monitor the laws and Rules which regulate horse racing.

- Monitor the use of animals in experiments and take action when animals are subjected to unnecessary pain and suffering or are unnecessarily used in any form of scientific or commercial venture.
- Inspect slaughterhouses, both municipal and private, to ensure that the - BSI rules, the PCA Act and the Rules made there under and other relevant Acts are being adhered to. Legal action be taken against slaughterhouses where rules are being violated.
- Regulate and oversee the transport of animals to such slaughterhouses is carried out in a legal and humane manner.
- Advise the animal husbandry departments with regard to the working of their veterinary and husbandry centers.
- Prohibit the use of animals such as birds, monkeys, mongooses, snakes, bears etc for street entertainment. All such animals will be confiscated and rehabilitated.
- Prohibit all blood sports such as dog fighting, cockfighting, ram fighting, snake - mongoose fighting etc.
- Prohibit the introduction of air or any chemical substance by injection into milch animals. Dairies that are using illegal chemicals/hormones/antibiotics etc. be shut down and legal action taken against them.
- Ban the release of impounded animals from shelters and pounds for biomedical research laboratories, training and educational institutions and pharmaceutical houses. Any animal welfare organization that voluntarily gives or sells animals to these institutions be blacklisted and proceeded against legally.
- Work towards the licensing and taxation of all dog and cat sellers.
- Check on factory farms and those animal husbandry institutions and commercial ventures that are involved in the raising, housing and selling of animals/birds for the purpose of meat. Any violations of the BSI and Prevention of Food Adulteration Act specifications will result in action taken.
- Encourage the use of natural pesticides and cow dung, cow urine for purposes of fertilizer, alternative sources of energy, medicinal purposes etc.
- Regulate the export/import of species and advise the government on the banning of particular species.
- Respond to complaints about the misuse and abuse of pets by their owners.
- Encourage and take whatever steps necessary for the improvement in the design of vehicles so as to lessen the burden on draught animals.
- Take action in respect of any activity that concerns the amelioration of pain and suffering to animals. H. H The Board would ensure participation of its representatives in the various committees and activities of the Planning Commission, Ministry of Agriculture Animal Husbandry, Ministry of Commerce and other Ministers including Animal Welfare for effectively co-ordination in suggesting and advising methodology for implementation of the above objectives, more particularly constitutional and statutory mandates for protecting from killing and slaughtering the animals. Its main thrust is in respect of the prohibition of dog killing and cattle killing and to provide and prohibit cruelties to all other animals also. It would also seek appropriate funding budget allotments and help from all the Ministries in addition to the parent Ministry of Animal Welfare, wherein it would like to evolve and participate in the Ministry's decisions of funding in all respects direct or indirect.

MODULE-4: ACTS AND RULES OF ANIMAL WELFARE - I

Learning outcome

This module focusing on the Acts and Rules are in force for Animal Welfare. This gives a detailed discussion on the rules related to animal welfare. The important laws and its uses in the prevention of cruelty against animals is major area.

ACTS AND RULES

Acts

- [No.59 of 1960 - The Prevention of Cruelty to Animals Act, 1960.](#)
 - The Prevention of Cruelty to Draught and Pack Animals Rules, 1965
 - The Prevention of Cruelty to Animals (Licensing of Farriers) Rules, 1965
 - The Performing Animals Rules, 1973
 - The Transport of Animals, Rules 1978
 - The Prevention of Cruelty to Animals (Application of Fines) Rules, 1978
 - The Prevention of Cruelty to Animals (Registration of Cattle Premises) Rules, 1978
 - The Prevention of Cruelty (Capture of Animals) Rules, 1972

Rules

- [S.O. 1995 \(E\) \[27/07/2010\] - Prevention of Cruelty to Animals \(Slaughter House\) Amendment Rules 2010.](#)
- [S.O.714\(E\), \[26/03/2010\] - Animal Birth Control\(Dogs\) Rules, 2009.](#)
- [S.O.1818\(E\), \[23/10/2006\] - Breeding of and Experiments on Animals \(Control and Supervision \)Amendment Rules, 2006.](#)
- [S.O.1817\(E\), \[19/10/2006\] - Reconstitution of Committee for the Purpose of Control and Supervision of Experiments on Animals\(CPCSEA\).](#)
- [S.O.42\(E\), \[14/01/2006\] - Breeding of and Experiments on Animals \(Control and Supervision \)Amendment Rules, 2005.](#)
- [S.O.1256\(E\), \[24/12/2001\] - The Animal Birth Control \(Dogs\) Rules, 2001.](#)
- [S.O.267\(E\), \[26/3/2001\] - The Performing Animals \(Registration\) Rules, 2001.](#)
 - [S.O.35\(E\), \[8/1/2002\] - The Performing Animals \(Registration\) Amendment Rules, 2001.](#)
- [S.O.271\(E\), \[26/3/2001\] - The Prevention of Cruelty to Animals \(Establishment and Regulation of Societies for Prevention of Cruelty to Animals\) Rules, 2001.](#)
- [S.O.270\(E\), \[26/3/2001\] - The Prevention of Cruelty to Animals \(Slaughter House\) Rules, 2001.](#)
- [S.O.268\(E\), \[26/3/2001\] - The Prevention of Cruelty to Animals \(Transport of Animals on Foot\) Rules, 2001.](#)
- [S.O.134\(E\), \[15/2/2001\] - The Breeding of and Experiments on animals \(Control and Supervision\) Amendment Rules, 2001.](#)
- [S.O.1074, \[15/12/1998\] - The Breeding of and Experiments on Animals \(Control and Supervision\) Rules, 1998.](#)
- [S.O.732\(E\), \[26/8/1998\] - The Experiments on Animals \(controls and Supervision\) \(Amendment\) Rules, 1998.](#)
- [S.O.No.1056, \[13/3/1979\] - The Prevention of Cruelty \(Capture of Animals\) Rules, 1972.](#)
- [No.14-20/76-LD.I, \[30/11/1978\] - The Prevention of Cruelty to Animals \(Registration of Cattle](#)

[Premises\) Rules, 1978.](#)

- [No. 18-6/70-LDI, \[23/3/1978\] - The Transport of Animals Rules, 1978 .](#)
 - [S.O.269\(E\), \[26/3/2001\] - The Transport of Animals \(Amendment\) Rules, 2001.](#)
- [No.14-21/76-LD.I, \[15/2/1978\] - The Prevention of Cruelty to Animals \(Application of Fines\) Rules, 1978.](#)
- [No.35-4/72-LD.I, \[22/5/1973\] - The Performing Animals Rules, 1973.](#)
- [No.9-18/62-LD, \[23/3/1965\] - The Prevention of Cruelty to Animals \(Licensing of Farriers\) Rules, 1965.](#)
- [No.9-18/68-LD, \[23/3/1965\] - The Prevention of Cruelty to Draught and Pack Animals Rules, 1965, amended 1968.](#)

THE PREVENTION OF CRUELTY TO ANIMALS ACT, 1960

The Prevention of Cruelty to Animals Act, 1960 (59 of 1960) (26th December, 1960)

An ACT

- To prevent the infliction of unnecessary pain or suffering on animals and for that purpose to amend the law relating to the prevention of cruelty to animals.

The Prevention of Cruelty to Animals Act, 1960 (59 of 1960)

As amended by Central Act 26 of 1982. **Arrangement of sections**

Chapter 1 - Preliminary

- Short title, extent and commencement.
- Definitions
- Duties of persons having charge of animals.

Chapter II - Animal Welfare Board Of India

- Establishment of Animal Welfare Board of India.
- Constitution of the Board.
 - Reconstitution of the Board.
- Term of Office and conditions of services of members of the Board.
- Secretary and other employees of the Board.
- Funds of the Board.
- Functions of the Board.
- Power of Board to make regulations.

Chapter III - Cruelty To Animals Generally

- Treating animals cruelty.
- Penalty for practising Phooka or doom dev.
- Destruction of suffering animals.

Chapter IV - Experimentation Of Animals

- Experiments on animals.
- Committee for control and supervision of experiments on animals.
- Sub-Committee.
- Staff of the Committee.
- Duties of the Committee and power of the Committee to make rules relating to experiments on animals
- Power of entry and inspection.
- Power to prohibit experiments on animals.
- Penalties.

Chapter V - Performing Animals

- "Exhibit" and "Trained" defined.
- Restriction on exhibition and training of performing animals.
- Procedure for registration
- Power of court to prohibit or restrict exhibition and training of performing animals.
- Power to enter premises
- Offences
- Exemptions.

Chapter VI - Miscellaneous

- Saving as respects manner of killing prescribed by religion.
- Power of court to deprive person convicted of ownership of animal.
- Presumptions as to guilt in certain cases.
- Cognizability of offences.
- Powers of search and seizure.
- Search warrants.
- General Power of Seizure for examination.
- Treatment and care of animals.
- Limitation of prosecutions.
- Delegation of powers.
- Power to make rules.
- Rules and regulations to be laid before Parliament.
- Persons authorised under section 34 to be public servants.
- Indemnity.
- Repeal of Act 11 of 1890.

**THE PREVENTION OF CRUELTY TO ANIMALS(SLAUGHTER HOUSE)
RULES,2001**

Q 28) What does law provides with respect to the slaughtering houses?

- A) There are certain rules provided under The Prevention Of Cruelty To Animals (Slaughter House) Rules, 2001.
- Section 2(c) of the above mentioned Act defines 'slaughter house' as a slaughter house wherein 10 or more than 10 animals are slaughtered per day and is duly licensed or recognized under a Central, State or Provincial Act or any rules or regulations made thereunder.

Q 29) What is the law regarding the slaughtering of animals?

- A) Section 3(1) of the above mentioned Act provides that animals cannot be slaughtered except in the recognized and licensed houses. Also, Section 3(2), prohibits slaughtering of any animal
 - which is pregnant or
 - has an offspring less than three months old, or
 - the animal which is under the age of three months or
 - has not been certified by a veterinary doctor that it is in a fit condition to be slaughtered.

Q 30) Is slaughtering of animal apart from the slaughter house forbidden?

- A) Wherever there is a Government slaughter-house, the slaughter cannot be done anywhere else. If there is no government slaughter house in that area then killing can only take place in licenced slaughter houses which should be situated where they are not public nuisance and an environmental hazard. These slaughterhouses have to follow the Municipal Corporation laws and the ISI regulations. Lambs or any other animals cannot be slaughtered in slums, in roadside ramshackle meat shops or in dhabas or in private houses.

Q31) Is it crime to sell camel's meat?

- A) Camel's meat is not notified item of food, as per the provisions of the prevention of cruelty to animals act,1960. Presently, provision is available, only for slaughtering cattle, goat, sheep and pig, within the Corporation limits. There is no qualified Veterinary Surgeon, who can certify the fitness of a camel or the suitability of its meat for consumption by human beings, or a licensed person to slaughter a camel. There are no licensed persons within the Corporation limits for the sale of camel's meat also. The license to sell beef will not enable the sale of camel's meat.

**THE EXPERIMENTS ON ANIMALS(CONTROL AND SUPERVISION)
RULES,1968**

Q 32) What does the law says regarding the conducting of experiments on animals?

A) Section 4 of the Experiments on Animals (Control and Supervision) Rules, 1968 lays down certain conditions regarding the conducting of experiments which are as follows :

1. Experiments should be performed with due care and humanity
2. Experiments shall be performed in every case by or under the supervision of persons duly qualified, in a laboratory adequately equipped and staffed for the purpose and under the responsibility of the person performing the experiment.
3. Minimum number of animals shall be used in an experiment
4. Experiments involving operative procedure more severe than simple inoculation or superficial venesection shall be performed under the influence of anaesthetic of sufficient power to prevent the animal feeling pain and it shall remain so throughout the experiment.
5. The experiment shall not be performed for the purpose of attaining or retaining manual skill.
6. Experiment shall not be performed by way of an illustration of lecture in schools or colleges
7. Experiments shall not be performed as a public demonstration except for advancement of knowledge
8. The substance known as Urari or Curari or any such paralysant shall not be used or administered for the purpose of any experiment except in conjunction with anaesthetic of sufficient depth to produce loss of consciousness;

Q 33) Is it illegal to sell animals for experiments?

- A) Yes, it is illegal to sale animals for experiments. Section 4A of The Experiments on Animals (control and Supervision) puts restriction on sale, etc., of animals for experiments. It says that no officer, employee or agent of any animal-control authority shall see, give transfer, trade, supply or otherwise provide any animal coming into his or her possession to any animal dealer, commercial kennel, pet shop, laboratory, educational institution or other person for the use in research, product development testing, education, biological production or other scientific, biomedical or veterinary purposes. Also the hospital, educational institution, laboratory or any person is prohibited to purchase or accept any dog or cat not purposely bred for research from any animal-control authority, commercial kennel, pet shop or animal dealer for use in research, product development, testing, education, biological production, or other scientific, biomedical or veterinary purposes.

MODULE-5: ACTS AND RULES OF ANIMAL WELFARE - II

Learning outcome

This module is discussing the following acts and rules.

- The Prevention of Cruelty to Animals (Transport of Animals on Foot) Rules, 2001
- The Breeding of and Experiments on animals (Control and Supervision) Amendment Rules, 2001
- The Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998
- The Experiments on Animals (controls and Supervision) (Amendment) Rules, 1998
- The Prevention of Cruelty (Capture of Animals) Rules, 1972
- The Prevention of Cruelty to Animals (Registration of Cattle Premises) Rules, 1978
- The Transport of Animals Rules, 1978
- The Transport of Animals (Amendment) Rules, 2001
- The Prevention of Cruelty to Animals (Application of Fines) Rules, 1978

MODULE-6: COMMITTEE FOR THE PURPOSE OF CONTROL AND SUPERVISION ON EXPERIMENTS ON ANIMALS

ROLE AND FUNCTION OF CPCSEA

- The Committee for the Purpose of Control and Supervision of Experiments on Animals (CPCSEA) has been constituted by the Government of India, under Sections 14 to Chapter IV of the Prevention of Cruelty to animals Act, 1960, for the purpose of controlling and supervising experiments performed on animals.

Duty of CPCSEA

- It is the duty of the committee to take all such measures as may be necessary to ensure that the animals are not subjected to unnecessary pain or suffering, before, during or after the performance of experiments on them.
- To achieve this purpose, the CPCSEA may, through notification in the Gazette of India, make such rules as it may think fit in relation to the conduct of such experiments.
- The CPCSEA will : Register institutions/establishments/breeders carrying out experiments on animals breeding animals. Call for and receive reports and other information as required from concerned persons/ institutions breeders Ensure that the experiments are carried out by qualified individuals and with full responsibility of the person in charge of the institution.
- Monitor and inspect the housing of animals of breeders/establishments and ensure that it is as per specified standards. Give permission to registered institutes/establishments for conducting experiments on animals. Regulate experiments on animals as per stipulated conditions and standards.
- Monitor the transfer and acquisition of animals for experiments. Participate in the meetings of the Institutional Animal Ethics Committees. Ensure that experiments are performed with due care and humanity and that as far as possible experiments involving invasive procedures/surgery are performed under the influence of some anaesthetic of sufficient power to prevent the animals feeling pain.
- Ensure that animals which in the course of the experiment under the influence of anaesthetic are so injured that their recovery would involve serious suffering are euthanised as per specified norms Ensure that experiments on animals are avoided whenever it is possible to do so and propagate the principles of 3 Rs that is to REDUCE, REDEFINE, and REPLACE the use of animals in experiments.
- Ensure that as far as possible experiments are not performed merely for the purpose of acquiring manual skill Ensure that the animals intended for the performance of the experiments are properly looked after both before and after experiments Ensure that required records are maintained with respect to experiments performed on animals.

CPCSEA GUIDELINES FOR LABORATORY ANIMAL FACILITY

Goal

- The goal of these Guidelines is to promote the humane care of animals used in specifications that will enhance animal well-being, quality in the pursuit of advancement of biological knowledge that is relevant to humans and animals.

VETERINARY CARE

- Adequate veterinary care must be provided and is the responsibility of a veterinarian or a person who has training or experience in laboratory animal sciences and medicine. Daily observation of animals can be accomplished by someone other than a veterinarian; however, a mechanism of direct and frequent communication should be adopted so that timely and accurate information on problems in animal health, behaviour, and well-being is conveyed to the attending veterinarian.
- The veterinarian can also contribute to the establishment of appropriate policies and procedures for ancillary aspects of veterinary care, such as reviewing protocols and proposals, animal husbandry and animal welfare; monitoring occupational health hazards containment, and zoonosis control programs; and supervising animal nutrition and sanitation.
- Institutional requirements will determine the need for full-time or part-time or consultative veterinary services.

QUARANTINE, STABILIZATION AND SEPARATION

- Quarantine is the separation of newly received animals from those already in the facility until the health and possibly the microbial status of the newly received animals have been determined. An effective quarantine minimizes the chance for introduction of pathogens into an established colony. A minimum duration of quarantine for small lab animals is one week and larger animals is 6 weeks (cat, dog, monkey, etc.)
- Effective quarantine procedures should be used for non-human primates to help limit exposure of humans to zoonotic infections.
- Regardless of the duration of quarantine, newly received animals should be given a period for physiologic, psychologic and nutritional stabilization before their use. The length of time stabilization will depend on the type and duration of animal transportation, the species involved and the intended use of the animals.
- Physical separation of animals by species is recommended to prevent interspecies disease transmission and to eliminate anxiety and possible physiological and behavioral changes due to interspecies conflict.
- Such separation is usually accomplished by housing different species in separate rooms; however, cubicles, laminar-flow units, cages that have filtered air or separate ventilation, and isolators shall be suitable alternatives.
- In some instances, it shall be acceptable to house different species in the same room, for example, if two species have a similar pathogen status and are behaviorally compatible.

SURVEILLANCE, DIAGNOSIS, TREATMENT AND CONTROL OF DISEASE

- All animals should be observed for signs of illness, injury, or abnormal behavior by animal house staff. As a rule, this should occur daily, but more-frequent observations might be warranted, such as during postoperative recovery or when animals are ill or have a physical deficit. It is imperative that appropriate methods be in place for disease surveillance and diagnosis
- Unexpected deaths and signs of illness, distress, or other deviations from normal health condition in animals should be reported promptly to ensure appropriate and timely delivery of veterinary medical care. Animals that show signs of a contagious disease should be isolated from healthy animals in the colony. If an entire room of animals is known or believed to be exposed to an infectious agent (e.g. *Mycobacterium tuberculosis* in non-human primates), the group should be kept intact and isolated during the process of diagnosis, treatment, and control. Diagnostic clinical laboratory may be made available.

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• MODULE-7: WELFARE OF ANIMAL USED FOR EXPERIMENTATION

ANIMALS USED FOR EXPERIMENTATION

- Right now, millions of mice, rats, rabbits, primates, cats, dogs, and other animals are locked inside cold, barren cages in laboratories across the country. They languish in pain, ache with loneliness and long to roam free and use their minds.
- Instead, all they can do is sit and wait in fear of the next terrifying and painful procedure that will be performed on them. The stress, sterility and boredom causes some animals to develop neurotic behaviors such as incessantly spinning in circles, rocking back and forth and even pulling out their own hair and biting their own skin. They shake and cover in fear whenever someone walks past their cages and their blood pressure spikes drastically. After enduring lives of pain, loneliness and terror, almost all of them will be killed.
- More than 100 million animals every year suffer and die in cruel chemical, drug, food and cosmetic tests, biology lessons, medical training exercises, and curiosity-driven medical experiments. Exact numbers aren't available because mice, rats, birds and cold-blooded animals—who make up more than 95 percent of animals used in experiments—are not covered by even the minimal protections of the Animal Welfare Act and therefore go uncounted. To test cosmetics, household cleaners, and other consumer products, hundreds of thousands of animals are poisoned, blinded, and killed every year by cruel corporations. Mice and rats are forced to inhale toxic fumes, dogs are force-fed pesticides, and rabbits have corrosive chemicals rubbed onto their skin and eyes. Many of these tests are not even required by law, and they often produce inaccurate or misleading results; even if a product harms animals, it can still be marketed to you.
- Millions of animals also suffer and die for classroom biology experiments and dissections, even though modern alternatives have repeatedly been shown to teach students better, save teachers time and save schools money.
- Each of us can help save animals from suffering and death in experiments by demanding that our alma maters stop experimenting on animals, by buying cruelty-free products, by donating only to charities that don't experiment on animals, by requesting alternatives to animal dissection and by demanding the immediate implementation of humane, effective non-animal tests by government agencies and corporations.

CRUELTY TO ANIMALS IN LABORATORIES

- Imagine living inside a locked closet without any control over any aspect of your life. You can't choose when and what you eat, how you will spend your time, whether or not you will have a partner and children, or who that partner will be. You can't even decide when the lights go on and off.
- Think about spending your entire life like this, even though you have committed no crime. This is life for an animal in a laboratory. It is deprivation, isolation, and misery.
- On top of the deprivation, there are the experiments. U.S. law allows animals to be burned, shocked, poisoned, isolated, starved, drowned, addicted to drugs, and brain-damaged. No experiment, no matter how painful or trivial, is prohibited – and pain-killers are not required. Even when alternatives to the use of animals are available, the law does not require that they be used—and often they aren't.
- Animals are infected with diseases that they would never normally contract, tiny mice grow tumors as large as their own bodies, kittens are purposely blinded, rats are made to suffer seizures, and primates' skulls are cut open and electrodes are implanted in them. Experimenters

force-feed chemicals to animals, conduct repeated surgeries on them, implant wires in their brains, crush their spines, and much more.

- After enduring these terrifying, painful procedures, animals are then usually dumped back into a cage without any painkillers. Video footage from inside laboratories shows animals who cower in fear every time someone walks by their cages. They don't know if they will be dragged from their prison cells for an injection, blood withdrawal, a painful procedure, surgery, or death. Often they see other animals killed right in front of them.
- No animals are safe from experimentation—primates, dogs, rats, mice, rabbits, pigs, and cats are just a few of the animals that are routinely used in these tests.

INFORMATION ABOUT ANIMAL TESTING

- As a consumer, you have a lot of power to help put an end to animal testing. By purchasing cosmetics and other products that are not tested on animals and by supporting compassionate charities, you send a powerful message to corporations and organizations that they will not receive your hard earned money as long as they continue conducting inhumane tests on animals.
- Hundreds of thousands of animals are poisoned, blinded, and killed every year in outdated product tests for cosmetics, personal-care products, household-cleaning products and even fruit juices.
- Although more than 1,100 companies have banned all animal tests forever, some corporations still force substances into animals' stomachs and drip chemicals into rabbits' eyes. These tests are not required by law, and they often produce inaccurate or misleading results—even if a product has blinded an animal, it can still be marketed to you.
- Fortunately, scientists have developed sophisticated product tests that are faster, cheaper, and far more accurate than blinding and poisoning tests, which were developed in the 1920s. Human cell cultures and tissue studies (in vitro tests) and artificial human “skin” and “eyes” mimic the body's natural properties, and a number of computer virtual organs serve as accurate models of human body parts.
- Please vow never to buy products from companies that use animals.
- And remember to support only compassionate charities when you write that end-of-the-year check! Some health charities ask for donations to help people with diseases and disabilities yet spend the money to bankroll horrific experiments on dogs, rabbits, rats, mice, primates, hamsters, pigs, ferrets, frogs, fish, guinea pigs, sheep, birds, and other animals.
- Instead of ravaging animals' bodies for cures for human diseases, compassionate charities focus their research where the best hope of treatment lies: with humans. They understand that we can improve treatments through up-to-date, non-animal methods, and they fund only non-animal research, leading to real progress in the prevention and treatment of disease.

- **ANIMAL TESTING IS BAD SCIENCE: POINT/COUNTERPOINT**

Animal experimenters want us to believe that if they gave up their archaic habit, sick children and other disease and accident victims would drop dead in droves. But the most significant trend in modern research in recent years has been the recognition that animals rarely serve as good models for the human body.

Studies published in prestigious medical journals have shown time and again that animal experimenters are often wasting lives—both animal and human—and precious resources by trying to infect animals with diseases that they would never normally contract. Fortunately, a wealth of cutting-edge, non-animal research methodologies promises a brighter future for both animal and human health. The following are some statements supporting animal experimentation followed by the arguments against them.

"Every major medical advance is attributable to experiments on animals."

This is simply not true. An article published in the esteemed *Journal of the Royal Society of Medicine* has even evaluated this very claim and concluded that it was not supported by any evidence. Most animal experiments are not relevant to human health, they do not contribute meaningfully to medical advances and many are undertaken simply of out curiosity and do not even pretend to hold promise for curing illnesses. The only reason people are under the misconception that animal experiments help humans is because the media, experimenters, universities and lobbying groups exaggerate the potential of animal experiments to lead to new cures and the role they have played in past medical advances.

"If we didn't use animals, we'd have to test new drugs on people."

The fact is that we already *do* test new drugs on people. No matter how many animal tests are undertaken, someone will always be the first human to be tested on. Because animal tests are so unreliable, they make those human trials all the more risky. The Food and Drug Administration (FDA) has noted that 92 percent of all drugs that are shown to be safe and effective in animal tests fail in human trials because they don't work or are dangerous. And of the small percentage that are approved for human use, half are relabeled because of side effects that were not identified in animal tests.

"We have to observe the complex interactions of cells, tissues, and organs in living animals."

Taking a *healthy* being from a completely different species, artificially inducing a condition that he or she would never normally contract, keeping him or her in an unnatural and distressful environment, and trying to apply the results to naturally occurring diseases in human beings is dubious at best.

,Physiological reactions to drugs vary enormously from species to species. Penicillin kills guinea pigs but is inactive in rabbits; aspirin kills cats and causes birth defects in rats, mice, guinea pigs, dogs, and monkeys; and morphine, a depressant in humans, stimulates goats, cats, and horses. Further, animals in laboratories typically display behavior indicating extreme psychological distress, and experimenters acknowledge that the use of these stressed-out animals jeopardizes the validity of the data produced.

"Animals help in the fight against cancer."

Since President Richard Nixon signed the Conquest of Cancer Act in 1971, the "war on cancer" in the United States has become a series of losing battles. Through taxes, donations, and private funding, Americans have spent almost \$200 billion on cancer research since 1971. However, more than 500,000 Americans die of cancer every year, a 73 percent increase in the death rate since the "war" began.

"Science has a responsibility to use animals to keep looking for cures for all the horrible diseases that people suffer from."

While funding for animal experimentation and the number of animals tested on continues to increase, the United States still ranks 49th in the world in life expectancy and second worst in infant mortality in the developed world. While rates of heart disease and strokes have shown slight declines recently—because of lifestyle factors such as diet and smoking rather than any medical advances—cancer rates continue to rise, while alcohol- and drug-treatment centers, prenatal care programs, community mental health clinics, and trauma units continue to suffer closures because they lack sufficient funds.

"Many experiments are not painful to animals and are therefore justified."

No experiment, no matter how painful or trivial, is prohibited – and pain-killers are not even required. Even when alternatives to the use of animals are available, the law does not require that they be used—and often they aren't. Because the Act specifically excludes rats, mice, birds and cold-blooded animals, more than 95 percent of the animals used in laboratories are not subject to the minimal protections provided by federal laws. Because they are not protected by the law, experimenters don't even have to provide mice and rats with pain relief.

"We don't want to use animals, but we don't have any other options."

Human clinical and epidemiological studies, human tissue- and cell-based research methods, cadavers, sophisticated high-fidelity human patient simulators and computational models are more reliable, more precise, less expensive, and more humane than animal experiments. Progressive scientists have used human brain cells to develop a model "microbrain," which can be used to study tumors, as well as artificial skin and bone marrow. We can now test irritancy on protein membranes, produce and test vaccines using human tissues, and perform pregnancy tests using blood samples instead of killing rabbits. Animal experiments don't persist because they are the best science, they persist because of experimenters' personal biases and archaic traditions.

"Don't medical students have to dissect animals?"

Nearly 95% of U.S. medical schools—including Yale, Harvard and Stanford—do not use any animals to train medical students and experience with animal dissection or experimentation on live animals is not required or expected of those applying to medical school. Medical students are trained with a combination of didactic methods, sophisticated human patient simulators, interactive computer programs, safe human-based learning methods and clinical experience. Today, one can even become a board-certified surgeon without harming any animals. Some medical professional organizations like the American Board of Anesthesiologists even require physicians to complete simulation training—not animal laboratories—to become board-certified.

"Animals are here for humans to use. If we have to sacrifice 1,000 or 100,000 animals in the hope of benefiting one child, it's worth it."

MODULE-8: WILDLIFE PROTECTION AND WELFARE

Learning outcome

Wildlife conservation is the preservation, protection, or restoration of wildlife and their environment, especially in relation to endangered and vulnerable species. All living non-domesticated animals, even if bred, hatched or born in captivity, are considered wild animals. Wildlife represents all the non-cultivated and non-domesticated animals living in their natural habitats. Our world has many unique and rare animals, birds and reptiles. However the pressure of growing population in different parts of the world has led to the increasing need of using land for human habitations and agriculture. This has led to the reduced habitat of many wild animals.

Hence Ministry of Environment and Forests, GoI implements wild life protection act into force to conserve the wildlife in India. This module mainly focusing on this area.

INTRODUCTION

- Perhaps the greatest challenge that we face in fighting cruelty is changing the way that humans view animals. Each year, millions of animals are killed because they are considered pests. Beavers, bats, geese, deer, pigeons, mice, raccoons, snakes, and squirrels are among those animals who most often suffer horrific deaths because some consider them a nuisance, but the list also includes bears, coyotes, ducks, foxes, mountain lions, prairie dogs, rabbits, and even wolves.
- Because of rapid urban and rural development and agricultural land use in the U.S., wild animals are forced to live in closer proximity to humans. Although they prefer natural, undisturbed areas, evicted and homeless wild animals have no choice but to seek food and shelter in buildings, parks, and even downtown areas. When they do, homeowners, communities, municipalities, and state agencies often hire nuisance-wildlife control operators (e.g., trappers, exterminators, pest-removal companies) or the U.S. Department of Agriculture Wildlife Services to use cruel, archaic methods that rely on inhumane tools such as steel-jaw traps and poisons to resolve perceived conflicts with these animals.
- Ducks and geese are shot, gassed, and even bludgeoned to death with baseball bats. Beavers are caught and drowned in body-gripping traps or are sealed up alive in their own homes. Squirrels, birds, mice, rats, and other small animals are ensnared by sticky traps and suffer when skin, fur, and feathers are ripped from their bodies as they struggle to free themselves from the powerful adhesive. Animals of all kinds are often discarded with the traps and left to die from stress, injuries, self-mutilation, dehydration, or starvation.
- Not only are these methods cruel, they are also largely ineffective because, without modifying areas to exclude newcomers, more animals simply move in to replace those who were killed, resulting in a perpetual, vicious kill cycle.

PROTECTION OF WILD LIFE IN NATURE AND CAPTIVITY

Wildlife Conservation Division

- This Division deals with all matters relating to national parks and wildlife sanctuaries not covered by the Project Tiger and the Project Elephant Divisions.
- The X Five Year Plan outlay for works relating to this Division is of the order of Rs 485 crores. The Division also acts as a nodal point for the Wildlife Institute of India and the Central Zoo Authority, which are autonomous bodies under the administrative control of the Government of India.
- The two organizations receive support from the Government of India in the form of grants processed by the Wildlife Division. The details on these two bodies are given separately.
- The Division also handles the Centrally Sponsored Scheme “Development of National Parks and Sanctuaries” and the Central Sector Scheme “Strengthening of Wildlife Division and Consultancies for Wildlife Conservation.”
- The Wild Life (Protection) Act, 1972 has been amended with effect from 1 April 2003. The amendments include, inter-alia, provisions that flow from the National Wildlife Action Plan adopted by the country in 2002.
- Two new categories of protected areas, namely Conservation Reserves and Community Reserves, have been incorporated in the amended Act to facilitate coverage of all biogeographic zones, forest types and wild species of flora and fauna, and peoples’ involvement in establishment and management of such protected areas.
- The amendment has also facilitated the issue of certificates of ownership to the bona-fide possessors of animals, animal articles, trophies, etc., derived from animals listed in Schedule I and part II of Schedule II of the Act, who had not been able to declare their possessions earlier.
- The law has become stricter for the offenders. It also makes clear that PAs which have areas that have had rights extinguished under any legislation, will be deemed to be finally notified (Sections 26-A(b) and Explanation U/S 35(8) of the Act)

CONSERVATION OF NATIONAL PARKS AND SANCTUARIES

- The Government of India through a Centrally Sponsored Scheme “Development of National Parks and Sanctuaries” provides the financial assistance to national parks and sanctuaries managed by the State Governments.
- The scheme provides 100% Central assistance on items of works of non-recurring nature. There are a few identified items of recurring nature which are essential and which need support for a few years. The scheme provides assistance on such items on a 50% sharing basis, the matching share coming from the State Government concerned.
- Under the scheme, an assistance of Rs 72.28 crores was provided to the States during the IX Five Year Plan. The outlay for the X Five Year Plan is Rs 350 crores, which includes the merged schemes for Eco-development and Tribal Rehabilitation.

STRENGTHENING OF WILDLIFE DIVISION AND CONSULTANCIES

- Under this Centrally Sponsored Scheme the infrastructural and conservational requirements of the Wildlife Division are met. This Division handles the works of the four sub-ordinate offices of the Deputy Directors, Wildlife Preservation located at Kolkata, Chennai, Mumbai and Delhi, with their supporting offices at Amritsar, Guwahati and Cochin.
- The function of these offices is to monitor and take measures to check the international trade in wildlife and wildlife articles passing through the ports of entry into and exit from the country. Besides, research proposals from independent research agencies and institutions on applied aspects of wildlife conservation, are also provided support from this head.
- There are 10 ongoing research projects, dealing mainly with applied wildlife conservation undertaken by various organizations including the BNHS (4), Institute of Environment Education and Research, Pune (1), University of Patna (1), Garhwal University (1), Nature Environment and Wildlife Society (1), Gene Campaign (1) and the Chilika Development Authority (1). The subjects covered relate to wildlife habitats in the Dangs, Rajaji National Park, Western Ghats, ecological studies on the Gangetic Dolphin, Irravady Dolphin, forest spotted owl, vultures, spot-billed pelican, endangered wildlife in West Bengal and genetic diversity in the Western Ghats.
- The duration of research projects varies between one year and three years and the total support asked for is Rs 88.34 lakhs. Nine more are in the pipeline. Organization of meetings, workshops, events, awards, etc., is also covered under this scheme. An amount of Rs 10 crores is provided as outlay for this Scheme for the X Five Year Plan.

• MINISTRY OF ENVIRONMENT AND FOREST - WILDLIFE AND CONSERVATION

The Government of India enacted Wild Life (Protection) Act 1972 with the objective of effectively protecting the wild life of this country and to control poaching, smuggling and illegal trade in wildlife and its derivatives. The Act was amended in January 2003 and punishment and penalty for offences under the Act have been made more stringent.

ACT

1. No. 16 of 2003, [17/1/2003] - The Wild Life (Protection) Amendment Act, 2002
2. The Indian Wildlife (Protection) Act, 1972, amended 1993.

BILL

1. Draft Wild Life (Protection) Amendment Bill 2010
2. Statement of Objects and Reasons
3. Explanatory Note

RULES

1. Recognition of Zoo Rules, 2009. English, Hindi
2. S.O.1092(E), [22/9/2003] - The National Board for Wild Life Rules, 2003.
3. S.O.445(E), [18/4/2003] - The Declaration of Wild Life Stock Rules, 2003.
4. G.S.R.350(E), [18/4/1995] - The Wildlife (Specified Plant Stock Declaration) Central Rules, 1995.
5. G.S.R.349(E), [18/4/1995] - The Wildlife (Specified Plants - Conditions for Possession by Licensee) Rules, 1995.
6. G.S.R.348(E), [18/4/1995] - The Wildlife (Protection) Rules, 1995.
7. Recognition of Zoo Rules, 1992.
8. G.S.R.328(E), [13/4/1983] - The Wildlife (Protection) Licensing (Additional Matters for Consideration) Rules, 1983.
9. G.S.R.29(E), [25/1/1973] - The Wildlife (Stock Declaration) Central Rules, 1973.
10. G.S.R.198(E), [9/4/1973] - The Wildlife (Transaction and Taxidermy) Rules, 1973.

NOTIFICATIONS

1. S.O.2180(E), [4/9/2010] - Amendments to Constitution of the National Board for Wild Life.
2. S.O.802(E), [16/05/2007] - The National Board for Wild Life Rules, 2007.
3. S.O.1093(E), [22/9/2003] - Constitution of the National Board for Wild Life.
4. S.O.1091(E), [22/9/2003] - Coming into force of section 6 of the Wild Life (Protection) Amendment Act, 2002 (16 of 2003).
5. S.O.446(E), [18/4/2003] - Delegation of Powers of section 58E of the Wild Life (Protection) Act, 1972 (53 of 1972).
6. S.O.447(E), [18/4/2003] - Delegation of Powers of section 54 of the Wild Life (Protection) Act, 1972 (53 of 1972).
7. S.O.332(E), [28/3/2003] - Coming into force of all the provisions except section 6 of the Wild Life (Protection) Amendment Act, 2002 (16 of 2003).
8. S.O.1085(E), [30/9/2002] - Amendments to Schedule I and Part II of Schedule II of the Wild Life (Protection) Act, 1972 (53 of 1972).
9. S.O.1197(E), [5/12/2001] - Amendments to Schedule I and Schedule IV of the Wild Life (Protection) Act, 1972 (53 of 1972).
10. S.O.665(E), [11/7/2001] - Amendments to Schedule I and Schedule III of the Wild Life (Protection) Act, 1972 (53 of 1972).
11. S.O.474(E), [28/5/2001] - Additions to Schedule I of the Wild Life (Protection) Act, 1972 (53 of 1972).

t. The Ministry has proposed further amendments in the law by introducing more rigid measures to strengthen the Act. The objective is to provide protection to the listed endangered flora and fauna and ecologically important protected areas.

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GUIDELINES FOR APPOINTMENT OF HONORARY WILDLIFE WARDENS

The Need

1. People's participation and support is crucial for nature and Wildlife conservation, One of the important ways of enlisting such support is by involving the community leaders and other persons of standing, who have the interest as well as the capacity to render assistance for this cause. Such assistance can be very useful in control over poaching for this clandestine trade in wild animals or their articles, identification of relatively less known wildlife refuges needing protection, carrying the message of conservation to the people living in and around the sanctuaries and national parks, and related matters. This objective can be accomplished if really suitable public men are identified, duties and appointed Honorary Wildlife Wardens, with their responsibilities, and powers clearly defined.

Legal Status

2. Section 4 of the Wildlife (Protection) Act. 1972 empowers the State Government to appoint

(a) a Chief Wildlife Warden

(b) Wildlife Wardens; and

(c) such other officers and employees as may be necessary for the purposes of the Act.

Honorary Wildlife Wardens can be appointed under sub-section (c) of section 4 of the Act. Under section 59 of the aforesaid Act, such Honorary Wildlife Wardens shall be deemed to be public servants within the meaning of section 21 of the Indian Penal Code.

Criteria for Selection

3. It is very important that the right persons are selected for appointment as Honorary Wildlife Wardens. Every State has a Chief Wildlife Warden and it is mainly his duty to recommend the names of suitable person for this purpose. However, in order to assist him in this regard as well as to introduce a measure of wider participation, the members of the State Wildlife Advisory Board should be requested to suggest suitable names, especially from their own areas.

4. The following criteria should be kept in mind while assessing the suitability of a person as an Honorary Wildlife Warden:

(a) Genuine concern for Wildlife conservation.

(b) Personal record free of involvement in any activity detrimental to the interests of nature and wildlife conservation.

Any person involved in commercial exploitation of Wildlife should not be considered.

(c) Capacity to render help to the official machinery.

(d) Local standing which make him/her effective, especially in conveying the conservation message.

5. An important point to bear in mind is the identification of areas particularly prone to poaching, e.g. forests in the vicinity of urban centres and cantonments or close to sanctuaries and national park. Likewise, centres of clandestine trade in wildlife and products thereof should be identified; so also areas where damage to people or their property from wild animals is heavy. Selection of persons as Honorary Wildlife Wardens must be related to such problem areas because it is these areas which need priority attention and where public participation is needed most.

Procedure for Appointment

- Under Section 6 of the Wildlife (Protection) Act, 1972, every State and Union Territory has a Wildlife Advisory Board to aid and advise the Government in matters connected with the protection of wildlife. The appointment of an Honorary Wildlife Warden should be generally with the recommendation of this Advisory Board. The Chief Wildlife Warden should submit the proposals for this purpose at the meetings (s) of the Board and then seek the orders of the Government.
- While recommending any person for such appointment, the criteria laid down in paras 4 and 5 above must be kept in mind by the Board. The appointment of an Honorary Wildlife Warden should, in the first instance, be generally for a period of one year. Thereafter on the recommendation of the Wildlife Advisory Board, it may be renewed for a period not exceeding 2-3 years at a time.
- The Wildlife Advisory Board of each State/Union Territory should review the functioning of the scheme of Honorary Wildlife Wardens at least once every year.
- The appointment order of an Honorary Wildlife Warden should clearly specify the jurisdiction, which should normally be a district or a few districts, in the area where the person resides. However, there is no objection to making members of the State Wildlife Advisory Board Honorary Wardens for larger areas.
- Each Honorary Warden should be issued an identity Card having his signature and photograph duly attested by the Chief Wildlife Warden. The Chief Wildlife Warden should also give each Honorary Warden a small booklet containing the Wildlife (Protection) Act and the Rules made thereunder as well as the duties, responsibilities, and power of an Honorary Wildlife Warden.
- The State Government may, at its discretion, terminate the appointment of an Honorary Wildlife Warden at any time, without assigning reasons.

Duties and Responsibilities

13. The main duty and responsibility of an Honorary Wildlife Warden is to assist whole heartedly the State organization responsible for wildlife conservation work, especially with regard to the following matters :

- (a) Control of poaching and clandestine trade in wild animals and products / articles thereof
- (b) Detection and prosecution of offences under the Wildlife (Protection) Act and the Rules made thereunder.
- (c) Preventing damage to the habitat of wildlife.
- (d) Identification and selection of areas suitable to be declared as sanctuaries, national parks, closed areas, etc; as well as measures for their proper protection.
- (e) Measures for dealing with the problem of damage by wild animals to life and property, including the assessment and payment of compensation, etc.
- (f) Carrying the message of conservation to the people and enlisting public support for nature and wildlife conservation. The effort should be specially directed towards the communities living in or near the declared wildlife reserves.
- (g) Any other matter connected with the protection of wildlife, which may be entrusted by the Wildlife Advisory Board or the Chief Wildlife Warden of the State, from time to time.

Powers

14. In accordance with sub-section (3) of section 4 of the Wildlife (Protection) Act, 1972, an Honorary Wildlife Warden appointed under sub-section (2) (c) of section 4 shall be subordinate to the Chief Wildlife Warden of the State and under section 59 of aforesaid Act, he shall be deemed to be a public servant within the meaning of section 21 of the Indian Penal Code. Protection for action taken in good faith is provided under section 60 of the Act.

15. With a view of making the Honorary Wildlife Wardens useful and effective it is necessary that the following specific powers under the Wildlife (Protection) Act, 1972 should be delegated to them:

- (a) Power to inspect records of licences under section 47 (b) of the Act;

(b) Powers of entry, search, seizure, and detention under section 50 for prevention and detection of offences under the Act.

16. Suitable Honorary Wildlife Wardens could be authorised also to file complaints in courts in accordance with section 55 of the Wildlife (Protection) Act, 1972. Normally, however, an Honorary Wildlife Warden should bring the offence detected by him to the notice of the Wildlife Warden having jurisdiction for making proper investigation and lodging a complaint in the Court as laid down in section 55 of the Act.

17. Apart from the above, the State Government may delegate any other power under the aforesaid Act, as it may consider necessary.

General

18. Just as it is expected that the Honorary Wildlife Wardens should assist the State Wildlife organization, it is equally essential that the Chief Wildlife Warden and the whole State Machinery responsible for the protection of wildlife should take all possible steps to associate the Honorary Wildlife Wardens in their work. This can be achieved best by fostering a spirit of mutual trust and confidence.

19. No staff or vehicle support can be provided to Honorary Wardens as a matter of course. However, if the circumstances warrant, the departmental staff should provide all possible help and assistance. Instructions to this effect should be issued by the State Government to all concerned officers in the field.

20. It is also appropriate that the actual expenses incurred by an Honorary Warden on travel by public transport for carrying out the duties assigned to him should be reimbursed by the State Government. In addition, all actual expenses incurred in the detection of an offence under the Wildlife (Protection) Act, which leads to successful prosecution may be reimbursed after due verification.

21. The State Government should recognise outstanding work or service rendered by any Honorary Warden. Such recognition can be by way of a letter of commendation, or a certificate signed by the Minister in charge of the Department, or the membership of the State Wildlife Advisory Board. Cash grants could also be considered in suitable cases.

MODULE-9: PERFORMING ANIMALS AND ITS WELFARE

Learning outcome

This module focusses on the protection of performing animals,

- to provide sanctuary to the abused, abandoned and retired captive wildlife,
- to enforce the best standards of care for all captive wildlife,
- to preserve the wild species and their habitat
- and to promote public education about the captive wildlife issues.

ANIMALS USED FOR ENTERTAINMENT

- Chimpanzees, bears, tigers, elephants, and other animals aren't actors, spectacles to imprison and gawk at, or circus clowns. Yet thousands of these animals are forced to perform silly, confusing tricks under the threat of physical punishment; are carted around the country in cramped and stuffy boxcars or semi-truck trailers; are kept chained or caged in barren, boring, and filthy enclosures; and are separated from their families and friends—all for the sake of human "entertainment." Many of these animals even pay with their lives.
- Animals are used extensively in the entertainment industry, including in circuses; zoos and pseudo-sanctuaries; marine parks; the exotic "pet" trade; advertisements, television shows, and movies; cruel "sports" such as bullfighting, rodeo events, and horse racing; and more. Businesses that exploit animals exist to make money, so the animals' needs are usually put last.
- Bears, elephants, tigers, and other animals used in circuses do not voluntarily ride bicycles, stand on their heads, balance on balls, or jump through rings of fire. To force them to perform these confusing and physically demanding tricks, trainers use bullhooks, whips, tight collars, muzzles, electric prods, and other painful tools of the trade. When they're not performing, elephants are often kept shackled by two legs, and lions, tigers, bears, primates, and other animals are forced to eat, sleep, and relieve themselves in tiny cages.
- Zoos, which sentence intelligent, social animals to live in cages for our fleeting distraction and amusement, aren't much better. Extremely bored, depressed, and deprived of everything that is natural and important to them, many animals in zoos literally lose their minds—a condition called "zoochosis." Animals with zoochosis engage in neurotic behaviors such as pacing, spinning, and bobbing their heads. They also mutilate themselves, become overly aggressive, throw feces, and engage in other abnormal behaviors that are not seen in the wild.
- Animals used in rodeos, horse racing, dog-sled racing, and other cruel "sports" are forced to run for their lives. When they aren't being used in competitions, they are usually kept chained or in cages or stalls. Those who don't "make the cut" are often casually discarded by being sent to slaughter or are destroyed.

CIRCUS ANIMALS

- Wild animals have no place in a crowded city, and certainly not in a cage. They are genetically adapted to a particular life style, and have their own habits of eating food, marking their territory, socializing, etc. The concept of a zoo ruins all this natural behavioral aspects, under the pretext of education. Worst is the case of 'circus' animals, who are not only deprived of their natural lifestyle, but they are also made to walk on their hind legs, ride a motor-cycle, etc. It has been amply proved

that while animals cannot speak or complain, they can most certain feel the humiliation, through which they are put, by forcing them into such acts before large crowds. The monkeys are closest to humans and have intelligence of a related child. When we cannot imagine putting a retarded child through such trauma, how can we permit such treatment to an animal that is equally intelligent, but totally helpless?

- The circus trade is inherently sinful and gives wrong signals to children. They teach children to look at fellow living creatures merely as objects of pleasure and entertainment. There is absolutely no education in all this.
- Animals working in a circus are tortured all their lives. They are forced to perform on stage for an audience who is least bothered of their pain. Wild animals like tigers, lions, elephants, etc., are forced to perform shows and look fierce to keep the audience captured with awe. No one ever notices the tight collars, the whips lashing on their bodies, beatings and deprivation of food. Have you ever gone backstage and had a look at the living conditions that these animals are kept in? It is against their wild nature to jump around prancing like a kitten. And when an animal cannot take the torture anymore, he attacks the abusers. These animal abusers are then labeled victims and the poor animal is shot dead. The old, sick and weak animals who cannot perform anymore are sold off to people who too are looking for a fast buck. These animals are killed in the most gruesome way, one cannot even imagine. Their fur, body parts, bones are sold off illegally. Few are sold off illegally in pet trade to people who have no idea about their care. These poor animals are either abandoned or further abused by the ignorant pet owners.

PROTECTION AND WELFARE OF PERFORMING ANIMALS

Care and Management

- To responsibly provide quality, long-term care to the growing number of animals at sanctuaries, it is vitally important for the organization to be financially strong, now, and well into the future.

Elephants

- Elephants are highly sensitive, intelligent animals who rely on an intact, functioning social order to prevent stress and maintain normal physical and mental health. Profound disruptions in routine and changes in environment and companions can cause trauma to individual elephants.
- The removal of an individual elephant from familiar surroundings and companions is a highly traumatic experience which can cause physical and psychological problems and, occasionally, death. All proposed elephant moves should be carefully studied and alternative solutions explored before a final decision is made to relocate individual elephants.
- If a situation requires removal of an elephant to provide a better quality of life, careful planning and coordination of effort is essential. Trailer training before transport is critical for the safety of the elephant, and may take months to achieve. Additional construction is often necessary to facilitate the process and should be completed before the elephant is moved.

Other animals considered as performing animals are

- Bears
- Larger felines
- Smaller felines
- Small primates

PRERFORMING ANIMALS RULES, 1973

In exercise of the powers conferred by section 38 read with section 37 of the Prevention of Cruelty to Animals Act, 1960 (59 of 1960), the Central Government hereby makes the following rules, namely:

1. Short Title and commencement

- These rules may be called the Performing Animals Rules, 1973.
- They shall come into force in any State on such date as the State Government may, by notification in the Official Gazette, appoint.

2. Definitions

In these rules, unless the context otherwise requires

- "Act" means the Prevention of Cruelty to Animals Act, 1960.
- "performing animal" means any animal which is used at, or for the purpose of any entertainment to which the public are admitted through sale of tickets.
- "prescribed authority" means the State Government or such other authority as the State Government may, by general or special order, specify in this behalf.
- "schedule" means a schedule appended to these rules.

3. Application for registration

- Every application by a person desirous of exhibiting or training any performing animal for registration under the Act shall be in the form and shall contain the particulars set out in the first Schedule.
- Every such application shall be made to the prescribed authority within whose jurisdiction the applicant ordinarily resides and, if he has no fixed place of residence, the application shall be made to such authority as the Central Government may by order specify in this behalf.

4. Fee and Registration

- Every application for registration shall be accompanied by a fee of rupees twentyfive which may be paid either in cash or in such other manner as may be specified by the prescribed authority.

5. Form of Certificate of Registration

- The certificate of registration to be issued by the prescribed authority shall be in the form set out in the Second Schedule.
- Every registration shall be given a serial number according to the order in which it is made, and the serial number shall be inserted in the certificate of registration issued to the applicant.

6. Register

- Every person to whom a certificate of registration is issued under these rules shall have his name entered in a register which shall be kept in the form set out in the Third Schedule.

7. Inspection of Register

- The register kept under these rules shall be open to inspection during office hours on any working day on payment of a fee of two rupees and any person may take extract therefrom or may require the prescribed authority to issue to him a certified copy of any entry therein on payment of a fee of five rupees.

8. Application for variation of entries in register

- Every application under sub-section (5) of section 23 of the Act for the variation of any particulars entered in the register shall be in the form set out in the Fourth Schedule and when any particulars are varied the existing certificate of registration shall be cancelled and a new certificate issued to the applicant.

9. Issue of duplicate copies of certificates

- Any person whose name is registered under these rules may, on proof by him that the original certificate of registration has been lost or destroyed and on payment of a fee of rupees five, be given a duplicate copy of the certificate of registration which for the purpose of these rules shall have the same effect as the original certificate of registration.

10. Copies of certificates etc. to be sent to the Animal Welfare Board of India

- The prescribed authority shall cause a copy of every certificate of registration or duplicate thereof or a new certificate issued under these rules to be sent to the Animal Welfare Board of India established under the Act, as soon as may be after it is issued.

FIRST SCHEDULE

FIRST SCHEDULE

(SEE RULE 3)

Form of Application

I, the undersigned, do hereby apply for registration under the Performing Animal Rules, 1973 and do hereby declare the following particulars to be true and complete to the best of my knowledge and belief.

Signature

Date

Address to which certificate of

registration is to be sent.

.....

..

PARTICULARS

1. Full name of applicant(in block letters)
2. State name (if any used in India)
3. Nationality
4. Either (a) address of fixed place of

residence in India, or (b) permanent

postal address in India to which letters

addressed to the applicant may be forwarded.

1. Address or addresses (if any) in India, other than temporary addresses while on tour, at which applicant trains or intends to train performing animals. (If none, write, "None".)
2. State whether previously registered under the Performing Animals Rules, 1973. If so, state the number and date of certificate of registration.
3. (i) Kinds of performing animals

proposed to be To be trained To be exhibited Kind Number Kind Number

(a) trained,

(b) exhibited, Stating number of each kind Kind Number

(ii) Trained animals already available for being exhibited.

8. Describe briefly the general nature* of the performance or performances in which the performing animals are to be exhibited or for which they are to be trained, mentioning any apparatus which is used for the purpose of the performance.

Note

* The description must be sufficient to give a general idea of what is done by the animals taking part in the performance, and should state the approximate duration of the performance, the number of times for which it is usually to be given in one and the same day, and the number of animals of each kind taking part in the performance. It need not give details which would divulge any professional secret.

SECOND SCHEDULE (SEE RULE 5)**Certificate of Registration**

This is to certify that the person to whom the under- Serial Number of Entry mentioned particulars relate has this day been registered in Register under the performing Animals Rules, 1975 with the registration authority for the

Name of the place:

Date:

Signature of Clerk or registration authority:

PARTICULARS

Name of trainer or exhibitor	Nationality	Either (a) Address of fixed place of residence in India or (b) Permanent postal address in India to which letters addressed to the trainer or exhibitor may be forwarded	Address or addresses at which the performing animals are to be trained	Particulars of any previous registration	Kinds of performing animals	Description of general nature of performance	Date of Registration	Particulars of any order of Court made under section 24 of the Prevention of Cruelty to Animals Act, 1960

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THIRD SCHEDULE (SEE RULE 6)

Form of Register

Name of trainer or exhibitor	Nationality	Either (a) Address of fixed place of residence in India or (b) Permanent postal address in India to which letters addressed to the trainer or exhibitor may be forwarded	Addresses or addresses at which the performing animals are to be trained	Particulars of any previous registration	Kinds of performing animals		Description of general nature of performance	Date of Registration	Particulars of any order of Court made under section 24 of the Prevention of Cruelty to Animals Act, 1960
					To be trained	To be exhibited			
					Kind No.	Kind No.			

1	2	3	4	5	6	7	8	9	10

FOURTH SCHEDULE (SEE RULE 6)

Form of application for variation of particulars entered in register

Application to have the particulars entered in register with respect to the applicant varied.

To

The Prescribed Authority,

Full Name of applicant : (in block letters)

Number and date of certificate of registration

I return herewith my certificate of registration under the Performing Animals Rules, 1973, and I hereby apply to have the particulars entered in the Register with respect to be varied, as follows and for the reasons given below :

I also request that my existing certificate may be cancelled and a new certificate of registration may be issued to me.

Signature

.....

Address

.....

.....

.....

.....

Note: No fee is payable for the issue of a new certificate of registration.

Notified in Gazette of India, Part 11 Section 3, Sub-section (ii) , Vide Govt. of India, Ministry of Agriculture No. 35-4/72-LD I dated 22nd May 1973.

**INSTRUCTIONS AND CONDITIONS FOR ISSUE OF CERTIFICATE OF
REGISTRATION BY AWBI UNDER THE PERFORMING ANIMALS
(REGISTRATION) RULES 2001 UNDER THE PCA ACT 1960**

- Producer of the Film in which animals are used should apply to the AWBI
- Application should reach AWBI 30 days before shooting with the animals.
- The prescribed proforma is also available in the website www.awbi.org.
- Furnish details of animals to be used and their performance sequences.
- Furnish ownership certificate for the animals.
- If they are elephants, snakes or wildlife, ownership certificate from the wild life authorities to be obtained. If the shooting is to be held in the states other than the state from which certificate is obtained then valid transit pass is to be provided.
- Recent health certificate for the animals to be used should be attached from a VCI registered veterinary doctor.
- The place (location), date and time of the proposed shoot should be mentioned in the application. The shooting must be completed within six weeks after pre shoot approval. In case the shooting does not materialize within the specified period, a fresh set of health certificates should be submitted.
- A demand draft for Rs. 500/- payable at Chennai drawn in the name of Animal Welfare Board of India should be enclosed.
- Applications by email and fax will not be considered till the original application with signature is received in the AWBI.
- Incomplete applications, details of performance and details of animals if not correctly given, certification will not be entertained.
- If animals in the background in the village scenes, or birds flying or animals grazing or animals/carts moving on the roads are seen the health certificate and ownership certificates can be waived.
- If performers are in the carts drawn by animals then the certificates are mandatory.
- Scenes depicting cruelty, causing hurt to animals, birds and animals in cages, making animals perform unnatural act are not permitted. The above scenes will be removed at the time of screening and legal action will be initiated and NOC will not be issued.
- Racing of animals, fights (Cock, bull, sheep/goat, snake and mongoose) 'Jallikkattu' like activities will not be permitted.
- Tripping, falling and throwing of animals, fighting scenes with animals will be denied certification and invite legal action.
- When large groups of animals are used proper medical facilities will be provided at the site of shooting and a representative of SPCA or Animal welfare organization should be present.
- Use of deafening noise, blasts, naked fire and such scenes should be avoided as it may cause panic and harm the animals and people.
- Fighting scenes in the farm houses / (dairy, poultry, and sheep) farms will not be permitted.
- A sub-committee screens the applications for issuing approvals for shooting (pre-shoot and post shoot approval)
- Application submitted for post shoot alongwith the CD should not depict/involve cruelty scenes, failing which AWBI will be constrained to initiate legal proceedings in addition to direct removal of above scenes.
- Incomplete application will not be entertained.
- Application received in original till the previous day of the meeting will be screened.

- After the shooting of the film, the clippings of the animal scenes should be sent in duplicate for the Sub-Committee to scrutinize confirming to the scenes/actions specified in the application. This should be accompanied by the post shoot health certificate in the prescribed proforma. Any scenes found in the CD not confirming in the original application will not be permitted and the pre-shoot permission, registration certificate will be cancelled under the provisions of the Performing Animals (Registration) Amendment Rules 2002. One of the approved CDs will be forwarded to CBFC to indicate the scenes approved and animals permitted by AWBI.
- The exhibition of the following species of animals have been banned under the Rules vide Notification No. G.S.R.619(E) dated 14.10.1998 - Tiger, panther, lion, bear and monkey which may be kept in mind before planning for the film shooting.
- After 1-12-2006, Certificate of Registration will not be issued for the films which have been completed without obtaining pre-shoot permission from AWBI. This has been widely publicized through National Newspapers on 04-11-2006. This facility was extended upto 31.03.07 for the benefit of film producers. Henceforth w.e.f. 01.04.07 applications for post shoot NOC without pre-shoot permission will not be accepted. NOC will be sent alongwith CD to the Censor Board by AWBI for issuing the Censor Certificate.
- 2 copies of CDs must contain the audio track for screening.
- The location with full address of the Censor Board where the film will be submitted for Censorship Certificate should be given in the time of post shoot NOC application to enable us to send NOC and CD to the concerned.
- Any regional language movies other than Hindi should have subtitles in English
- Details should be sent about the relevance of using animals in the sequence
- If there are several animal shots, then it should be taken in 1 CD as continuous shots
- Submission of application is not sufficient for filming (animal shooting) to be done. The film shooting with animals can only be done after formal approval of AWBI and after prior intimation of date, time and exact location of film shooting is communicated to the Board.
- If details of shooting (date, place, time of shooting with sufficient time) is not intimated to the Board, as required, the Board reserves the right not to issue NOC

MODULE-10: PROTECTION AND WELFARE OF WORKING ANIMALS

Learning outcome

This module focusses on the welfare of Working animals, particularly the pack and drought animals. Bullocks are commonly used for farming practices in agricultural sector of India. Mechanization of the farming practices slowly replaces the drought animals from farming practices.

gistration.

WORKING ANIMALS

- A working animal is an animal, usually domesticated, that is kept by humans and trained to perform tasks. They may be close members of the family, such as guide or service dogs, or they may be animals trained strictly to perform a job, such as logging elephants. They may also be used for milk, a job that requires human training to encourage the animal to cooperate. Some, at the end of their working lives, may also be used for meat or other products such as leather. Such animals are sometimes called draft animals or beasts of burden.
- The history of working animals may predate agriculture, with dogs used by our hunter-gatherer ancestors. Around the world, millions of animals work in relationship with their owners. Domesticated species are often bred to be suitable for different uses and conditions, especially horses and working dogs. Working animals are usually raised on farms, though some are still captured from the wild, such as dolphins and some Asian elephants.
- People have found uses for a wide variety of abilities found in animals and even in industrialized society many animals are still used for work. The strength of horses, elephants and oxen is used in pulling carts and logs. The keen sense of smell of dogs and, sometimes, rats are used to search for drugs and explosives as well helping to find game while hunting and to search for missing or trapped people. Several animals including camels, donkeys, horses and dogs are used for transport, either riding or to pull wagons and sleds. Other animals including dogs and monkeys provide assistance to blind or disabled people.
- On rare occasions, wild animals may be not only tamed, but trained to perform work, though often solely for novelty or entertainment purposes, as such animals tend to lack the traits of trustworthiness and mild temper that characterize the true domesticated working animal. Conversely, not all domesticated animals are working animals. For example, while cats may perform work catching mice, it is an instinctive behavior, not one that can be trained by human intervention. Other domesticated animals, such as sheep, or rabbits, may have agricultural uses for meat, hides and wool, but are not suitable for work. Finally, small domestic pets such as most birds (other than certain types of pigeon) or hamsters are generally incapable of performing work other than that of providing simple companionship.

TYPES OF WORK

1. Types of work
 1. Riding animals or mounts
 2. Pack animals
 3. Harness animals
2. Animals used for their senses or instincts
 1. Hunting
 2. Searching for people
 3. Assistance animals
 4. Herding
 5. Gathering plants

PACK ANIMALS

- A pack animal or beast of burden is a working animal used by humans as means of transporting materials by attaching them so their weight bears on the animal's back; the term may be applied to either an individual animal or a species so employed. The term pack animal is sometimes used in contrast to draft animal, which is a working animal that typically pulls a load behind itself (such as a plow or a wheeled cart) rather than carrying cargo directly on its back.
- Many ungulate species are traditional pack animals, including elephants, camels, the yak, reindeer, goats, water buffalo and llama, and many of the domesticated Equidae (horse family).
- The term is not routinely applied to humans carrying loads on their backs except to make a pejorative point about the injustice of so employing them, or about the privation that usually occasions accepting such work without explicit coercion. (The 1978 Rolling Stones song "Beast of Burden" refers to a sense of abuse, accepted within a romantic relationship.) Nevertheless, from a physical point of view, certainly many considerations apply equally to human and other pack animals, without considering the range of social conditions ranging through slaves, abused women and children, Himalayan and African natives employed as expedition porters, vacationing students whose duties as staff of mountaineering huts include packing heavy loads of supplies up steep slopes, and purely recreational hikers and backpackers including both short-trip ones, and long-trip backpackers who court injury and emaciation in carrying their heavy loads.
- Another unconventional form of pack animal may be the dogs that are brought along on hikes carrying their own supply of drinking water and snacks on their backs, whether to provide them more exercise, or in pursuit of a hiker's ethic of "everyone carries his own gear".

PROTECTION AND WELFARE OF WORKING ANIMATE

Animals in agriculture

- Half a century ago, family farms were prevalent. Animals grazed on pasture, breathing fresh air, and feeling sunshine on their backs. During inclement weather, they were sheltered in straw-bedded barns.
- In contrast, the rearing of farm animals today is dominated by industrialized facilities that maximize profits by treating animals not as sentient creatures, but as production units. Raised by the thousands, at a single site, animals are confined in such tight quarters that they can scarcely move, let alone behave normally. Such production creates what appears to be “cheap” meat, eggs, and dairy products. But what at the cash register appears inexpensive in fact costs dearly to farm animals, the environment, rural and traditional farming, human health, and food quality.
- Over 9 billion chickens, pigs, cattle, turkeys, sheep, goats, ducks, and geese are bred, raised, and killed for food annually in America. Each is a social, feeling individual capable of experiencing pleasure but realistically the vast majority are familiar with deprivation, fear, and pain.
- Of concern to Animal Welfare are all aspects of a farm animal’s life including breeding, growing, transport, and slaughter. Each phase offers the opportunity for cruelty or compassion.
- Animal Welfare works in myriad ways to halt the inhumane and irresponsible practices intrinsic to industrial agriculture, and seeks to replace them with methods which are both humane and economical. These alternative methods take the feelings of the animals into account, and they ensure each species can express

MODULE-11: PET AND COMPANION ANIMAL WELFARE

- **Learning outcome**
- The common types of pets are domesticated pets. They are mammals and bird species. But wild animals also considered as pets like Exotic mammals, birds, reptiles, amphibians, fish, arthropods and mollusks. This module focusses on the pet animals and their welfare in domestic environment. In addition this module also focusses on the laws related to pet animals.
- normal behavior.

PET

- A pet is an animal kept for companionship and enjoyment or a household animal, as opposed to wild animals or livestock, laboratory animals, working animals or sport animals, which are kept for economic or productive reasons. The most popular pets are noted for their loyal or playful characteristics, for their attractive appearance, or for their song. Pets also generally seem to provide their owners with non-trivial health benefits; keeping pets has been shown to help relieve stress to those who like having animals around. There is now a medically-approved class of "therapy animals," mostly dogs, that are brought to visit confined humans. Walking a dog can provide both the owner and the dog with exercise, fresh air, and social interaction.

CRUEL PRACTICES

- While some lucky animal companions are treated as members of the family (as they should be!), many others experience nothing but suffering, abuse, and sadness. Some forms of abuse and neglect of animals are illegal, but in many states, animals have no legal protection from "standard" practices such as being chained to a post all day and night; having their toes, ears, or tails cut off for human convenience or preference; or being forced to wear collars that deliver painful shocks in response to perfectly normal behavior such as barking.
- Chaining dogs, while unfortunately legal in most areas, is one of the cruelest punishments imaginable for social animals who need and deserve companionship, exercise, and mental stimulation. It can also turn dogs into ticking time bombs. Many people, especially children, have been bitten, mauled, or killed by chained dogs. Similarly, keeping dogs in crates or cages prevents them from satisfying all their needs and is just a way for guardians to ignore and warehouse their dogs until they get around to taking care of them properly.
- Birds don't belong to cages, either. Bored, lonely, denied the opportunity to fly or stretch their wings, and deprived of companionship, many birds become neurotic—pulling out their own feathers, bobbing their heads incessantly, and repeatedly pecking at the bars of their cages.
- Declawing, another cruel practice performed only for the convenience of the cat's owner, is a painful mutilation that involves 10 individual amputations—not of the nails but of the ends of the toes themselves (bone and all). The long-term effects of declawing include skin and bladder problems and the gradual weakening of cats' legs, shoulders, and back muscles. Declawing is both painful and traumatic, and it has been outlawed in Germany and other parts of Europe as a form of cruelty.

- Cruel, unnecessary surgeries that are performed on dogs, including ear-cropping, tail-docking, and debarking, are so painful and traumatic to dogs that they are banned in many countries, but they remain legal in the U.S.
- "Training" devices such as shock collars, electric fences, and prong collars rely on painful punishment and negative reinforcement, causing dogs to live in fear of being electrocuted or choked for normal behavior such as crossing invisible lines, barking, jumping onto surfaces within their own homes, and pulling on the leash during walks.
- Hoarding of animals exists in virtually every community. Formerly referred to as "collectors" and regarded as well-intentioned people who were overwhelmed by the animal overpopulation crisis, hoarders are now seen in a very different light. New information has shown that the problem is far more serious than having too big a heart. The consequences for hoarders, their human dependents, animals, and the community are extremely serious—and often fatal for animals.

HOUSING AND HUSBANDRY

- Dogs are kept in kennels by a variety of organizations. Examples include: rescue organizations, hunt kennels, quarantine kennels, boarding kennels, police, and assistance organizations. In addition dogs (mainly beagles) are bred and kept in kennels for biomedical research and for nutritional studies. A useful general source of information for housing and care of laboratory dogs is *hur*. Traditionally kennels were often small, designed to house one dog, and were often devoid of anything to occupy the dog or provide variety. Such environments resulted in timid fearful dogs or dogs with repetitive behavior disorders such as stereotypies, or self injuring behavior. Studies in both shelters and research establishments have shown that the longer dogs spend in sub-optimal environments, the more likely they are to show abnormal or undesirable behaviors. These dogs are likely to make poor research subjects; moreover, since dogs may spend considerable periods in kennels the welfare impact can be considerable. Over the last 15 years or so, there have been some important changes in the conditions considered acceptable for the breeding and keeping dogs used in research and these have also influenced standards for other types of kenneling. There are several recent documents that provide comprehensive reviews and recommendations based on applied research and the natural history of the dog
- One of the most important changes in the kennelling of research dogs has been in the reduction of single housing. For many years social housing has been the default system in the UK. In the UK minimum enclosure dimensions are sufficient to house one or two dogs, therefore making pair housing financially advantageous. A similar strategy has now been adopted in the recent revision of Council of Europe recommendations (Council of Europe, 2006), and social housing is becoming ever more widely accepted throughout the research community. Even where dogs have to be isolated during dosing in regulatory studies, or feeding during nutritional studies, dogs can still be housed in pairs or groups for the rest of the time. In Europe, it is common for dogs on GLP toxicology studies to be socially housed, but separate them for feeding so food consumption can be better measured. This trend is less common, however, in the United States. Nonetheless, have shown that it can be done, and as UK companies have to comply with either the same or similar regulatory requirements, a more general adoption of social housing would seem to be a relatively easy gain to be made in the USA. Good kennel design is critical in allowing the adoption of social housing and modular designs, allowing animals to be run together as desired, are a useful flexible means of

achieving this. Social housing does bring an increased risk of injury, however, and husbandry regimes should be designed to monitor aggression.

PET HEALTH

Pets can add fun, companionship and a feeling of safety to life.

- Before getting a pet, think carefully about which animal is best for your family.
- What is each family member looking for in a pet?
- Who will take care of it?
- Does anyone have pet allergies?
- What type of animal suits your lifestyle and budget?

Once a pet is owned it should be kept healthy. It should be taken to a veterinarian if

- Loss of appetite
- Drinking a lot of water
- Gaining or losing a lot of weight quickly
- Strange behavior
- Being sluggish and tired
- Trouble getting up or down
- Strange lumps

are noticed.

EXERCISE, HUMAN CONTACT, SOCIALIZATION, AND TRAINING

Some dogs clearly enjoy exercise, but when companion animals are exercised they also experience exploration, territory marking and socializations all of which are likely to add to the hedonistic experience of the activity. The benefits of exercise periods will depend on the exploratory and social opportunities offered to the dogs. Human contact can be very important for them . Often less than an hour a day , like 45 minutes has been shown to be beneficial. Dogs used in research can be trained to cooperate with husbandry and research procedures. Dogs given more human contact time are less nervous, (good for their welfare), and more cooperative and easier to handle (good for the staff and researcher).

NON-SOCIAL ENRICHMENT

Provision of chews, platforms, visual sight lines are worked out keeping the dogs' natural behavior and sensory modalities in mind. Dog appeasement pheromone (DAP) can be given as a means of reducing the stress when placed in unfamiliar circumstances, on dogs taken to veterinary surgeries, and in shelters .

NUTRITION AND HEALTH AND BEHAVIOR

Many dogs are highly motivated by food (chews that taste of food seem to be preferred by dogs), Overweight has risks of musculoskeletal, cardiovascular problems and diabetes. Malnutrition, which, if experienced during development, can lead to behavioral as well as physical abnormalities. Much attention is given by pet food manufacturers to ensuring that their diets are palatable, as owners are often influenced in their purchasing decisions by their dog's reaction to food. Very palatable diets can increase the risk of aggression in kennels . Dogs have been used in studies on the effects of diet on behavior and cognitive function as a model of human degenerative disorders as well as to study the effects of diet on companion animals.

NOISE

Dogs generate a great noise, which can be a potential health problem for humans working in kennels, and given the greater sensitivity of dogs to sound, may be a welfare issue for the dogs themselves. Buildings and husbandry routines should be designed so as to reduce noise. Group housing is associated with less noise production which could well be a significant benefit in the research setting.

MODULE-12: ANIMAL WELFARE IN COMMERCIAL LIVESTOCK FARMING PRACTICES

Learning outcome

This module mainly focusses on the commercial livestock farming practices and the associated welfare rules and laws. In India commercial livestock farming includes dairy practice and sheep and goat rearing. In minor horse rearing, and pig rearing also takes place. Poultry farming nowadays has become an industrial farming practice.

ANIMAL WELFARE IN COMMERCIAL LIVESTOCK FARMING PRACTICES

Feed and water housing systems

- Animals should receive a daily diet adequate in composition, quantity, and containing appropriate nutrients to maintain good health, meet their physiological requirements, and avoid metabolic and nutritional disorders. Feed should be palatable and free of contaminants, moulds, and toxins.
- It should be noted that food and water requirements vary with feed composition, physiological state, stage of growth, size and condition, pregnancy, lactation, exercise and activity, and climate. Access to feed should be at intervals appropriate to the physiological needs of the animals, and at least once daily.
- Animals should have an adequate daily supply of water that is palatable and not harmful to their health.
Food and water should be provided in such a way that all animals have an opportunity to feed or drink without undue competition and injury.
- Animals on highly concentrated diets may also require access to bulky or high fiber feed in order to satisfy hunger. Medicated or enriched food and water should only be used on professional advice, or when it is not detrimental to animal health and welfare.
- Reserves of food and water should be maintained to allow for interruption to supply.

Housing systems

- Animal accommodation should be designed, constructed, and maintained to allow all animals space to stand, turn around, stretch, sit, and/or lie down comfortably at the same time. They should also allow all animals to directly interact with herd or flock mates, unless isolated for veterinary reasons.
- Stocking densities should be low enough to prevent excessive temperatures and humidity; competition, stress, and aggression between animals, and abnormal behavior; and to enable good litter management.
- All animals should have access to a clean and dry place. Floor litter must be kept free of excessive moisture, and be loose and friable in the case of broiler chickens. All surfaces and flooring should be non-slip, without sharp projections or edges likely to cause injury, and provide for the animal to bear weight on the entire sole of the foot.
- Housing should be constructed of fire-resistant materials and electrical and fuel installations planned and fitted to minimize fire risk. Fire fighting equipment and smoke detectors should be

installed with sufficient exits to enable evacuation of the building in an emergency. There should be sufficient drainage to protect animals from flooding. Automated feeding and watering systems should allow all animals the opportunity of access to sufficient feed and water without undue competition (including intimidation, bullying and aggression) likely to cause injury or distress.

- Feeding and watering systems should be designed, constructed, placed, and maintained to prevent contamination or spoiling, and minimize spillage. All automated systems supplying food and water; removing waste; and controlling temperature, lighting, and ventilation should be checked and maintained regularly, and backup systems should be available in case of failure.
- Natural or artificial light (of an intensity of at least 20 lux) should be available in all buildings for a minimum of eight hours daily, and there should be a period of darkness sufficient to allow proper rest.

Air quality should be maintained by removing excessive heat and moisture, minimizing transmission of airborne infectious agents, preventing the build up of noxious or harmful waste gases, and to reasonably control humidity and minimize dust particles.

- Effluent and waste should not be allowed to build up where it leads to discomfort and compromised welfare.
- Animals should be protected from extreme temperatures or abrupt temperature fluctuations, cold draughts, and from predators, vermin, and excessive noise.
- Animals with access to, or living, outdoors should have access to shade and shelter and protection from predators.

THE PREVENTION OF CRUELTY TO DRAUGHT AND PACK ANIMALS RULES, 1965

Q 21) What is the maximum load for draught animals?

A) Below is the table, stating the specific animals meant for drawing a vehicle of the kind with the load of excess weight.

TABLE I

1) Small bullock or Small buffalo	Two wheeled vehicle-	1000 kilograms
	a) If fitted with ball bearings	750 kilograms
	b) If fitted with pneumatic tyres	500 kilograms
	c) If not fitted with pneumatic tyres	
2) Medium bullock or Medium buffalo	Two wheeled vehicle-	1400 kilograms
	a) if fitted with ball bearings	1050 kilograms
	b) if fitted with pneumatic tyres	700 kilograms
	c) if not fitted with pneumatic tyres	

3) Large bullock or Large buffalo	Two wheeled vehicle-	1800 kilograms
	a) if fitted with ball bearings	1350 kilograms
	b) if fitted with pneumatic tyres	900 kilograms
	c) if not fitted with pneumatic tyres	
4) Horse or mule	b) if fitted with pneumatic tyres	750 kilograms
	c) if not fitted with pneumatic tyres	500 kilograms
5) Pony	b) if fitted with pneumatic tyres	600 kilograms
	c) if not fitted with pneumatic tyres	400 kilograms
6) Camel	Two-wheeled vehicle	1000 kilograms

Q 22) What is the maximum load of certain pack animals?

A) Below is the Table provided specifying animals allowed to carry loads along with the excess weight .

TABLE II

1) Small bullock or buffalo	100 kilograms
2) Medium bullock or buffalo	150 kilograms
3) Large bullock or buffalo	175 kilograms
4) Pony	70 kilograms
5) Mule	200 kilograms
6) Donkey	50 kilograms
7) Camel	250 kilograms

Q 23) What are the powers provided to police officers under The Prevention of Cruelty to Draught and Pack Animals Rules,1965?

A) Section 11 of the Act, provides that if police officer above the rank of a constable feels that the rule of reasonable excess weight with regard to the draught and pack animals is getting violated, he may, ask the owner or the other person in charge of such animal to take the animal or the vehicle or both to the weighbridge for the purpose of determining the weight of the load which animal has been or is drawing or carrying.

And if the owner incharge of the aforesaid animals refuses to comply with the demand of the police officer , the policeman has every right to take the animal or the vehicle or both to the weighbridge and get it measured. And as soon as any weight is determined under this rule, the owner or other person in charge of the said animal shall be given a statement in writing signed by the police officer as to the weight so determined and any other information relevant for the purpose.

Q 24) What are the general conditions for the use of draught and pack animals?

A) Section 6 of The Prevention of Cruelty to Draught and Pack Animals Rules, 1965, lays down the general conditions for the use of draught and pack animals. No person is allowed to use any animal for drawing any vehicle or carrying any load :

- For more than nine hours in a day in the aggregate;
- For more than five hours continuously without a break or rest for the animal;
- In any area where the temperature exceeds 37 degree C (99 degree F) during the period between 12 noon and 3 p.m.

Q25) What are other relevant provisions with regard to The Prevention of Cruelty To Draught And Pack Animals Rules,1965?

A) Following are the important sections that should be known. These are:

- Section 7:- Animals to be disengaged after work:- No person shall continue to keep in harness any animal used for the purpose of drawing vehicles, after it is no longer needed for such purpose.
- Section 8:- Use of spiked bits prohibited:- No person shall, for the purpose of driving or riding an animal or causing it to draw any vehicle or for otherwise controlling it, use any spiked stick or any other sharp tackle or equipment which causes bruises, swellings, abrasions or severe pain to the animal.
- Section 9:- Saddling of horses:- No person shall cause a horse to be saddled in such a way that the harness rests directly on the animal's withers without there being sufficient clearance between the arch or the saddle and the withers.

BULLOCKS

Bullocks work relentlessly throughout the day, in sun and rain, through the heavy traffic on the streets. They are beaten mercilessly and kicked, to make them walk faster or to run. On top of it, they have no shelter where they can rest after a hard day's work. They usually spend the night on the street tied to their own carts, where they have no protection from the rain. Their resting place is such that they cannot even sit down, as the area where they are tied is absolutely dirty with their own muck. So all their life they face the brunt of a hot sun and heavy rain. Being people of limited means, the bullock cart "owners" do not take good care of their animals that slave for them. They rarely take them to vets, and instead treat them

with their own herbal medicines. They never vaccinate them, with the result during the monsoon season they suffer from Foot & Mouth Disease or HS & BQ (Haemorrhagic Septicemia & Black Quarter) - both very painful conditions. The saddest part of the story is that after working so hard all their lives, they never get any retirement. When they become old and unproductive, they land in the slaughter houses. This is how we treat these animals, who work without any expectation of wages, bonus, weekly holidays, etc. All they ask for is a little kindness and compassion.

To help these animals what we need to do is:

- It must be made compulsory for all these cart owners to carry fitness certificates of the bullocks issued by SPCA. That will make it necessary for every cart "owner" to bring his animal to SPCA for a check up, in order to obtain such a certificate.
- The oil companies that use bullocks to cart kerosene, must provide the animals with a decent shed to house them when they are not working. These oil companies must treat these animals as they would treat their contract workers, and provide them with basic needs like shelter, food, water and medical aid.
- The bullock carts should have rubber tyres thereby reducing the neck load on the bullock.

SHEEP HUSBANDRY

SHELTER AND ENVIRONMENT

- In cold climates, sheep may need shelter if they are freshly shorn or have baby lambs. Freshly shorn hoggets can be very susceptible to wet, windy weather and can quickly succumb to exposure. Sheep have to be kept dry for one to two days before shearing so that the fleece is dry enough to be pressed and to protect the health of the shearers.

HEALTH CARE

- Lambs receive their first antibodies via their mother's colostrum in the first few hours of life., they get the first vaccine at six months followed by subsequent boosters.
- If farmers work with animal nutritionists and veterinarians sheep can be kept healthy. Lambs may be castrated and have their tails docked for easier shearing, cleanliness, and to help protect them from fly strike. Shearers or farmers need to remove wool from the hindquarters around the anus so that droppings do not adhere. In the southern hemisphere this is called dagging or crutching.

WATER, FOOD AND AIR

- Sheep need fresh water from troughs or ponds, except in some countries, such as New Zealand, where there is enough moisture in the grass to satisfy this requirements.
- They are usually given feed twice a day from troughs or are allowed to graze in a pasture.
- Sheep are most comfortable when the temperature is moderate, so fans may be needed for fresh air if sheep are kept in barns during hot weather. In Australia, sheep in pasture are often subjected to temperatures of 40 °C (104 °F) and higher without deleterious effects.

MODULE-13: WELFARE OF ANIMALS DURING TRANSPORTATION

- **Learning outcome**

- Transportation of animals in the present situation is attracting animal welfare organizations to concentrate on welfare during transportation. This module particularly focusses on the laws related to transportation of livestock.

WELFARE OF ANIMALS DURING TRANSPORTATION

- Animals to be transported shall be healthy and in good condition and such animals shall be examined by a veterinary doctor for freedom from infectious diseases and their fitness to undertake the journey; provided that the nature and duration of the proposed journey shall be taken into account while deciding upon the degree of fitness.
- An animal which is unfit for transport shall not be transported and the animals who are new born, diseased, blind emaciated, lame, fatigued or having given birth during the preceding seventy two hours or likely to give birth during transport shall not be transported.
- Pregnant and very young animals shall not be mixed with other animals during transport.
- Different classes of animals shall be kept separately during transport.
- Diseased animals, whenever transported for treatment, shall not be mixed with other animals
- Troublesome animals shall be given tranquilisers before loading during transport.
- Animals shall be transported in their on-farm social groups (established atleast one week prior to journey).
- While transporting animals all rules must be followed, and they must have a valid certificate from a veterinarian. The rules require stopping for water and feed, and that only 6 large animals in a lorry can be transported or about 12 for smaller calves.
- Any violation of these laws is termed illegal.
- The Vehicle transporting them should be large enough to carry animals comfortably and the animals should not be packed and jammed inside. The animals should also be protected from the weather.
- The permissible loading in a truck is only 4 buffaloes or 40 sheep/goats, and truck having wheel base below 142 inches, shall not carry more than five cattle without calves or four with calves.
- In the case of trucks whose wheel base is over 142 inches shall not carry more than six cattle without calves or five with calves.
- Animals in tempo are not allowed. Only four passengers excluding the driver are allowed on a tonga or a total of 325 kgs.

Rules and Acts

- Transport of Animals (Amendment) Rules, 2009
 - Transport of Animals, Rules, 1978.
- Whereas certain draft rules further to amend the Transport of Animals Rules, 1978 were published as required by sub-section (1) of section 38 of the Prevention of Cruelty to Animals Act, 1960 (59 of 1960) under the notification of the Government of India in the Ministry of Social Justice & Empowerment number S.O. 1164 (E) dated 26th December 2000 in the Gazette of India. Extraordinary, Part 11, Section 3. Sub Section (ii) dated the 27th December, 2000 inviting objections and suggestions from all persons likely to be affected thereby, before the expiry of the

period of sixty days from the date on which copies of the Gazette containing the said notification are made available to the public.

- And, whereas copies of the said Gazette were made available to the public on the 1st January 2001.
- And, whereas no objection or suggestion has been received from the public in respect of the said draft rules by the Central Government.
- Now, therefore in exercise of the powers conferred by sub-section (1) and (2) of section 38 of the Prevention of Cruelty to Animals Act 1960 (59 of 1960), the Central Government hereby makes the following rules further to amend the Transport of Animals Rules 1978, namely
 - These rules may be called the **Transport of Animals (Amendment) Rules, 2001.**
 - They shall come into force on the date of their publication in the Official Gazette.

TRANSPORT OF MONKEYS

Monkeys are to be transported in suitable wooden or bamboo cages. The following two sizes of cages shall be used during the transportation of monkeys through rail.

1. 910 x 760 x 510 mm- This cage shall contain not more than twelve monkeys, weighing between 1.8 and 3.00 kilograms each or ten monkeys weighing between 3.1 and 5.0 kilograms each
2. 710 x 710 x 519 mm- This cage shall contain not more than ten monkeys weighing between 1.8 and 3.00 kilograms each or eight monkeys weighing between 3.1 and 5.00 kilograms each.

Also, not more than one cage shall be placed over the other and gunny packing shall be placed between two cages, when one is placed over the other.

But when the monkeys are transported by air the following two sizes of cages shall be used.

1. 460 x 460 x 460 mm- This cage shall contain not more than ten monkeys weighing from 1.8 to 3.0 kilograms each or four monkeys weighing from 3.1 to 5.0 kilograms each
2. 760 x 530 x 460 mm :- This cage shall contain not more than ten monkeys weighing from 1.8 to 3.0 kilograms each or eight monkeys weighing from 3.1 to 5.0 kilograms each.

TRANSPORT OF CATTLE

3. When cattle is to be transported by rail an ordinary goods wagon shall carry not more than ten adult cattle or fifteen calves on broad gauge, not more than six adult cattle or ten calves on meter gauge, or not more than four adult cattle or six calves on narrow gauge. And while transporting cattle by goods vehicle, only six cattle should be loaded.

TRANSPORT OF EQUINES

- For the transport of equines by rail, ordinary goods wagon when used for transportation shall carry not more than eight to ten horses or ten mules or ten donkeys on broad gauge and not more than six horses or eight donkeys on meter- gauge.

- And if equines are to be transported by good-vehicles each vehicle should not carry more than four to six equines.
- For the transport of equines by sea, horses may normally be accommodated in single

TRANSPORT OF SHEEP AND GOATS

Following is the chart of transport of sheep and goats through railway wagon, for different gauge.

Broad Gauge

- In the area of wagon less than 21.1 Square Metres number of sheep or goat allowed is 70

Metre Gauge

- In the area of wagon 21.11 square Metres and above number of sheep or goat allowed is 100

Narrow Gauge

- In the area of Wagon less than 12.5 Square Metres the number of sheep or goat allowed is 50
- In the area of wagon 12.5 Square Metres and above the number of sheep or goat allowed is 60

And the Goods vehicle of capacity of 5 or 4.5 tons, which are generally used for transporting animals, shall not carry more than forty sheep or goats.

- stalls and mules in pens, each pen holding four to five mules.

TRANSPORT OF POULTRY BY RAIL, ROAD AND AIR

- Containers are used for transporting poultry by rail, road and air. There is specific number of different kind of poultry to be transported in containers.
- Kind of Poultry Number in a container

Month old chickens	24
Three-month old chickens	12
Adult stock(excluding geese and turkeys)	12
Geese and turkeys	10 youngs 2 growings 1 grown up

Chicks	80
Poult	60

TRANSPORT OF PIGS BY RAIL OR ROAD

In transport of pigs by road, good vehicles which are generally used for transporting of animals shall not carry more than twenty pigs.

And while transporting pigs by rail, no railway wagon shall accommodate more than the number of pigs as provided below:

Broad Gauge

- In the area of wagon less than 21.1 Square Metres number of pigs allowed is 35

Metre Gauge

- In the area of wagon more than 21.1 Square Metres number of pigs allowed is 50

Narrow Gauge

- In the area of wagon less than 12.5 Square Metre number of pigs allowed is 25
- In the area of wagon more than 12.5 Square Metre the number of pigs allowed is 30
- Not Allowed

SPECIAL CONSIDERATION IN TRANSPORT TO OVERLOADING

- Overloading of animals amounts to treating of animals cruelly under Section 11 of The Prevention Of Cruelty to Animals Act, 1960. And if the above general conditions with regard to the transportation of animals are not met, the animals should be immediately unloaded and sent to the nearest animal shelter. And since the offence so committed falls under the Section 11 of The Prevention of Cruelty To Animals Act, 1960, the offender (in the case of a first offence) will have to pay fine which shall extend to fifty rupees and if it is the case of second offence or subsequent offence committed within three years of the previous offence, he will be fined with not less than twenty-five rupees but which may extend to one hundred rupees or with the imprisonment for a term which may extend to three months or with both. Also, in the case of second offence, the offender's vehicle is confiscated, and he will never be allowed to keep an animal again.

MODULE-14: ANIMAL WELFARE DURING NATURAL CALAMITIES AND DISASTER MANAGEMENT

- **Learning outcome**
- This module focusses on animal welfare during natural calamities and in disasters. After completion of this module the learner might have confidence to manage animal welfare in emergency situations and will be able to develop an emergency kit for animal welfare.
- allowed to keep an animal again

ANIMAL WELFARE DURING NATURAL CALAMITIES AND DISASTER MANAGEMENT

Safety of pets during storms

- Pets need special attention during storms.
 - Keep cats indoors all day, every day all year long.
 - During the monsoon season, keep dogs indoors. Thunderstorms easily frighten many dogs that might dig or jump from the yard to escape. High winds can blow open gates, knock over fences, and even topple brick walls, making it easy for a dog to run away.
 - Be sure your pets are confined safely indoors when you leave in the morning. Many storms roll in late in the day when people are at work; violent storms can crop up quickly!
 - Confine pets to a safe, secure room. Turn on the TV or a radio to help drown out scary noises. Make sure pets have water, bedding and favorite toys.
 - Be sure all pets have current identification in case they run away. Recommends a microchip complemented by a collar and I.D. tag.
 - The I.D. tag should include two telephone numbers, such as a home number and a work, pager or cellular number. If you have recently moved or changed phone numbers, be sure to update all tags!
- **Thunder and blowing wind** can scare even the bravest of animals and they usually try and escape, usually under the bed or behind the closet. But there may be some pets that suffer from storm anxiety.
- The *symptoms can include biting, excessive chewing, hiding, pulling out own hair and shaking*. The case may be a mild one but there are cases where it turns severe which is why it needs to be effectively taken care of.
- Some breeds of dogs are more prone to chronic anxiety namely Bull dogs, German Shepherd, Dalmatians and most breeds of terriers. There are many anti-anxiety drugs available with vets that can help ease the fear and stress.
- Do make sure your pet has proper identification so that if he runs away, you can get him back safe and sound.

PREPARING FOR A DISASTER: PLANNING FOR PETS AND LIVESTOCK

- Disasters, such as hurricanes, tornadoes and floods don't just affect people – they also affect pets and other animals. Planning ahead in the event of a natural disaster can protect the safety and well being of you and your pet.

CREATE A DISASTER KIT

Pack extra pet care and transportation items in an easy to grab kit, including:

- Extra collars, tags and leashes for all pets and extra pet food with a manual can opener if needed
- A supply of stored drinking water
- Toys or blankets the pet will find familiar
- Paper towels, plastic bags and disinfectant for waste clean-up
- Copies of your pet's medical and vaccination records
- Extra supplies of any medications your pet is currently taking

IF YOU NEED TO EVACUATE

- Take your pets with you whenever possible (only service animals are allowed in Red Cross shelters)
- Identify "pet friendly" hotels
 - Board with friends/relatives in a safe area.
 - Check with your local animal shelter.
 - Leave in plenty of time – you may not be able to take your pet at the last minute.
- Identify your pets, include your address, phone number and the phone number of a friend outside of the disaster range. Have photos for identification purposes.
- To transport your animals safely:
 - Condition your animals to being in a cage/carrying case/pen/trailer.
 - Keep animals on a strong leash/harness.
 - Take three to five days' worth of supplies – food, water, high water-content fruits/vegetables, medication, cat litter, "comfort toys."
 - Birds/lizards – blanket to keep cage warm/plant mister to hydrate feathers.
 - Snakes – pillowcase to transport/heating pad for warmth/water bowl to soak.
 - Pocket pets (hamsters/gerbils) – cage/bedding material/water bottles.

IF YOU MUST LEAVE YOUR PETS BEHIND

- Leave them untied in an interior room with adequate air and no windows – such as a bathroom.
- Purchase a self-feeder in advance and leave enough food and water for at least three days. Leave faucet dripping with drain open.
- Leave favorite bed and toys.
- Place notice on front door with location and type of pets, their names and your contact phone number.
- NEVER leave animals tied up outside.

DISASTER PREPAREDNESS FOR LIVESTOCK OWNERS

During a disaster, the behavior and activities of livestock can change dramatically. Advice on disaster planning and precautions is available from several sources, your local emergency management office, animal control center and area veterinarians.

- **EVACUATE LIVESTOCK WHENEVER POSSIBLE.** Arrangements for evacuation, including routes and host sites, should be made in advance. Alternate routes should be mapped out in case the planned route is inaccessible. Evacuation sites should have or be able to readily obtain food, water, veterinary care, handling equipment and facilities.
- Trucks, trailers, and other vehicles suitable for transporting livestock (appropriate to the type of animal) should be available, along with experienced handlers and drivers to transport them. Whenever possible, the animals should be accustomed to these vehicles in advance so they're less frightened and easier to move.
- If evacuation is not possible, a decision must be made whether to move large animals to available shelter or turn them outside. This decision should be determined based on the type of disaster and the soundness and location of the shelter (structure). All animals should have some form of identification that will help facilitate their return.
- Your disaster plan should include a list of emergency phone numbers for local agencies that can assist you if disaster strikes - - including your veterinarian, state veterinarian, local animal shelter, animal care and control, county extension service, local agricultural schools and the American Red Cross. These numbers should be kept with your disaster kit in a secure, but easily accessible place.

DISASTER RISK MANAGEMENT PROGRAMME

- A Disaster Risk Management Programme has been taken up in 169 districts in 17 multi-hazard prone States with the assistance from UNDP, USAID and European Union. Under this project, the States are being assisted to draw up State, district and Block level disaster management plans; village disaster management plans are being developed in conjunction with the Panchayati Raj Institutions and disaster management teams consisting of village volunteers are being trained in various preparedness and response functions such as search and rescue, first aid, relief coordination, shelter management etc. Equipment needs for district and State Emergency Operation Centres have been identified by the State nodal agencies and equipment is being provided to equip these EOCs. Orientation training of masons, engineers and architects in disaster resistant technologies have been initiated in these districts and construction of model demonstration buildings will be started soon.
- Under this programme Disaster Management Plans have been prepared for 8643 villages, 1046 Gram Panchayat, 188 blocks and 82 districts. More than 29000 elected representatives of Panchayati Raj Institutions have already been trained, besides imparting training to members of voluntary organizations. About 18000 Government functionaries have been trained in disaster mitigation and preparedness at different levels. 865 engineers and 425 architects have been trained under this programme in vulnerability assessment and retrofitting of lifeline buildings. 600 master trainers and 1200 teachers have already been trained in different districts in disaster preparedness and mitigation. Disaster Management Committees consisting of elected representatives, civil society members, Civil Defence volunteers and Government functionaries have been constituted at all levels including village/urban local body/ward levels. Disaster Management Teams have been constituted in villages and are being imparted training in basic

functions of first aid, rescue, evacuation and related issues. The thrust of the programme is to build up capabilities of the community since the community is invariably the first responder. During the recent past, it has been experienced that the capacity building of the community has been very helpful even in normal situations when isolated instances of drowning, burns etc. take place. With the creation of awareness generation on disaster mitigation, the community will be able to function as a well-knit unit in case of any emergency. Mock drills are carried out from time to time under the close supervision of Disaster Management Committees. The Disaster Management Committees and Disaster Management Teams have been established by notifications issued by the State Governments which will ensure that the entire system is institutionalized and does not disintegrate after the conclusion of the programme. The key points being stressed under this programme are the need to ensure sustainability of the programme, development of training modules; manuals and codes, focused attention to awareness generation campaigns; institutionalization of disaster management committees and disaster management teams, disaster management plans and mock-drills and establishment of techno-legal regimes.

MODULE-15: VETERINARY ETHICS AND JURISPRUDENCE

- **Learning outcome**

- This module focusses on the legal duties of veterinarians. In field practice the veterinarian must know how to face the legal issues coming under his control. It starts from the issue of health certificate, PM certificate, Cruelty to animals, common offenses against animals. This module exposes the learner to forensic and state medicine laws in Veterinary Jurisprudence.

VETERINARY ETHICS AND JURISPRUDENCE

- Veterinary jurisprudence, veterinary forensic medicine and veterinary legal medicine are considered synonyms.
- It pertains to their (animals) protection: unraveling crimes against them.
- In a way it protects the animals, which in turn may protect the individuals and the society. It makes to understand the Principles of Code and Ethics for the professionals. It deals with the questions which related to injury and death of the animal brought to the veterinarian for taking up the matters in the Court of law. It includes line of treatment and diagnosis. It helps the vets to approach and handle the accident and poisoning cases in the field.
- It is essential to have fair knowledge in the subjects of veterinary science to give evidence in Veterinary - legal problems. He must be well aware of the Acts affecting his privileges and obligations in the veterinary practice.
- Hence, a thorough knowledge of the legal procedures in the criminal courts is essential for veterinary practitioner.

LEGAL DUTIES OF THE VETERINARIANS

- To make a veterinarian know and recognize his exact duties, privileges and responsibilities in veterinary-legal cases both as witness and as an expert advisor it is very essential that he should have a fair knowledge of all the subjects of veterinary sciences.
- He must be well acquainted with the animal related Acts and the legal procedures in the criminal courts.
- Veterinarian has the following responsibilities.
 - Conducting postmortem examination of the veterinary-legal cases.
 - Investigations of common offences against animals.
 - Investigations in case of malicious and accidental poisoning.
 - Investigations in case of frauds in the sale of livestock and livestock products.
 - Issuing health certificates.
 - To get the real culprits punished and help in providing justice and fair treatment to man and animal and also to save innocent people/persons from the false accusations of crime.
 - To prevent cruelty to animals.
 - Application and enforcement of state of law meant for animals. (Prevention of cruelty to Animals, Cattle Trespass Act)
 - To protect the interests of society at large and prevent the unethical practices in relation to animals in the society.
 - In fact everything in which recourse to law is necessary in relation to the veterinary profession.

- The evidence is recorded as follows
 - **Examination in chief**
 - This is the first examination of a witness by the party which calls him.
 - In Government prosecution cases, the prosecuting inspector as a rule, first examines the witness to elicit the principal facts concerning the case.
 - If the witness is summoned by a private party, he is first examined by the pleader of that party. Leading questions are not allowed. (Eg.) If a person x strike or with a stick
 - The proper forms of the question in a case of an assault are
 - When did this incident occur?
 - Where were you at that time?
 - What did you notice?
 - In that case the witness will narrate the whole incident of a person 'x' striking 'y' as he saw it.
 - **Cross examination**
 - This may be regarded as the most reliable procedure of testing the value of an evidence and is held by counsel for the accused who tries to elicit facts (or) demonstrate the possibility of theories, not necessarily inconsistent with the evidence the witness has given, but helpful to his own case.
 - Leading questions are permissible and witness should be very cautious in answering them.
 - He should not attempt to answer the questions unless he clearly and completely understands them.
 - There is no time limit to cross examination.

Re examination

- The prosecuting Inspector or counsel who conducts the examination in chief has the right to re-examine the witness to explain away any discrepancies that may have occurred during cross examination.
- The witness should not introduce any new subject without the consent of the judge or opposing counsel lest he should become liable to cross examination on the new point thus introduced.

Question put by the Judge, Juror or Assessor

- The Judge, Juror or Assessor may question the witness at any stage to clear up doubtful points.

FORENSIC AND STATE MEDICINE LAWS

- Application of basic science of judiciary and foretell on the grounds of scientific knowledge to detect crime and responsibility for the damage.
- It is a special discipline of veterinary procedural steps.

EXAMINATION OF BLOOD-STAINS

- Examination of blood-stains is of great importance for vetero-legal purposes. The blood-stains are examined to ascertain whether or not they consist blood; if so, to what species of animal the blood belongs, whether mammal, bird, fish or reptile.

- The examination of all kinds of stains is done by the Chemical Examiner. Before removing stains for analysis, a note should be made indicating exactly where they were found and their relative position.
- Stains from persons and animals are best removed with a piece of filter paper soaked in normal saline solution.
- The paper is left in contact until it is coloured, then removed and dried. In case of garments and weapons, the articles should be allowed to dry, then carefully and separately wrapped and packed without unnecessary handling.
- Liquid stains are taken in dry test tubes with the help of dropper or pipette and the test tube is sealed. The test tube is placed on a cracked- ice container. Moist stains are completely dried and handled as dry stains.
- Removable stained objects must be sent to the laboratory. Blood-soaked soil is taken up completely and is sent in a glass container. If stains lie on non-removable objects, such as wall and floors, a representative section which contains the stain, is removed by digging out and sent in a container.
- In exceptional cases stains can be scrapped up by a knife. The examination of blood-stains may be carried out by the following methods:
 - Physical examination
 - Chemical examination
 - Microscopic examination
 - Spectroscopic examination
 - Serological examination

COMMON OFFENCES AGAINST ANIMALS AND LAWS RELATED TO THESE OFFENSES

Common offences against animals

- It is more or less obligatory for a veterinarian to be familiar with the laws for protection of animals, more so because of his training and profession which inculcate a spirit of sympathy and understanding towards our dumb companions, the domestic animals, who deserve all sympathy and kindness.
- The common offences against animals are
 - mischief,
 - bestiality and
 - cruelty.

Mischief

- This includes killing, poisoning or maiming an animal. Poisoning is the commonest method of mischievous killing. Abrus precatorius seed, arsenic, aconite root, snake venom, datura leaves, and seeds of yellow oleander are common poisons used in the mischievous killing of the animals.
 - Accidental poisoning may occur in animals by linseed or jowar consumption.
 - Mischief is punishable under sections 428 and 429: I.P.C.

Maiming

- It means making an animal permanently useless by the use of violence. This type of offence is also common and its aim is to harm the owner when his animal damages the crop or other property. The common forms of hurting animals and rendering them useless are:
- Fracture of bone
- Cutting tendons of legs and neck
- Injury to udder in milch animals
- Tearing of the vagina or rectum by introducing sharp or blunt object
- Punctured wounds, etc.

Bestiality

- Bestiality means carnal intercourse with man, woman or animal, against the order of nature. This type of crime is quite frequently found in India due to the following reasons:
 - The common belief among illiterate people that intercourse with she-donkey is a remedy for gonorrhoea.
 - Excessive sexual desire with little opportunity for natural intercourse.
 - Young villagers who go out to graze cattle in fields far away from human eyes, are incited, owing to loneliness and the proximity of the animals, to commit this crime, People having some mental abnormalities.
 - The human male is generally the active agent and the passive agent a goat, donkey, mare, cow, or even hen. Examination of animal for bestiality
- Bestiality is punishable under Section 377, I.P.C. The offenders are usually caught red-handed.
- The vagina of the animal should be examined for evidence of injury and the suspect examined for marks of injuries caused by the kicks teeth or claws of the animal.
- The surrounding hair of the animal should be examined for presence of human spermatozoa (it is very important to note whether the spermatozoa found are of the same animal or not and for this purpose the presence or absence of heat in the female animal is also a guide).
- The presence of organisms of gonorrhoea in the vagina of the animal is a definite sign of bestiality.
- The clothes of the suspect will smell of urine or faeces of the animal as after a sexual act animals have a habit of urinating.

Cruelty

- This offence includes basting, overloading, using a diseased animal for work, starvation, "Phuka", etc. These acts are punishable under the Prevention of Cruelty to Animals Act, 1960.

EXAMINATION OF LIVING AND DEAD ANIMALS IN CRIMINAL CASES

- The object of the examination is to ascertain the cause and the manner of death in all deaths from criminal violence, or in sudden deaths from unknown causes.
- Without proper examination it is not possible to ascertain cause of death in all cases of criminal violence, and sudden deaths due to unknown causes, even when apparently the animal appears to have died from natural causes.
- It is not uncommon to find that cases of apparently natural deaths or of death due to contagious disease have actually died of poisoning or violence, which caused rupture of internal organs without any external lesions. If it is the question of an injury, or death following an injury, it is necessary not only to describe in detail the injury, but also to indicate the approximate time of its occurrence, nature of the weapon used, and manner of causing injury.

- Every possible cause should be investigated before venturing an opinion as to the cause of death. Not only should the post-mortem lesions, both external and internal, be carefully examined, but all circumstantial evidence given and the surroundings, such as the field, or the land, etc., where the animal was found dead should be described.
- The history, as given by the owner, should be noted briefly and signed by the owner, or left thumb impression (right thumb impression, if a woman) taken if he or she is illiterate.
- In our climatic conditions, it is necessary to perform the post-mortem as early as possible to avoid decomposition of the carcass and its mutilation by carrion animals such as dogs, vultures, hyaenas, etc., because safeguarding these carcasses under village conditions is not always easy.
- Needless to say that the post-mortem report should be carefully drafted in a logical order without any exaggeration, so that a true picture of the case can be stated on oath in the witness-box.
- The proforma for recording a post-mortem report must be filled in triplicate at the time of the post-mortem examination, signed, timed and dated. It would be useful to maintain a vetero-legal register at various institutions to provide guidance to newcomers.
- Where only bones are available, it is necessary to record the detailed description of the important bones, which would provide sufficient information to identify the species, the sex and age if possible of the animal.
- This is specially important in the case of cattle and buffaloes on account of ban on slaughter of cows of all ages and of bullocks below 15 years of age. Any malformation should be specially noted. The odour and feel of bones should be noted carefully for assessing the time of death, as recent bones have a peculiar odour and a greasy feel.

CRUELTY TO THE ANIMALS

- Cruelty to animals is the infliction of suffering or harm upon animals, other than humans, for purposes other than self-defense. More narrowly, it can be harm for specific gain, such as killing animals for food or for their fur. Diverging viewpoints are held by jurisdictions throughout the world.
- Broadly speaking, there are two approaches to the issue. The animal welfare position holds that there is nothing inherently wrong with using animals for human purposes, such as food, clothing, entertainment, and research, but that it should be done in a humane way that minimizes unnecessary pain and suffering.
- Animal rights theorists criticize this position, arguing that the words "unnecessary" and "humane" are subject to widely differing interpretations, and that the only way to ensure protection for animals is to end their status as property, and to ensure that they are never used as commodities. Laws concerning animal cruelty are designed to prevent needless cruelty to animals, rather than killing for other aims such as food, or they concern species not eaten as food in the country involved, such as those regarded as pets.

BESTIALITY (ZOOPHILIA)

- Zoophilia, from the Greek ζῷον (zōion, "animal") and φιλία (philia, "friendship" or "love"), also known as zoosexuality, is the practice of sex between humans and non-human animals (bestiality), or a preference or fixation on such practice. A person who practices zoophilia is known as a zoophile.
- Although sex with animals is not outlawed in some countries, it is not explicitly condoned anywhere. In most countries, such acts are illegal under animal abuse laws or laws dealing with crimes against nature.

MODULE-16: SOUNDNESS

- **Learning outcome**

- This module deals with the soundness in animals. Soundness in horses, cattle and dogs are important in the selection of animals.

SOUNDNESS IN HORSES

- The structural development of the limbs, size and body proportions that make up an individual horse's conformation can influence its balance and movement during exercise. The strength of its limbs and conformation in turn, can influence its overall soundness and ability to perform in competition.
- Conformation and soundness are interrelated in determining the movement, performance and overall quality of a horse during its competitive lifetime.
- Soundness can be classified in a number of ways relative to the horse's freedom from lameness but it can also relate to the horse's suitability for a particular purpose, such as eventing, jumping, polo, cross-country, hack and dressage competition, which can include other measures of soundness for the purpose, such as boldness, temperament and flow of the stride.
- Soundness can also be influenced by factors such as the type of working surface, shoeing, dietary balance, body weight and the exercise program, especially as a horse ages. Accidental injuries such as lower limb lacerations, excessive bone modelling responses and abnormal growth and balance in the hooves can all influence the relative soundness for the purpose. A horse with a minor joint or tendon injury, for example, may not be suitable for upper level equestrian sports, but its soundness as a hack will not be compromised. As horses age, natural 'wear and tear' and other degenerative changes, that are not always related or influenced by conformation, can occur and restrict its suitability for long term training and competition. It is always important to select a horse that is sound and strong if you plan to compete over a few years, rather than try to rehabilitate a horse from the race track which has broken down due to tendon and joint problems and carry the legacy of unsoundness into retirement and equestrian competition.

HOW TO PROTECT YOUR HORSE'S DELICATE LEGS

- Even though horses are big, strong animals, their legs are surprisingly delicate. The muscles and tendons that make up the legs can get injured quite easily. Your horse might stumble and cut his front leg with his back hoof. Or he might land awkwardly after a jump and strain a tendon. Constant pounding on hard surfaces can also damage legs.
- Leg injuries can be serious and many take months (or even years) to heal. Some injuries never heal properly and you could be left with a lame horse.
- Of course, you can't cover your horse's legs in bubble wrap (well, maybe you could but he would look pretty strange!), but here are 10 things you can do that may help your horse stay sound.

BOOT UP

- If you're going to jump or gallop, put brushing boots or splint boots on your horse. These are padded boots that go on all four legs that protect your horse's legs in case he knocks his front legs

with his hind hooves, which can cause bruises or cuts. Padded boots also give extra support to the muscles and tendons in a horse's leg.

- If a horse is traveling at fast speeds, a hind hoof may "overreach" and tread on the heel of a front leg, causing a nasty cut that may take forever to heal. Prevent these cuts by putting rubber overreach boots on his front legs.

AVOID HARD GROUND

- Try to avoid riding your horse on hard ground. If the ring is rock hard, stick to a walk or ride out in the fields. Riding on hard ground causes concussion in your horse's legs as he bangs on the ground with his hooves. This concussion can injure or weaken the tendons and bones in his legs and can cause arthritis or navicular when he gets older.
- If you board at a barn, it's the owner's responsibility to make sure the ring surface is kind on your horse's legs. The ring should be raked or dragged regularly. If the ring is always hard, move your horse to a barn where the owner cares about horses' legs.

DAILY CHECKS

- Take a look at your horse's legs every day. If he lives in a pasture, walk out and run your hands down his legs to feel for lumps and bumps that might need treating. You should be picking out your horse's hooves on a daily basis, so check his legs at the same time.

TREAT SWELLINGS OR LAMENESS QUICKLY

- If you spot an injury on your horse's leg, spray cold water on it to remove blood so you can get a closer look. If the wound is serious, call the veterinarian right away because it might need stitching. If the bleeding doesn't stop, place a bandage over the cut and use your hand to apply pressure to the wound until the vet arrives.
- If you notice a swelling on a leg, feel it right away. If it feels hot, hose it with cold water for about 15 minutes, then check it again. Cold water can reduce swelling. If the leg still feels hot, call the vet and explain the situation. He may tell you to put an ice pack on the swelling, or he may say keep on hosing until he can get to your barn to examine your horse.

HANDLE YOUNG LEGS WITH CARE

- Young horses have bones that are still growing and developing, so it's important that you don't work them too hard at an early age. If a youngster is lunged on hard ground or galloped too much he may develop arthritis later on. You may also damage his delicate tendons and muscles.

- If you back a horse at two, stick to slower speeds and don't do too much with him until he's a bit older. Many sport horse people start horses under saddle when they turn three. And it's not a great idea to jump a horse until he's at least four because you can put too much stress on his young bones

REGULAR SHOEING AND TRIMMING

- Schedule your farrier on a regular basis—try to get him or her to trim or shoe your horse every five or six weeks. You don't want shoes falling off or hoof cracks getting bigger and bigger. If your horse has lameness issues, a good farrier will be able to forge therapeutic shoes for him. These are special shoes that may help your horse stay sounder.

JOINT-CARE SUPPLEMENTS

- If your older horse is starting to show signs of arthritis (stiffness or lameness), you could try putting him on a joint-care supplement. These supplements contain joint lubricants such as glucosamine hydrochloride, chondroitin sulfate or hyaluronic acid. Ask your veterinarian if she thinks a particular supplement would be beneficial to your horse.

WARM UP GRADUALLY

- Don't just jump on your horse and start galloping around the arena or through the fields. Just like humans, horses need to warm up for exercise gradually or they can strain muscles or tendons. Always walk for five or 10 minutes before you start jogging or loping. And remember to cool down your horse as well. Spend about five or 10 minutes walking before you take him back to the barn and untack him.

SUFFICIENT BEDDING

- Don't skimp on straw or shavings in your horse's stall or pen. Horses can injure their legs while lying down or getting up if they are kept on hard concrete floors. Horses can also get painful sores

OUTSIDE LIVING

- A horse's leg joints are far healthier if he spends most of his time outdoors in a field, able to mosey where he likes. If he stays in a stall all the time, he can get stiff.
- Young horses, especially, should spend loads of time outdoors. If they spend too much time stuck in a stall their bones won't grow and develop normally and this can cause lameness or arthritis when they get older.

- Older, arthritic horses seem to be less ouchy and stiff if they spend lots of time outdoors as well.
- on their hocks if there isn't enough bedding in their pens or stalls.

BREEDING SOUNDNESS EXAMINATION OF THE DOG

Identification

- The identification is not only important for the medical records, but is essential in the case of a pre-purchase examination.
- Failure to positively identify an animal could cause legal problems in the future.

History

- Has the dog ever been bred?
- Has he ever sired a litter? If has bred, but never sired a litter, you may consider congenital infertility.
- If he has sired litters, and is now infertile you may consider acquired infertility.
- Frequency of use
- Every other day breeding is usually acceptable.
- Breeding more than that may be overusing the dog.
- Has there been any exogenous drug therapy (testosterone etc.)?
- What is the dog's habitat?
- Is he in a kennel?
- Is he housed with bitch? If so, the bitch may be dominant.
- Is he 'mommies little boy', who has not had any canine contact?
- Has he had any exposure to sex?
- Has he been punished for showing sexual interest?
- Has his libido changed?
- Has he ever been shown?
- Does he have any endocrine disease?

Physical Exam

General exam

- Look for signs of endocrine problems.
- Check for any congenital disorders that would make him an unsuitable potential breeder.
- Reproductive Exam

MODULE-17: LEGAL ASPECTS OF EXAMINATION OF INJURIES AND POST-MORTEM EXAMINATION

- **Learning outcome**
- PM examination is one of the major Veterinary Ethics and Jurisprudence practices in the field

POST - MORTEM EXAMINATION OF VETERO - LEGAL CASES

- Postmortem in these cases has to be conducted in order to ascertain
 - The time since death
 - The cause of death.
- practice of every veterinarian. This module focusses on the Vetro-Legal aspects of PM examination.

RULES FOR PM EXAMINATION

- To be done by written order from the police officer or the Executive Magistrate.
- Carefully read the police report first.
- Examination to be done in day light. (Before sunset / after sunrise).
- Examination to be thorough and complete.
- All the details to be noted in the PM reports on the spot of the PM examination.
- The notes and the report to be sent to the Court must tally with each other (Police and owner).
- Should have fair knowledge of the normal and pathological appearance of viscera.
- Time and date of arrival of carcass to be noted.
- No unnecessary delay in conducting PM.
- No unauthorised person should be allowed to be present at the time of PM except investigating police officer.

INSTRUMENTS REQUIRED

Chemicals

- Preservatives – common salt and alcohol.
- Fixatives -10% Formalin, 50% glycerin, liquid paraffin, mercuric chloride, chromates, osmic acid etc.

RUMINANTS

- Keep the left side down in order to keep the rumen out of way.
- Right fore and hind legs are severed. Disarticulate right hind limb.
- Incision from jaws to the perineal region.
- Skin of right side reflected.
- Abdominal wall is removed.
- Rumen, reticulum, omasum and abomasum are opened in situ and examined.
- Expose thorax as in the case of horse.

- Lungs and heart removed with trachea and oesophagus.
- Urogenital organs are removed.

PROCEDURE FOR PM EXAMINATION

External examination

- Note the general condition of the carcass.
- Note the species age, breed and sex of the animal.
- Examine the body surface for injuries, burns, swellings etc.
- Look for the discharge from the natural orifices.
- Examine the visible mucous membranes.
- Examine the external genitalia.
- All bone for fracture and joint dislocation.

Internal Examination

- Necropsy procedure: Equine
- Place the horse on the right side down.
- Make an incision through skin in the ventral midline from chin to anus dorsolaterally around the mammary glands or penis.
- Skin of the left side of neck, thorax and abdomen is reflected.
- Left fore and hind limbs are severed by cutting all muscles and attachments
- Finish skinning thorax and abdomen to the top of the spinous process of the vertebrae
- Incise midline from xiphoid to pubis and pubis to lumbar region through abdominal wall.
- Pull upward and forward the large flap of abdominal wall and cut diaphragm dorso-ventrally close to its attachments
- Abdominal viscera are removed in the following order.
 - Great colon and caecum.
 - Left kidney
 - Left adrenal
 - The small colon
 - The small intestines
 - Liver
 - Right kidney
 - Right adrenal
 - The stomach
- Thoracic cavity – by cutting ribs along their ventral ends from thoracic inlet to last rib.
- Remove the thoracic organs by cutting oesophagus, trachea and other attachments lift the heart and lungs out.
- Cut the symphysis of pubis. Examine - genital organs, rectum and urinary bladder.

SWINE

- Place the carcass on its back on the table (supine position).
- Incisions are made to drop the limbs horizontally.
- Remove a strip of the ventral body wall from chin to the pubis regions.
- At sternal incisions passing through the costal cartilages.

- The anterior end of the sternal is raised with the left hand so that the heart is not cut.
- Ventral abdominal wall is incised and left to the pubis.
- Thoracic cavity by breaking ribs.
- Thoracic organs removed.
- First remove the spleen and omentum.
- Push the intestines left side in order to examine rectum.
- Cut the oesophagus posterior to diaphragm.
- Liver, stomach and intestines removed together.
- Remove adrenal glands before kidneys.
- Kidney by tearing its peritoneal covering.
- Seminal vesicles and other accessory organs in boar.

CANINE AND FELINE

- Place the dorsum upon the table.
- Head and neck straightened.
- Incision from mandibular symphysis to pubis.
- Mammary gland examined.
- Separate the skin.
- Legs pulled to the sides.
- Cut the coxo – femoral joints.
- Incision from xiphoid cartilage to the pubis.
- Make transverse incision.
- Abdominal cavity is exposed.
- Expose thoracic cavity by cutting sternal ribs/cartilage.
- Examine the organs in situ.

EXAMINATION OF ORGANS

Lungs

- Lungs and associated lymph node inspected.
- Open the trachea and its branches.
- Incision to different lobes.
- Squeeze the cut edges and examine for pus, blood or edematous.
- Examine diaphragmatic lobe for parasites.

Hearts

- Examine pericardium, pericardial fluid and epicardium.
- Remove pericardium.
- Locate the septum and open right ventricle by incision.
- Pass the knife through tricuspid valve into the pulmonary artery and open it.
- Similar procedure followed to left side heart.
- Pulmonary vein opened by passing knife through bicuspid valve.

Brains

- Examine for the presence of tumors, cysts, abscesses, parasites, haemorrhages etc.

Digestive tract

- Parenchymatous organs - Liver, spleen, kidneys, testes, adrenals.

SUBMISSION OF SPECIMENS IN SUSPECTED CASES OF POISONING

Suspected poison	Required material in order of importance
Arsenic (Acute)	Liver, kidneys, stomach contents
Arsenic (Chronic)	Hair, liver, urine
Alkaloids	Liver, urine, brain, stomach contents.
Copper	Liver
Cyanide (HCN)	Stomach contents, liver, oxalated blood
Chlorinated insecticides	Fat, liver, stomach contents
OPC	Oxalated blood, whole blood, stomach contents
Lead (Acute)	Kidney, Liver, urine
Lead (Chronic)	Hair, liver, kidneys, urine
Mercury	Liver, kidney, stomach contents, intestinal contents
Nitrate/nitrite	Stomach contents, whole blood
Phosphorous	Stomach contents, Liver, oxalated blood
Phenol – cresols	Liver, stomach contents, kidney
Rodenticides	Stomach contents, liver, urine
Strychnine	Stomach contents, urine, liver, brain, kidneys
Sodium chloride	Oxalated whole blood brain, stomach contents, liver

QUANTITIES OF SPECIMENS TO BE SENT**Tissue / organs and quantity**

Blood	30 to 50 ml
Brain	Entire
Fat	200 gm

Hair	5- 10 gm
Intestinal contents	500-1000 g
Kidneys	One
Liver	500-1000 g
Stomach contents	500 – 1000g in LA and available contents in SA
Urine	All available

- Visceras to be preserved for a period of 6 months and then destroyed after obtaining the permission from the magistrate.

CONTAINERS FOR THE MATERIAL TO BE PRESERVED

- Wide mouth glass bottle (2 litre) with air light stoppers (arsenic free).
- Bottle should have
 - Case no.
 - Nature of the contents.
 - Place and date of preservation etc.
 - Signature of veterinarian done - PM.

PRESERVATIVE

- Alcohol – suitable for viscera (Except cases of alcohol, phenol, by other volatile substance)
- Saturated common salt.

Preservation by alcohol

- Whole organ in dipped into alcohol.
- Saturated common salt
- Solid common salt

Preservation of blood sample

- Sodium fluoride – 20 mg / ml of blood.
- Sodium citrate (5 mg) and mercuric chloride mixture (0.1 mg) / ml of blood.
- Sealing of containers and other material

LETTER TO THE CHEMICAL EXAMINER

- Species sex, age, identification mark and location of lesions.
- Impression of the seal used in closing the bottle and description of seal.

- Separate necropsy report.
- Details of the purpose for which analysis is required.
- List of articles forwarded and mode of transmission.
- Well packed and labeled.
- Send 25 ml of preservative in separate bottle as sample preservative.
- Send copy of inquest.

COLLECTION AND SUBMISSION OF SPECIMENS FOR HISTOPATHOLOGICAL EXAMINATION

- Fixing agents - formaldehyde, alcohol, mercuric chlorides, chromates, osmic acid, picric acid, etc.
- Important is 10% formalin.
- Specimen should be representative of the lesion with normal tissues - 0.5 -1.00 cm thick and 10 times volume of fixatives.

POST-MORTEM ARTIFACTS

- Changes introduced in the dead animals after death.
- Not related to the physiology state of body or tissues either health or disease to which the animal might be exposed prior to death should be differentiated.
- Unless it will lead to misinterpretation of
 - Wrong cause of death
 - Wrong manner of death
 - Undue suspicion of criminal offences
 - Miscarriage of justice

PM ARTIFACTS - CLASSIFICATION

- [Putrefaction](#)
- [Environmental artifacts](#)
 - Burning
 - Corrosion
 - Maceration
- [Third party artifacts](#)
 - Animals, birds, insects, maggots etc.
 - Emergency treatment just prior to death and after death
 - Deliberate mutilation
- [Miscellaneous artifacts](#)

Putrefaction: kept under sunlight < 2 hour - severe decomposition takes place.

- *Example*
 - Distension of abdomen, and thorax
 - Profusion of tongue
 - Bulging of eye balls.
 - Swelling of lips, nose, eyelids, vulvar lips, udder etc.
 - Presence of sero sanguineous fluid in and around.
 - Mouth, nose, vulvar opening etc.

- PM – vomition of ruminal contents
- PM – defecation and urination

Environmental artifacts

- *Postmortem burning*: unusual places affected, no tissue reaction.
- *Post – mortem corrosion*: Acids, caustic soda, industrial effluent etc. and seen on body surface touching ground.
- *PM – maceration*: Prolonged physical contact of the body parts and bones with water, mud, etc.

Third party artifacts

- Birds ants, flies, etc - Ears, eyes, nose, mouth, anus, vulva, udder.
 - *Example*
 - Haematoma around jugular vein, sub-cutaneous fluid accumulation at the sites of injection, laceration and obstetrical handling marks in urogenital tract, rumen puncture wound etc. are some of the marks left after emergency treatment.
- Deliberate mutilation of body – to mask the identify of the animal.
 - *Example*
 - Cutting ear bearing tattoo mark / tag.
 - Cutting of tail.
 - Damage of scar mark etc.

Miscellaneous

- Dragging marks on the lower surface of the body.
- Rope marks on the fetlock region around horns, neck etc.

POST MORTERM APPEARANCE OF SHOCK

- Signs severe injuries
 - Fracture
 - Crushing of tissues
 - Extensive lacerations and Avulsions
- Generalised visceral congestion
- Oedema of the lungs
- Dark red fluid blood
- Petechial haemorrhage in serious membranes and parenchymatous organs.

MODULE-18: COLLECTION AND DESPATCH OF MATERIALS FOR CHEMICAL EXAMINATION**• Learning outcome**

- In veterinary clinical services the sample collection and dispatch of collected material from one area to the lab for diagnostic and investigative purposes is very important in case that material is related to an outbreak or legal case. This module will educate the learner on the same in an effective manner for vetero-legal cases.

PROCEDURES ON SUSPICION

- Take careful history, which may indicate changes of feed composition, exposure to poison or administration of potentially toxic preparation.
- Make a careful examination of the environment to look for the potential pathogens.
- Note the symptoms carefully and the time of their occurrence.
- When there is illness of large number of animals at a time and showing the same symptoms and post-mortem lesions.
- Carefully examine dead animals for signs of struggling, frothy nasal discharge etc.
- For correct diagnosis, it is essential to have the proper collection, preservation and chemical analysis from a forensic laboratory. The following table gives the summary of the samples to be collected in different suspected conditions.

PROCEDURE FOR COLLECTION OF MATERIAL FOR CHEMICAL EXAMINATION

Collection of sample	Tissues or fluids should be as fresh as possible
Sample amount needed	
In small animals	Whole unopened cadaver in case of dog, cat, poultry, etc.
In large animals (tissue, one, ingesta)	100 gms
Water	100 ml
Whole blood	15 ml
Hair	5-15 gms
Brain	50 ml
Feed	1 kg
Preservation of sample	
Preferable preservative	• 95% ethanol @ 1ml/g of tissue • Avoid use of formalin as it interferes with many tests.

Storage of sample	Refrigerator or deep freeze
Container requirement	- Plastic bags or jars; Glass containers - Avoid jars with metal screws when metal poisoning is suspected
Precautions	Do not wash the sample with water before packing and dispatching

SUBMISSION OF SPECIMENS IN SUSPECTED CASES OF POISONING

Suspected poison	Material required in order of importance
Antimony	Liver, thyroid, kidney, urine, milk, stomach and gut content
Arsenic (acute)	Liver, kidneys, stomach and gut contents, urine, feed
Arsenic (chronic)	Hair, liver, urine, spleen, altered organs
Alkaloids	Liver, urine, brain, stomach and gut contents
Ammonia	Whole blood or serum, urine, rumen contents * Send frozen samples * In rumen contents, may add 1-2 drops of saturated HgCl_3
Barbiturates	Blood, brain, liver, adipose tissue
Cadmium	Kidney, liver, hair
Copper	Liver, blood, kidney, faeces, urine
Cyanide (HCN)	Stomach and gut contents, liver, muscles, oxalated blood, brain, suspected feed. * Should be rushed to laboratory in frozen airtight container * Liver HCN decrease rapidly, so many times muscle analysis gives good results
Fluorides	Altered parts of bone, teeth, urine, stomach, contents, liver, kidney, feed and/or drinking water
Insecticides (chlorinated)	Fat, liver, stomach contents, brain * Use only glass containers
Insecticides (organophosphorus)	Oxalated blood, whole blood, liver, stomach and gut contents
Lead (acute)	Kidneys, liver, urine, whole blood

Lead (chronic)	Hair, liver, kidneys, bone, faeces, urine
Mercury	Liver, kidneys, intestinal contents, stomach contents, muscle, brain, faeces
Molybdenum	Feed, liver, kidneys, altered organs (bone), hair * The feed is examined / analysed for its Mo:Cu contents. Liver contains low amount of copper
Nitrate and Nitrite	Stomach and gut contents, whole blood, urine, suspected feed * Should be dispatched soon as both nitrate and nitrite alter rapidly and escape detection
Oxalates	Fresh forage, kidneys (Do not macerate; freeze, fixed in formalin)
Phenol and cresol	Liver, kidneys, stomach and gut contents
Phosphide	Stomach and gut contents, suspected material (grain, flour etc.) (Dispatch as soon as possible as phosphide breaks down rapidly and hydrogen phosphide produced may escape)
Phosphorus	Stomach and gut contents, liver, oxalated blood, altered organs (lungs), vomitus, faeces
Rodenticides	Stomach contents, liver, urine
Selenium	Suspected feed, altered organs, whole blood, liver, hair
Sodium chloride	Oxalated and whole blood, brain
Sodium fluoroacetate	Stomach contents, liver
Strychnine	Stomach, gut contents, urine, liver, kidney and brain
Urea	Feed, stomach contents (See aminonia)
Warfarin	Whole blood, liver, feed
Zinc	Liver, kidney, pancreas, faeces, bone (humerus)
Zinc phosphide	Liver, gastric contents, feed (See phosphide)

**PROCEDURE FOR DISPATCHING OF
MATERIAL FOR CHEMICAL EXAMINATION**

- Complete the report forms for pathological specimens.
- Enter the name and address of the submitter.
- Enter full name and address of the owner of the stock.
- Describe the full history along with ante and post-mortem findings if any.

- A separate container should preferably be used for each case.
- Put a frozen coolant pad, into each container or pack these specimens in ice before dispatching.
- Specimens must be packed so that they do not contaminate each other, or leak in transit.
- Normally samples are sent through express rail or local transport arrangements.

SUBMISSION OF SPECIMENS IN POISONING SUSPECTED CASES

- In suspected cases of poisoning the specimens should be sent for chemical analysis.

Poison	Materials required in order of importance	
MINIMUM QUANTITIES OF SPECIMENS TO BE SENT FOR TOXICOLOGICAL EXAMINATION	Materials	Quantity
	Blood	30 – 50 ml
	Brain	Entire brain
	Fat	200 gm
	Hair	5 to 10 gm
	Intestinal contents	500 to 1,000 gm
	Kidneys	One
	Liver	500 to 1,000gm
	Stomach contents	500 to 1,000gm in Large Animals
	Urine	Availage contents in Small Animals
		All available
• Unless, the viscera and other articles are forwarded to the chemical examiner, they are to be preserved for a period of <u>six</u>months. After six months, they are to be destroyed after obtaining the permission from the magistrate.		

- **SUBMISSIONS OF SPECIMENS FOR HISTOPATHOLOGICAL EXAMINATION**

- Histo pathological examination in many cases is helpful in diagnosing the cause of death. Tissues for histo pathological examination should be preserved. The tissues must be fixed as soon as they are removed from the animal.
- The ideal fixing agent should quickly penetrate the tissue and preserve it for longer period.
- Different fixing agents are
 - Formaldehyde
 - Alcohol (Ethanol)
 - Mercuric chloride
 - Chromates
 - Osmic acid
 - Picric acid
- Most common fixative used is 10 per cent formalin.
- Specimen taken for H.P. examination should be representative of the lesion and should include some of the normal tissues.
- Specimen should be cut in size of 0.5 to 1cm thick and placed in the fixative which is 10 times the volume of the tissue.
- The labeled bottles are then packed carefully to prevent leakage or breakage.

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MODULE-19: PROVINCIAL AND CENTRAL ACTS RELATING TO ANIMALS

- **Learning outcome**

- In India, Ministry of Environment and Forests - Animal Welfare Division, Animal Welfare Board of India, National Institute of Animal Welfare and CPCSEA are major key players in drafting and implementing the laws related to veterinary ethics and jurisprudence. This module will educate the learner on this area and expose them to laws and rules related to animal welfare.

ACT

- No.59 of 1960 - The Prevention of Cruelty to Animals Act, 1960.

RULES

1. S.O. 1995 (E) [27/07/2010] - Prevention of Cruelty to Animals (Slaughter House) Amendment Rules 2010.
2. S.O.714(E), [26/03/2010] - Animal Birth Control(Dogs) Rules, 2009.
3. S.O.1818(E), [23/10/2006] - Breeding of and Experiments on Animals (Control and Supervision)Amendment Rules, 2006.
4. S.O.1817(E), [19/10/2006] - Reconstitution of Committee for the Purpose of Control and Supervision of Experiments on Animals(CPCSEA).
5. S.O.42(E), [14/01/2006] - Breeding of and Experiments on Animals (Control and Supervision)Amendment Rules, 2005.
6. S.O.1256(E), [24/12/2001] - The Animal Birth Control (Dogs) Rules, 2001.
7. S.O.267(E), [26/3/2001] - The Performing Animals (Registration) Rules, 2001.
8. S.O.35(E), [8/1/2002] - The Performing Animals (Registration) Amendment Rules, 2001.
9. S.O.271(E), [26/3/2001] - The Prevention of Cruelty to Animals (Establishment and Regulation of Societies for Prevention of Cruelty to Animals) Rules, 2001.
10. S.O.270(E), [26/3/2001] - The Prevention of Cruelty to Animals (Slaughter House) Rules, 2001.
11. S.O.268(E), [26/3/2001] - The Prevention of Cruelty to Animals (Transport of Animals on Foot) Rules, 2001.
12. S.O.134(E), [15/2/2001] - The Breeding of and Experiments on animals (Control and Supervision) Amendment Rules, 2001.
13. S.O.1074, [15/12/1998] - The Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998.
14. S.O.732(E), [26/8/1998] - The Experiments on Animals (controls and Supervision) (Amendment) Rules, 1998.
15. S.O.No.1056, [13/3/1979] - The Prevention of Cruelty (Capture of Animals) Rules, 1972.
16. No.14-20/76-LD.I, [30/11/1978] - The Prevention of Cruelty to Animals (Registration of Cattle Premises) Rules, 1978.
17. No. 18-6/70-LDI, [23/3/1978] - The Transport of Animals Rules, 1978 .
18. S.O.269(E), [26/3/2001] - The Transport of Animals (Amendment) Rules, 2001.
19. No.14-21/76-LD.I, [15/2/1978] - The Prevention of Cruelty to Animals (Application of Fines) Rules, 1978.
20. No.35-4/72-LD.I, [22/5/1973] - The Performing Animals Rules, 1973.

21. No.9-18/62-LD, [23/3/1965] - The Prevention of Cruelty to Animals (Licensing of Farriers) Rules, 1965.
22. No.9-18/68-LD, [23/3/1965] - The Prevention of Cruelty to Draught and Pack Animals Rules, 1965, amended 1968.

NOTIFICATIONS

23. G.S.R.619(E), [14/10/1998] - The Prevention of Cruelty to Animals (Restricted to Exhibit on Trained as a Performing Animals).

THE GLANDERS AND FARCY ACT - 1899

- An Act to consolidate and amend the law relating to Glanders and Farcy.
- Whereas it is expedient to consolidate and amend the law relating to Glanders and Farcy : It is hereby enacted as follows
- **Short title extent and commencement**
 - This Act may be called the Glanders and Farcy Act 1899
 - It extends to the whole of British India.
- **Definition of “diseased”**
 - In this act unless there is anything repugnant in the Farcy or any other dangerous epidemic disease among horses, which the Governor General in Council may by notification in the Gazette of India, specify in this behalf, either generally or in respect of any local area.
 - The provisions in this Act relating to horses shall apply also to Asses and mules.
- **Application of Act to local areas by local Government**
 - The local Government may, by notification in the local official Gazette apply this Act or any provision of this Act, so far as all or any of the diseases mentioned in or specified in a notification under section 2. sub section 1, are concerned to any local area to be specified in such notification within the Province.
- **Local Government to appoint the Inspects**
 - When this Act has been so applied to a local area, the local Government may by notification in the local official Gazette appoint such persons as it thinks fit to be Inspectors under this Act and to exercise and perform within the whole of the local area, of such portions thereof as it may prescribe the powers conferred and the duties imposed by this Act on such officers.
 - X.L.V of 1860
 - Every person so appointed shall be deemed to be a public servant within the meaning of the Indian Penal Code.
- **Power of entry and search**
 - Within the local limits for which he is so appointed, any such Inspector as aforesaid may subject to such rules as the Local Government may make in this behalf enter and search any field building or other place for the purpose of ascertaining whether there is therein any horse which is diseased.
- **Power of seizure**
 - Within such limits as aforesaid, the Inspector may seize Power of any horse, which he has reason, to believe to be diseased.

- **Horse to be examined by Veterinary Practitioner**
 - On any such seizure as aforesaid, the Inspector shall cause the horse seized to be examined as soon as possible by such Veterinary Practitioner as the Local Government may appoint in this behalf.
 - Provided that, when the Inspector is also a Veterinary Practitioner so appointed, he may make the examination himself.
 - For the purposes of the examination, the Veterinary Practitioner may submit the horse to any test or tests which the Local Government may prescribe.
- **Horse to be destroyed if found diseased: otherwise restored**
 - If the Veterinary Practitioner certifies in writing that the horse is diseased, the Inspector shall cause the same to be immediately destroyed.
 - Provided that, in the case of any disease other than Glanders or Farcy, horses certified to be diseased as aforesaid, may subject to any rules which the Local Government may make in this behalf, be either destroyed or otherwise treated or dealt with as the Veterinary Practitioner may deem necessary.
 - If, after completing the examination the Veterinary Practitioner does not certify that the horse is diseased, the Inspector shall at once deliver the same to the person entitled to the possession thereof.
- **When horse diseased, place where it has been to be disinfected etc.**
 - When any diseased horse has been in any building, shed or other enclosed place, or in any open lines, the Inspector may issue a notice to the owner of the building, shed place or lines or to the person in charge thereof, directing him to have the same disinfected and the internal fittings thereof or such other things found therein or near thereto as the Local Government may by rule prescribe, destroyed.
 - On the failure or neglect of such owner or other person as aforesaid to comply with the notice within a reasonable time, the Inspector shall cause the building, shed place or lines to be disinfected and the fittings or other things to be destroyed, and the expense (if any) thereby incurred may be recovered from the owner or other person as if it were a fine.
- **Owner or person in charge of diseased horse to give notice**
 - The owner or any person in charge of a diseased horse shall give immediate information of the horse being diseased to the Inspector or to such authority as the Local Government may appoint in this behalf.
- **Prohibition against removal without license, of horse which has been with diseased horse**
 - No person in charge of any horse which has been in the same field, building or place as, or in contact with, a diseased horse, shall remove such horse except in good faith for the purpose of preventing infection, or under a license to be granted by the Inspector and subject to the conditions of the license.
- **Vexatious entries, searches and seizures.**
 - Whoever, being an Inspector appointed under this Act, vexatiously and unnecessarily enters or searches any field building or other place, or seizes or detains any horse on the pretence that it is diseased, shall be punishable with imprisonment.
 - For a term which may extend to six months, or with fine which may extend to five rupees, or with both.
 - No prosecution under this section shall be instituted after the expiry of three months from the date on which the offence has been committed.
- **Penalty for refusing to comply with notice under section 9 or for removing horse contrary to section 11**
 - Whoever refuses or neglects to comply with any notice issued by the Inspector under section 9, or removes any horse in contravention of section 11, shall be punishable with

imprisonment for a term which may extend to one month, or with fine which may extend to fifty rupees, or with both.

- The Local Government may make rules to carry out the purposes and objects of this Act.
- In particular and without prejudice to the generality of the foregoing power such rules as aforesaid may -
 - regulate entries, searches and seizures by Inspectors under this Act ;
 - regulate the use of tests and the isolation of horses subjected thereto, and provide for recovering the expense of detaining, isolating and testing horses from the owners or persons in charge thereof, as if it were a fine;
 - regulate the destruction or treatment as the case may be of horses certified under section 8 to be diseased, and the disposal of the carcasses of diseased horse;
 - regulate the disinfecting of buildings and places in which diseased horses have been, and prescribe what things found therein or near thereto, shall be destroyed; and
 - regulate the grant of licenses under section 11 and the conditions on which those licenses shall be granted.
- All rules under this section shall be published in the local official Gazette and, on such publication, shall have effect as if enacted by this Act.
- In making any rule under this section, the Local Government may direct that a breach of it shall be punishable with imprisonment for a term which may extent to one month or with fine which may extent to fifty rupees, or with both.
- **Appointment of same person to be both inspector and Veterinary Practitioner**
 - Any Veterinary Practitioner may be appointed by the Local Government to be both Inspector and Veterinary Practitioner for all or any of the purposes of this Act or of any rule there under.
- **Protection to persons acting under Act**
 - No suit, prosecution or other legal proceeding shall lie against any person for anything which is, in good faith, done or intended to be done under this Act.

THE DOURINE ACT, 1910

- An Act to provide for the prevention of the spread of Dourine.
- **Short title and extent**
 - Whereas it is expedient to provide for the prevention of the spread of dourine; It is hereby enacted as follows:
 - This Act may be called the Dourine Act, 1910.
- **The definitions**
 - This section extends to the whole of British India: rest of this Act extends only to such areas as the Local Government may, by notification in the local official Gazette, direct.
 - In this Act, the expressions "inspector" and "veterinary particular" mean, respectively, the officers appointed as such under this Act, acting within the local limits for which they are so appointed.
 - The provision of this Act in so far as they relate to entire horses shall, if the Local Government, by notification as aforesaid, so directs, apply also to entire asses used for mule-breeding purposes.
- **Registration of horses**
 - The Local Government may, by notification as aforesaid, make such orders as it thinks fit directing and regulating the registration of entire horses maintained for breeding purposes.
 - Appointment of Inspectors and Veterinary Practitioners

- The Local Government may, by notification as aforesaid, appointment any persons it thinks fit to be inspectors, and any qualified veterinary surgeons to be veterinary practitioners, under this Act, and to exercise and perform, within any area prescribed by the notification, the powers conferred and duties imposed by this Act upon such officers respectively.
- ALVA of 1860
- Every person so appointed shall be deemed to be public servant within the meaning of the Indian Penal Code.
- **Powers of Inspectors**
 - An inspector may, subject to such rules as the Local Government may make in this behalf,-
 - Enter and search any building, field or other place for the purpose of ascertaining whether there is therein any horse which is affected with dourine; and
 - prohibit, by order in writing, the owner or keeper of any horse, which in his opinion is affected with dourine, from using such horse for breeding purposes, pending examination by the veterinary practitioner, (see Amendment Act, 1920).
- **Duties of Inspectors**
 - An inspector issuing an order under section 5, shall forthwith forward a copy of such order to the veterinary practitioner.
- **Inspection of horses**
 - A veterinary practitioner receiving a copy of an order forwarded under section 6 shall, as soon as possible after receipt of such copy, examine the horse mentioned therein, and may for such purpose enter any building, field or other place.
- **Powers of Veterinary Practitioner**
- A Veterinary Practitioner may cancel any order issued under section 5, clause, (b); or if on microscopical examination he finds that any horse is affected with dourine,
 - in the case of an entire horse, cause it to be castrated,
 - in the case of a mare, cause it to be branded in such manner as he may direct, or with the previous sanction of the Commissioner or such other officer as the Local Government may appoint in this behalf, cause it to be destroyed.
- **Compensation for horse destroyed, etc.**
- When any horse is castrated or destroyed under section 8, the market-value of such horse immediately before it became affected with dourine shall be ascertained; and the Local Government shall pay as compensation to the owner thereof-
 - in the case of a mare which has been destroyed, or of an entire horse which has died in consequence of castration, such market value, and,
 - in the case of an entire horse which survives castration, half the amount by which such value has been diminished owing to infection with dourine and castration.
- **Settlement of compensation**
 - A veterinary practitioner may award, as compensation to be paid under section 9 in respect of each horse castrated or destroyed under section 8, a sum not exceeding two hundred and fifty rupees.
 - If in the opinion of the veterinary practitioner the amount which should be paid as such compensation exceeds two hundred and fifty rupees, he shall report accordingly to the Collector, who shall decide the amount to be so paid.
- **Committee for hearing appeals**
 - The Local Government shall, by rules published in the local official Gazette, make provision for the constitution of a committee or committees for the hearing of appeals from decisions under section 10.
 - Such rules shall provide that not less than one member of any committee constituted there under shall be a person not in the employ of Government or of a local authority.

- **Appeals**
 - Any owner may, within two months from the date of a decision under section 10, appeal against such decision to the committee constituted in that behalf by rules made under section 11, and the decision of such committee shall be final.
- **Vexatious entries and searches**
 - Whoever, being an inspector appointed under this Act, vexatiously and unnecessarily enters or searches any field, building or other place, shall be punishable, with imprisonment for a term which may extend to six months, or with fine which may extend to five hundred rupees, or with both.
 - No prosecution under this section shall be instituted after the expiry of three months from the date on which the offence has been committed.
- **Rules**
 - The Local Government may make rules for the purpose of carrying into effect the provisions of this Act.
 - In particular, and without prejudice to the generality of Rules the foregoing power, such rules as aforesaid may.-
 - regulate entries searches and orders by inspectors under section 5; and
 - regulate the action to be taken by the veterinary practitioners under section 8; and
 - make provision for the payment of compensation to the owner of any mare branded under section 8.
 - All such rules shall be published in the local official Gazette, and, on such publication, shall have effect as if enacted in this Act.
 - In making any rule under this section the Local Government may direct that a breach of it shall be punishable with fine which may extend to fifty rupees.
- **Penalties**
 - Whoever uses or permits to be used for breeding purposes
 - any horse which has not been registered in accordance with the requirements of a notification under section 3. or
 - any horse in respect of which an order under section 5, clause (b) is in force, or
 - any mare which has been branded in pursuance of section 8, clause (b)
 - shall be punishable with fine which may amount, in the case of a first conviction, to fifty rupees, or in the case of a second or subsequent conviction, to one hundred rupees.
- **Protection to person acting under Act**
 - No suit, prosecution or other legal proceeding shall lie against any person for anything which is, in good faith, done or intended to be done under this Act.
 - II.ACT No. VIII of 1920 AMENDING THE ABOVE ACT.
 - (PASSED BY THE INDIAN LEGISLATIVE COUNCIL)
 - (Received the assent of the Governor-General on the 4th March, 1920.)
 - An Act to amend the Dourine Act, 1910.
- (Short title.)
- V of 1920. Whereas it is expedient to amend the Dourine Act, 1910; It is hereby enacted as follows:-
 - (Amendment of section 5, Act V of 1910.)
 - This Act may be called the Dourine (Amendment) Act, 1920.
 - V of 1920.
 - In section 5 of the Dourine Act, 1910 (hereinafter referred to as the said Act),
 - the word "and" at the end of clause (a) shall be omitted; and
 - after clause (b) the following clause shall be added, namely :-

- "(c) direct, by order in writing, the owner or keeper of any horse which, in the opinion of the Inspectors, is affected with dourine to remove it or permit it to be removed for the purpose of segregation to a place specified in the order, and such direction shall be sufficient authority for the detention of the horse in that place for that purpose."
- (Amendment of section 6, Act V of 1910.)
- In section 6 of the said Act the word and letter "clause (b)" shall be omitted. (Amendment of section 8, Act V of 1910.)
- In section 8 of the said Act
 - in clause (a) the word and letter "clause (b)" shall be omitted; and
 - in clause (b)
 - after the words "on microscopical examination" the words "or by other scientific test" shall be inserted;
 - for sub-clause (ii) the following shall be substituted namely :-
 - "(ii) in the case of mare, with the previous sanction of such authority as the Local Government may appoint in this behalf, or, if so empowered by the local Government, without such sanction, cause it to be destroyed."
- In sub-section (2) of section 14 of the said Act - (Amendment of section 14, Act V of 1910.)
- for clause (a) the following shall be substituted, namely :-
- "(a) regulate the exercise of the powers conferred on Inspectors under section 5" ; and the word "and" at the end of clause
- (b) and the whole of clause
- (c) shall be omitted. (Amendment of section 15, Act V of 1910.)
- For clause (b) and (c) of section 15 of the said Act, the following shall be substituted, namely:-
- "(b) any horse in respect of which an order under clause (b) or clause (c) of section 5 is in force."

MODULE-20: SUDDEN DEATH IN ANIMALS

- **Learning outcome**

- The animal death may happen due to various reasons. But sudden death in animals may be due to some unknown causes. This unknown causes may elicit suspects to animal owners. To rule out the causes the veterinarian has to rule out all possible causes. This module focusing on various reasons of sudden death in animals and its related management.

SUDDEN OR UNEXPECTED DEATH

- Sudden or unexpected death may occur from
 - Unnatural causes - Violence or poison
 - Natural causes
- Unnatural death - always to be investigated by the police.
- Natural death if they have occurred suddenly in apparently healthy animals and under suspicious circumstances, vetero legal investigations are needed. The possibility of death as a result of disease and injury together has to be kept in mind. In such cases, a veterinary practitioner should not certify the cause of death without conducting a postmortem examination even if there is a string evidence of the disease.

Etiology/causes

- Natural causes producing sudden death are
- Diseases of the cardio vascular system
 - Acute myocarditis - due to infections like enteric fever (Typhoid) Diptheria, pericarditis
 - Fathydegeneration of the myocardium Rupture of a myocardial infact or aneurysm.
 - Left ventricular failure - associated with Aortic valvular disease, hyperthyroidism.
 - Right ventricular failure - associated with chronic emphysema and lung diseases.
 - Diseases of pericardium - Pericarditis.
 - Congenital abnormal conditions of the heart in calves.

Classification of Death

- Vetero-legal deaths is ased upon the concept that the cessation of the vital function depends upon tissue anoxia, which is brought about by four ways.
- Defective oxygenation of the blood in the lungs (or) anoxic anoxia is produced by
 - Obstruction to the passage of air into the respiratory tract as in
 - Suffocation
 - Smothering and
 - Overlaying
 - Obstruction in the passage of air down the respiratory tract as in
 - Strangulation
 - Drowning
 - Chocking from impaction of a foreign body.
 - External compression of the chest and abdominal walls

- Eg. on being buried under earth (or) debris.
 - Primary cessation of respiratory movements causing respiratory failure
 - Eg. Narcotic poisoning, electrical shock
 - Breathing in a vitiated atmosphere containing excess of carbondioxide or inert gases.
- Reduced oxygen carrying capacity of the blood – Anaemic anoxia occurring in acute poisoning by
 - Carbon monoxide
 - Chlorates
 - Nitrites
 - Coal tar derivatives
- Depression of the oxidative processes in the tissues – histotoxic anoxia occur in acute cyanide poisoning
- Inefficient circulation of the blood through the tissues – stagnant anoxia – occurring in death from
 - Traumatic shock
 - Heat stroke and
 - Acute and corrosive poisoning
- All four types of anoxia produce circulatory failure and leads to death.
- The postmortem finding in all forms of death of vetero-legal nature can be divided into two types.
 - Basic pathological haemorrhages and
 - Special pathological changes depending upon the particular type of death. Eg. Local injuries to the neck in throttling a strangulation.
- The tissue anoxia induced, leads to circulatory failure. Hence, the fundamental pathological changes are uniform in all forms of death they vary in degree.
- Sudden primary cardiac failure cause rapid death and the absence of visceral congestion when compared with death occurring slowly.

CHIEF SIGNS OF DEATH

- An assessment of the various changes in the body, after death, enable us to arrive at an approximate idea as to the length of time that has elapsed since death. It is impossible to fix the exact moment of death.

Chief signs of death

- **Cessation of circulation and respiration**
 - Complete cessation of the circulation and respiration are the only reliable signs of death in early postmortem stages. In any case of doubt, the carcass should be left until some of the later signs appear which render the diagnosis absolute.
 - Hibernating animals are known to resuscitate to life after having remained/for some time in a condition in which the action of heart and lungs was in abeyance and the muscles stiff and motionless. This state of suspended animation may last for a few seconds to half an hour or more may be found in cases of hysteria, sunstroke, concussion, drawing frozen, coma, electric shock, tetanus, convulsions, surgical shock and anesthesia. The animal can be revived by cardiac massage or electric stimulator accompanied by artificial respiration.
 - Cessation of circulation can be tested by
 - Auscultation of the heart
 - Absence of a pulsation in the arteries
 - Cessation of respiration may be tested by

- Holding a bright looking-glass in front of the open mouth and nostrils become dim due to the condensation of warm moist air exhaled from the lungs if animal respire.
 - Hold a feather (or) cotton fibre in front of the mouth and nostril, if respiration has stopped, there will be no movement.
- **Changes in the eye**
 - Soon after death, the eye loses its luster. The corneal reflex is lost. Cornea becomes opaque and looks like dimmed glass. Such a condition may be present before death in uraemia and narcotic poisoning. Pupils may change their form and become oval, triangular or polygonal when pressure is applied by fingers on two or more sides of the eye balls of a dead animal. Pupils retain their round form in a living animal. Eyes look sunken due to reduction of intra-ocular tension.
- **Cooling of the body**
 - After death, the body starts losing its heat by conduction, convection and radiation and gradually attains the same temperature as that of its surrounding medium. The loss of heat from the body cannot be considered as a certain sign of death until the body has lost 15 to 20 degrees of the normal heat viz 80°F for a rectal temperature. The body surface will be cold in 8 to 12 hours. The rate of cooling of the body may be influenced by age, condition of the body, manner or mode of death and surroundings of the body.
 - Fat and large well-nourished bodies retain heat much longer than lean and weak bodies. In general small and emaciated animals lose their warmth rapidly in cold surroundings. Freezing temperature accelerates the loss of the body heat.
 - Cooling of the body is more rapid in deaths occurring from severe haemorrhage, chronic and wasting diseases, whereas the body stays warm for a long time, when death has resulted from Asphyxia as in lightning, suffocation or poisoning by carbon dioxide.
- **Cadaveric changes in the muscles**
 - After death, Muscular tissues of the body pass through 3 stages
 - Primary flaccidity or relaxation
 - Rigor mortis or cadaveric rigidity
 - Secondary flaccidity
 - **Primary flaccidity**
 - Soon after death, the whole muscular system begins to relax. Hence, we notice that the lower jaw falls, the eye lids lose their tension, the extremities become soft, and flaccid and the joints flexible. But, the muscles are contractile and react to external stimuli, mechanical or electrical due to their retaining molecular life after somatic death. This stage lasts for 3 to 6 hours, but the average is 2 or 3 hours.
 - **Rigor mortis or cadaveric rigidity**
 - Is also known as death stiffening. Rigor mortis comes on immediately after the muscles have lost the power of contractility (both voluntary and involuntary muscles). Adenosine triphosphate plays a fundamental role in the normal contraction, relaxation of the living muscle and it also determines the physical condition of the muscle protein, myosin. Myosin protein consists of two protein myosin and actin. Their combined form is called actomyosin, which possesses the property of contractility and relaxation in the presence of adenosine triphosphate which is closely bound to the muscle protein and potassium chloride (KCl).
 - **Secondary Flaccidity**
 - Secondary flaccidity is characterized by the relaxation of the muscle which is synchronous with the onset of putrefaction. The muscles do not respond to any electrical stimuli.
 - **Putrefaction or Decomposition**

- It is the last stage in the resolution of the body. It is a slow process which is brought about by the action of ferments, produced by living saprophytic micro-organisms which resolve the complex organized tissues of the body into simpler inorganic compounds. Aerobic and gas forming anaerobic micro organisms in the living animal are found in large numbers in the alimentary cannal but within a short time after death are found scattered in all the tissues and organs and even in the blood. Unless special care is taken to prevent the spread of micro-organisms, they may cause putrefaction of the carcass.
- **Factors responsible for the process of putrefaction**
 - Temperature: Most favourable temperature for putrefaction is 100°F.
 - Access of air: Presence of air promotes putrefaction and vice versa.
 - Moisture: Presence of moisture in the air promotes putrefaction. Absence of moisture in the air promotes mummification.
 - Age: Young animals carcass putrefy more rapidly. Old animals carcass do not decompose very rapidly.
 - Cause of death: Putrefaction is rapid in sudden death. Poisoning by Arsenic and Antimony in animals retard the putrefaction changes.
- **Signs of putrefaction**
 - Distension of the abdomen due to the formation of gasses
 - Blood stained fluid from the mouth and nostrils
 - Liquefaction of the eye balls
 - Development of foul-smelling gases
 - Thoracic and abdominal cavities will burst, protruding out the stomach and intestines
 - Conversion of the tissues into a semifluid mass
- **Adipocere**
 - Adipocere is formed when the carcass is kept in moist condition. It is due to the gradual hydrogenation of pre-existing fats. The fat is converted into palmitic, stearic and hydro stearic acid. Mixture of these substances constitutes adipocere. Adipocere has a distinct rancid or sweetish smell which is quite characteristic. It floats in water and dissolves in ether and alcohol. Adipocere is formed first in the subcutaneous tissues and later in the adipose tissues elsewhere in the body. It does not occur naturally in the viscera or in the nonfatty tissues.
- **Mummification**
 - Desiccation or shriveling of the carcass from the evaporation of water is mummification. Mummification occurs, when the carcass is in dry surrounding like sandy soil or a cool dry place.
- **Consumption by insects and animals**
 - Insects and animals may begin to consume a body soon after death. Flies, maggots and beetles attach open areas of the body. They gather on soft body tissues. An entomologist's study of the insect larvae on a body can help estimate the time of death.

PM REPORT

- After conducting the PM, veterinarian discloses the opinion regarding the cause of death based on observations made by him during PM.
 - Report should be concise but with complete opinion.
 - Forward as soon as possible to the investigating officer.

- A word 'Probably' should not be used as it creates bad impression in the mind of Judge for being unable to come to a definite conclusion.

EXAMINATION OF DECOMPOSED ANIMAL

- Same lines of examination as in fresh autopsies.
- Should be complete.
- Precaution to be taken by veterinarian to protect himself from septic poisoning.
- External fetal injuries to be searched for cause of death.
- Skin and epidermis has peeled off, under lying tissues to be examined.
- Body thoroughly examined for presence of foreign body such as bullet, piece of weapon, wire, wood etc.
- Bone fracture to be examined.

EXAMINATION OF MULTILATED CARCASS

- Mutilation of dead animals – Criminal Act.
- Done to destroy mark of identification.
- Identify the species, sex, age and colour marks etc.
- Time since death.
- Cause of death.
- Microscopic examination of blood, blood grouping, hair.
- Precipitation test for species identification.

RIGOR MORTIS

- Sets in
- 2-3 hrs after death in hot climates
- Relaxation occurs after 12 hrs
- Duration doubled in cases of winter climate.
- Delayed in
 - Well fed, well rested animal.
- Quick setting in
- Death due to heat stroke, racing animals, high environment temperature, weak emaciated animal due to tetanus.

Rigor mortis

- 1st stage: Acidic muscles (relaxation with Contractibility).
- 2nd stage: Complete rigidity.
- 3rd stage: Alkaline (Flaccidity with contractibility).

Setting of Rigor mortis

- From head, forelimbs (quarter) and hind limbs (quarter) and tail.

COOLING OF BODY

- 1st 3 ° F raise - during contraction after death
- After 2 – 24 hrs cooling occurs ,except in frost bite.
- Rate of cooling: is 1 ° C per 2.5 hrs. i.e., 3.5 ° F per hr for 1st 6 hrs.
- 1.5°F to 2.0°F/hr after 6 hrs.
- Hours since death = (Normal Temp – rectal temp)/ Rate of cooling / Hr (Multiplying rectal temperature with 1.31)

HYPOSTATIC CONGESTION

- Having following characters
 - Epidermal in situation.
 - Seen in dependent parts.
 - Uniform in colour.
 - It gives the position of the carcass to which side the animal was lying before death.
 - It starts ½ to 1 hour after death, marked in 6-8 hrs.
 - Fixed after 12 hrs.
 - Cherry or pink colour - CO or HCN poisoning. Dark brown – Phosphorus poisoning.
 - Putrified carcass- pseudomelanosis.

PUTRIFICATION

- Last stage of the resolution of the body.
- It is absolute and sure sign of death.
- Begins after 6-36 hrs.
- Due to aerobic and anaerobic bacteria present in body.
- *Clostridium welchii* – Proteolytic, sacchrolytic and lipolytic enzyme.
- Lecithinase – RBC (dissolves RBC) – PM Haemolysis
- Quick putrification occurs in
 - Increase temperature
 - Air/ Moisture
 - Young animal
 - Fatty animal
 - Due to poisoning, arsenic etc.
 - Summer months.

Order of putrification

- Stomach, intestines, spleen, liver, brain, heart, lungs, kidney, diaphragm, blood vessels, and uterus.

ASSESSING OF TIMINGS OF DEATH

- Colour changes:
 - After 6 -12 hrs in summer.
 - 1.3 days in winter.

- Green colour seen on the skin on iliac region face, front of neck and spreads all over abdomen by 12-18 hrs.
- Softening of parenchyma tissue:
 - Liquefaction of eye ball 6 -12 hrs
- Foul smell: 24 -48 hrs after death.
- Formation of blisters: 36-72 hrs.
- Softening of soft parts: 3-5 days.
- Appearance of maggots: 3-6 days.
- Time lapsed between death and PM
 - Loosening of the epithelium in omasum within few hour after death.
 - At death, the heart is in diastole i.e., left ventricle almost empty and right ventricle about half full with in first hour of death.
- Cornea of eye – initially glistening, pupils dilated. Later opaque films of dried secretions and pupil constricted.
- Little blood tinged frothy exudates from nostrils shortly after death.
- Warm body – shortly after death.
- Cold and stiff – long period has elapsed since death a some time.

DROWNING

- When atmospheric air is prevented from entering the lungs, caused by the submersion of a body in water, it is called drowning.
- Drowning is rarely employed as a means of mischievous killing of animals.
- In the vetero-legal cases, the veterinarian may be asked to confirm whether the cause of death was drowning or if there were other factors responsible for the death of the animal.
- By examination of the carcass it can be decided whether the animal was alive at the time of drowning or the dead animal was thrown into the water.

Signs suggestive of death by drowning

- External surface
 - Skin shows a corrugated appearance
 - If the animal was alive at the time of drowning, the injuries will have an inflammatory reaction.
- Respiratory system
 - A fine white lathery froth is seen at the mouth, nostril, trachea and bronchi.
 - Lungs are edematous and distended like ballons.
 - Presence of water in the lungs.
- Gastrointestinal tract
 - Large amount of water in the stomach containing objects present.
 - in the water at immersion (eg) Sand, Mud and Algae.
- Rigor mortis.
- Early Mode of death - In drowning, death is due to Asphyxia, shock and Fatal injuries.
- Carcass should be examined carefully for death due to other causes like poisoning, violence, Disease. When the signs of death by drowning are absent.
- Typical signs of drowning are seen in the body only if it is removed from water within few hours after death and examined immediately. These signs are modified with the time and may disappear all together when the body is removed from after putrefication.

DEATH FROM DROWNING

- Means death from prevention of atmospheric air from entering the lungs caused by the submersion of the body in any fluid medium.
- Submersion of mouth and nostrils alone with prevent the air from entering the respiratory tract.
- Complex patho-physiological events occur.
- May differ according to the fluid entering in to the lungs.

DEATH DUE TO FRESH WATER

- Large amount of water enter (hypotonic nature)
- Lungs
- Absorbed rapidly in to blood stream
- Hypervolemia
- Haemodilution
- Anoxia caused by dilution of haemoglobin associated with RBCs
- Ventricular fibrillation
- Haemolysis of erythrocytes

DROWNING IN SEA WATER

Drowning in sea water (Hypertonic)

- Rapid diffusion of salts into blood.
- Effusion of water from blood.
- Haemoconcentration and Pulmonary oedema due to entering of water into alveoli.

MODE OF DEATH IN CASE OF DROWNING

- Asphyxia
- Shock
- Fatal injuries
- **Signs which indicate the occurrence of death by drowning**
 - Fine leathery froth is seen at the mouth and nostrils.
 - Skin - corrugated appearance.
 - Early rigor mortis.
 - Lungs –distended like balloons.
 - Oedematous and spongy lungs.
 - Fine white froth in larynx, trachea and bronchial tubes.
 - Presences of water in lungs.
 - Large amount of water in stomach along with the objects present in the water like algae etc.
 - Injuries on the animal body- Live at death - inflammatory process.
 - Above feature are absent the carcass should be thoroughly examined for other causes. Eg. Poisoning, violence, diseases etc.

LIGHTNING STROKE

- Accidental deaths of animals may occur due to a lightning stroke and Electrocution

Lightning Stroke

- Common causes of lightning stroke are
 - Flashes of lightning during thunderstorms may prove fatal to animals
 - Big trees usually have well-developed root systems. During a thunderstorm, if the ground is damp, then electrification of the root system charges a wide surface area. Merely standing on electrified earth electrocutes animals.
- In Lightning stroke, if animals die following points should be taken for consideration
 - History of thunderstorm in the locality.
 - Evidence of the effects of lightning (eg) Damage to houses or trees.
 - Often, a lightning stroke will affect several animals in a herd.

Signs of Lightning

- Half-chewed food in the mouth is a significant finding in the case of a lightning stroke.
- Singeing and burning marks occur in about 90 per cent of cases of lightning. Mostly on the medial sides of the limbs but the hairs have a brownish discolouration and abnormal kinking.

DEATH FROM LIGHTENING STROKE AND ELECTROCUTION

- Common in animal.
- Causes of death from electric currents is not necessarily the same in all cases.
 - Low tension and intensity current
 - Kill by producing ventricular fibrillation.
 - Closing the current - inhibit the heart.
 - Opening the current - is followed by fibrillation.
 - High tension and intensity.
 - Kill by paralysing the bulbar respiratory centre.
- Flash of lightning
- Considerable air-pressure
- Death is immediate due to CNS affection
- Severe injuries
- Paralysing heart or respiratory center
- Flash – zigzag (easiest path of least resistance).
- Lightning always - DC.
- Forms ladder path from clouds to earth.
- 10,000 to 2,00,000 amperes & voltage up to 2,00,00,000.
- 1/1000 of second.
- Thunderous sound – high voltage and amperage generates tremendous heat cause blasts.
- Destruction zone - 30 mts.

CAUSES

- Linear lighting – group of animals affected.
- Tall trees with developed roots e.g. oak, poplar, elm, conifer, walnut, ash, beech.
- Faulty electric circuits.
- Metallic manager, electric pole, water or milk line electrified due to short circuit.
- Transmission wires breaks fall on animals.
- Transmission wires fall on ground.
- Geological composition - Loam, sand, clay, marble and chalk enhance conductivity.
- Rocky area – more exit.
- Chewing of electrical wires.
- 110 to 220 volts sufficient to kill adult cattle provided their contact with surface ground is strong enough.
- Good conduct when which is standing in water or on a damp floor.

SYMPTOMS

- Shock
- Fell dead with out struggles
- Less severe cases – unconscious and recover after few minutes
- Residual nervous symptom -depression, paraplegia and cutaneous hyper aesthesia
- Minor case – restless, nervous and kicking at the stanchion or dividing rail.
- Intermittent shock eat and drink with care
- Singeing and burning marks – medial sides of limbs.
- Hair brownish and abnormal kinking
- Half chewed food in the mouth.

CAUSES OF DEATH AND PM CHANGES

Causes of death

- Nervous shock
- Respiratory paralysis
- Ventricular fibrillation.

PM changes

- Rigor develops and passes quickly - early putrefication
- Bloody discharge from natural orifice
- Blood-dark and tends to clot slowly
- Distended rumen and swollen carcass
- Petechial haemorrhage throughout the body endocardium, meninges and CNS.
- Congested upper respiratory tract mucosae.
- Extravasation of blood in muscles and lymph nodes
- Longitudinal fracture of long bones.
- Singeing or burning marks on the feet or muzzle.

ELECTROCUTION

- Common causes are
 - Defective electric circuits in cow sheds.
 - Electric pole or water lines may be electrified due to a short circuit.
 - Transmission lines may break and fall on the animals.
 - Fallen transmission wires may electrify a pool of water and animals passing through it may be killed instantly.
 - Animals may chew the electric-cord and receive an electric shock.

Effect of electrocution on the body

- The clinical symptoms due to electrocution depend upon
 - The strength of the current
 - The efficiency of the contact and earth and
 - The length of exposure and the area in contact.
- Death by electrocution occurs due to
 - Ventricular fibrillation
 - Cardiac arrest
 - Respiratory failure.
- **Clinical symptoms of electrocution**
 - Minor
 - Animal will be restless, nervous and may kicking.
 - Animal which have received intermittent shock, may eat and drink with care.
 - Less severe
 - Animal becomes unconscious and may recover completely after a few minutes.
 - Some animals may show nervous symptoms such as depression, paraplegia.
 - These symptoms may persist or disappear gradually.
 - Severe
 - Animals may fall dead without struggle in most instances.
 - Singeing and burning marks occurs in most of the cases of lightning.
 - Hairs have a brownish discolouration and abnormal kinking.
 - Half-chewed food in the mouth is a significant finding in the case of a lightning stroke.
 - Varying degrees of shock may occur.
- **Postmortem changes in electrocution**
 - Circulatory system
 - Bloody discharge from all natural orifices.
 - Blood is dark and tends to clot slowly.
 - Petechial haemorrhages throughout the body, endocardium, meninges and CNS.
 - Skin & Musculo skeletal system
 - Extravasation of blood in muscles and superficial lymphnodes.
 - Longitudinal fractures of long ones.
 - Singeing or burning marks on the feet or muzzle.
 - Gastro Intestinal tracts
 - Distended rumen and swollen carcass.
 - Respiratory system

- Congested upper respiratory tract mucosal.
- Rigor mortis - Rigor mortis develops and passes quickly early putrefaction

SNAKE BITE

- Fatal snakebites are common in dogs.
- Because of their size, horses and cattle seldom die due to direct result of snake venom ut death may follow bites on muzzle, head or neck.

Type of Snake	Clinical symptoms
Elapines • e.g. Cobra, coral snakes. • They have short fangs and produce neurotoxin that paralyses the respiratory centre	• Pain and swelling are minimal. • Systemic neurologic signs predominate • Excitement with convulsions
Viperines • e.g. True vipers and pit vipers • They have long fangs and produce hemotoxin, necrotizing and anticoagulant	• Severe Intal tissue damage • Tissue becomes discoloured within few minutes. • Dark bloody fluid may ooze out of the fang wounds and coagulability of blood is completely lost.

Diagnosis of snake bite

- Clinical sings
- Identification of the snake

Differential diagnosis of snake bite

- Fracture
- Abscess
- Spider envenomations
- Allergic reactions to insect bites

SUFFOCATION

- Exclusion of air from the lungs by means other than compression of the neck is termed as suffocation. Suffocation may cause death.

Causes of suffocation

- Smothering or closure of the mouth and nostrils
 - Chocking or obstruction of the air passages from within
 - Inhalation of irrespirable gases
- **Smothering or closure of the Mouth and Nostrils**
 - Calves are often accidentally smothered by being overlaid by their dams. A common method of killing the calves is to close the mouth and nostrils.
- **Chocking or obstruction of the Air passages from within**
 - Mostly accidental, this may be due to
 - Presence of foreign bodies – piece of carrot, potato, corn, coin, cork, rag, roundworm, mud, cotton, leaves in the air passage.
 - It is not necessary that the foreign body should be of big size as to block the air passages completely. Even a small object blocking the lumen partially may cause death by spasm.
- **Inhalation of Irrespirable Gases**
 - Inhalation of gases such as CO₂, an anaesthetic used by mistake carbon monoxide, hydrogen sulphide or smoke from a burning house will produce suffocation.

Mode of death due to suffocation

- Usually death is due to asphyxia but it may be due to shock, when the heart stops by reflex action through the vagus nerves.
- Death occurs on an average from 10 to 15 minutes after complete withdrawal of air from the lungs. Death was almost instantaneous when the windpipe was blocked by a foreign body. Recovery may occur if treated within 5 minutes.

Postmortem Appearance

- **External**
 - Death may be due to the cause leading to suffocation or asphyxia.
 - Due to forcible application of the hand over the mouth and nostrils, bruises and abrasions may be found on the lips and edge of the mouth and alongside the nostrils. The inner surface of the lips may be found lacerated from the pressure on the teeth. Nasal septum may be fractured from pressure of the hand. These signs are very rare.
 - Bruises and Abrasions on the cheeks and malar regions or on the lower jaw, if there has been a struggle.
 - Rarely, Fracture or dislocation of cervical vertebral may occur if the neck has been forcibly wrenched in an attempt at smothering with the hand. If a soft cloth has been used to block the mouth and nostrils, there will be no signs of violence.
 - Signs due to Asphyxia – Suffocation
 - Eyes are open
 - Eye balls are prominent
 - Conjunctivae are congested

- Lips are livid
 - Tongue is protruded
 - Bloody froth comes out of the mouth and nostrils
- **Internal – P.M.changes – Suffocation**
 - Mud or any other foreign matter may be found in the mouth, throat larynx or trachea if suffocation caused by impacted of a foreign substance in the air passage. Foreign matter may also be found in the pharynx and oesophagus.
 - Mucous membrane of trachea is usually rightred, covered with bloody froth and congested.
 - Lungs are congested and emphysematous.
 - Punctiform subpleural ecchymoses (Tardieu's spots) are present at the root, ase and lower margins of the lungs may be present in asphyxial deaths from other causes. It is not characteristic of death by suffocation. Punctiform subpleural ecchymoses are also found on the thymus pericardium and along roots of the coronary vessels.
 - Lungs may be found normal, if death has occurred quickly.
 - Heart – Right side heart is full of dark fluid blood. Left side empty.
 - Blood does not readily coagulate. Wounds caused after death may bleed.
 - Brain – Generally congested
 - Liver, spleen and kidneys – found congested

ANEMIA

- Anaemia – due to sudden and excessive haemorrhage from wounds of the large blood vessels, (or) Internal organ such as lungs, spleen (or) from bursting of an aneurysm or a varicose vein.
- Shock resulting from
 - Sudden fright
 - Sudden blows on the heart or on the epigastrium
 - Drinking large quantity of cold water → when in a heated condition extensive injuries to the spine or other parts of the body
 - Sudden evacuation of the natural or pathological fluids from the body
 - Sudden pressure
 - Severe exposure to cold
- Asthenia due to deficient power of the heart muscle as in fatty degeneration of the heart or aorta regurgitation or poisoning.
- Exhausting diseases
 - **Symptoms**
 - Mucous membrane - Pallor
 - Eye – Dilated pupils, Dimness of vision
 - Restlessness
 - Air hunger Gasping respiration
 - Nausea or vomiting
 - Marked fall in Blood pressure due to reflex vasodilation

SYMPTOMS OF ANAEMIA

- Rapid pulse in Asthenia
- Pulse - Slow and Weak and Fluttering in Anaemia
- Slight delirium, insensibility and convulsions precede death

- In collapse, the animal retains consciousness though the condition attended with failure of the heart's action.

Post mortem

- Heart - Found to be contracted, chambers empty - when death due to anaemia
- Heart - Both chambers contain blood in case of death due to asthenia
- Lungs, brain and abdominal organs are found pale.

ASPHYXIA

- Asphyxia refers to an interference with the oxygen interchange in the lungs. It is used in the broader sense of anoxia and tissue asphyxiation produced when the oxygen and carbondioxide interchange between the blood in the capillaries and tissues cells in all parts of the body is prevented.
- Death from asphyxia take place, when the respiratory function stops due to lack of oxygen, before the heart ceases to act.

ETIOLOGY

- Mechanical obstruction in the respiratory tract by Foreign bodies, exudation, tumours, anaphylactic bronchospasm, suffocation and drowning – blocks the lumen of the respiratory tract from within.
 - Spasm of glottis occur due to mechanical irritation and irritant gases, forcible closure of mouth and nose by any means.
- Absence of sufficient oxygen at high altitudes. Presence of inert gases like carbon monoxide in the atmosphere.
- Stoppage of movements of chest from exhaustion of the respiratory muscles due to cold and debility.
 - Paralysis of the respiratory muscles due to the diseases (or) injury of the medulla (or) phrenic nerve (or) pneumogastric nerves.
 - Mechanical pressure on the chest or abdomen
 - Tonic spasm due to Tetanus or Strychnine poisoning
- Collapse of the lungs due to
 - Penetrating wounds of the thorax
 - Diseases – Pleurisy with effusion
 - Empgema
 - Pneumothrax
- Non-entrance of blood into the lungs as in embolism plugging the pulmonary artery.

STRANGULATION

- Is the closure of the air passages as a result of external pressure of the neck which prevents the ingress or regress of air during respiration.

Symptoms of asphyxia

- Asphyxia is manifested by symptoms divided into 3 stages
 - Dyspnoea
 - Convulsions
 - Exhaustion
- **Dyspnoea:** caused by the accumulation of carbondioxide in the blood due to deficient oxygenation of the RBC's carbondioxide stimulates the respiratory centre in the medulla and respiration becomes deep, hurried and laboured. The blood pressure rises and pulse become rapid. Animal bears an anxious look, eyes prominent and Lips are livid.
- **Convulsions:** Expiratory muscles of respiration become more active with spasmodic movements which are followed by convulsions of all the muscles of the body. Owing to venous and capillary stagnation, the m.m are deeply congested and cyanosed. Tongue is protruding consciousness is lost. The sphincters are relaxed.
- **Exhaustion:** It is the 3 stage. Respiratory centre is paralysed. Muscles become flaccid, completely insensibly, reflexes are lost, pupils are widely dilated.
 - Blood pressure ↓ falls
 - Respirations – prolonged sighing occur, at longer and longer intervals until they cease altogether and death ensues.
 - Pulse is scarcely perceptible. Heart beat continue for some minutes after respiration has ceased.

Post Mortem appearance

- **External**
 - Mucous membrane
 - Pale in slow Asphyxia
 - Distorted, congested & Blue in Sudden Asphyxia
 - Tongue - Protruded
 - Frothy and blood stained mucus at nostrils and from mouth
 - Rigor mortis – commences slowly but may be rapid in some cases
- **Internal**
 - Mucous membrane of the Trachea and Larynx - Cinnabar red and slight frothy mucus
 - Lungs
 - Dark and purple in colour
 - Gorged with dark venous blood
 - On being cut, lung exlude frothy, dark fluid blood.
 - Air sacs are distended or ruptured due to emphysema
 - Heart - Right cavity of the heart filled with dark – coloured imperfectly clotted blood and so in the pulmonary artery and the vena cava.
 - Heart – The left side cavity, aorta and the pulmonary veins are empty.

- In many cases, both sides of the heart are found to be full, if examined soon after death but after rigor mortis has set in, the heart is found contracted and empty. The tension in the abdomen presses on the inferior vena cava and drives the blood up into the heart.
- Similarly, the lungs are found heavier with blood if examined sometime after death, or the tension in the abdomen or contraction of the heart muscle will drive more blood into the lungs, irrespective of the cause of death.
- Brain – is congested and excess of serous fluid is found in the lateral ventricles. Cranial sinuses are filled with dark – coloured blood.
- Abdominal organs are found congested.
- Petechial haemorrhages or ecchymoses are seen under serous membranes of various organs due to rupture of capillaries caused by increased pressure. Petechial haemorrhages are observed under visceral pleurae, pericardium, thymus meninges of the brain and spinal cord, conjunctiva, epiglottis and under skin of the face and neck.

MODULE-21: VETERO - LEGAL ASPECTS OF DEATH

- **Learning outcome**

- Though sudden death in animals may be due to natural causes, it may also be suspicious. This module focusses on Vetero-Legal aspects of death and manner of death.

VETERO-LEGAL ASPECTS OF DEATH

- It may be defined as a complete and persistent cessation of respiration and circulation called somatic death and death of the tissue, molecular death.
- Death is classified as somatic or systemic and molecular. Somatic and systemic death is that state of the body in which there is complete loss of sensibility and ability to move and complete cessation of the functions of the brain, heart and lungs, the so-called 'tripod of life', which maintain life and health. Molecular death means the death of the tissues and cells individually, which takes place some time after the stoppage of the vital functions, and is accompanied by cooling of the body, the temperature of which is reduced to an equilibrium with the external world.

Manner of Death

COMA

- Traumatic injury brain by accidental collision fall from height animal interaction or a weapon.
- Cerebral haemorrhage, thrombosis, embolism, tumours, abscess and other inflammatory conditions of the brain. Spontaneous cerebral haemorrhage observed in cows at parturition.
- *Coenurus cerebralis* – (Gid, Sturdy, Staggers) – Sheep and Cattle.
- Poisoning: Eg - bromides, carbon monoxide, zinc phosphide, barbiturates etc.
- Uremia and heat stroke.

CAUSES OF DEATH

- Natural (or)
- Accidental death

Three primary modes of death

- Coma
- Syncope
- Asphyxia

Death may occur due to the combined effects of coma and asphyxia.

Coma

- Loss of consciousness.
- May be partial or complete.
- May be transitory lasting for a few minutes or may continue for some days or until the death takes place.

SYMPTOMS

- Sudden onset of unconscious.
- Dilatation of pupil.
- Unconscious with as without transient convulsions.
- Absence of papillary light reflexes.
- Slow and irregular respiration.
- Slow pulse.
- Bleeding from nose and ears.
- Involuntary passing of urine and feces.
- Hemiplegia or paralysis of individual cranial nerves.

SYNCOPE

- Circulatory failure
- Acute and progressive failure of the peripheral circulation is known as shock.
- Shock is “stagnant hypoxia”.
- **Haemorrhagic shock**
 - Common cause of shock in animals. Up to 20% blood loss animal can tolerate.
 - Hypovolemic before reinfusion
 - Normovolemic – after reinfusion
- **Traumatic shock**
 - Depressed state of vital function that results from severe traumatic injury.
 - Two kinds of traumatic shock
 - Primary or immediate
 - Secondary or delayed – due to loss of plasma occurred few hours after injury.
 - **Primary shock**
 - Overwhelming of vital centres in the medulla by nerve impulse originating in the injured area and conveyed to CNS.
 - Death within a few minutes from vasomotor collapse in primary shock.
- **Septic shock**
 - Fatal infection by pyogenic organisms following an injury if animal survives immediate complication of shock and haemorrhage.
 - Suppurative pathogenic bacteria.
 - Rupture of intestines
 - Injury to gastro intestinal tract
 - Reticulo endothelial system of a shocked animal – not having capability of detoxifying the bacterial toxins.
 - Endotoxin produces peripheral circulatory failure (exact mechanism is not known).
- **Thermal shock**
 - Exposure to extreme degree of heat or cold,

- It is are characterized by progressive hypovolemia,
- Haemoconcentration and
- Circulatory failure.
- **Anaphylactic shock**
 - Is an immunologic phenomenon.
 - Occurred between sensitized body cells and specific antigens.

SYMPTOMS

- Coldness of the skin.
- Subnormal temperature.
- Rapid and shallow respiration.
- Pulse is faint while rate is high.
- Pale / cyanotic mucous membranes.
- Reduced amplitude of heart sounds.
- Dilated pupil and lacrimation.
- Muscular weakness and recumbency.
- Oliguria and increased thirst.
- Veins are difficult to rise.

POST MORTEM APPEARANCE

- Injury to skull.
- Brain and its membranes are congested.
- Blood in the cranial cavity.
- Right heart full and left empty.
- Tumour, cyst or abscess in the cranial cavity.

ANOXIA

ETIOLOGY

- Either natural or traumatic causes.

Mechanic obstruction

- Nasal granuloma.
- Oedema of larynx.
- Acute cellulitis of the throat.
- Peripharyngeal cellulitis, and oedema.
- Laryngeal obstruction.
- Tumour / abscess etc
- Anaphylatic reactions following injection of foreign proteins or serum - fatal asphyxia.
- Paralysis of respiratory nerves, muscles or respiratory centre form injury or disease or poisoning.
- Disease of lungs or pulmonary circulation failure (lung disease, pleural effusion, pneumothorax, pulmonary embolism, collapse of the lung due to penetrating wounds of thorax and circulatory disturbance of the lung.

- Strangulation in the closure of the air passages by external pressure on the neck which prevents the ingress or egress of air during respiration. Crossing of the halter of two bulbs tied close together.
- Inhalation of irrespirable gases eg. CO, Hydrogen sulphide, sulphur dioxide, nitrous oxide and chlorine.
- Interference with the oxygen interchange in lungs and at tissue level.

SYMPTOMS

Four stages

- Dyspnoea - accumulation of CO₂ in blood affect respiratory centre – Rapid & shallow respiration, rapid pulse and cyanosis.
- Convulsions- clonic convulsion then tonic, opisthotonus spasm, pupil dilated, heart rate slows.
- Apnoea – unconsciousness.
- Final stage - respiratory paralysis heart may beat.

PM appearance

- Protruded tongue.
- Frothy blood stained fluid in nostrils.
- Congested conjunctiva.
- Laryngeal and tracheal mucosa congested.
- Lungs engorged and oedematous (frothy, dark, coloured blood, exudates).
- Petechial haemorrhage of pleura, pericardium, myocardium, endocardium and sometime of brain.
- Cavities of right side heart are engorged with dark coloured, imperfectly clotted blood.
- Left heart are comparatively empty.
- Congestion of abdominal viscera.
- Congested brain and excess serous fluid in the lateral ventricles.
- Cranial sinuses filled with dark – coloured blood.
- Blood dark in colour mostly fluid in the large veins.
- Microscopically lung tissue shows marked dilatation of both the capillaries and veins.

PM APPEARANCE

- Signs of severe injuries.
- Generalized visceral congestion.
- Oedema of lungs.
- Dark red fluid blood.
- Petechial haemorrhages.

EXAMINATION OF BLOOD STAINS

Important in veterolegal cases

- Blood stains present or not / which species of animal the blood belongs. E.g. Mammals, birds, fish or reptiles.
- Done by chemical examiner.
- Exact location of stains to be noted
- Stains from person and animals are best removed with a piece of filter paper soaked in NS and allowed to contact and remove and dry it.
- Garments and weapons allowed to dry and wrapped and packed with out unnecessary handling.
- Liquid stains are taken in test tubes with the help of dropper. Test tube placed in Ice box.
- Blood soaked soil sent in glass container walls and floors – digging and sent in a container.
- Stain can be scrapped up by knife.
 - Physical
 - Chemical
 - Microscopical
 - Spectroscopic
 - Serological

PHYSICAL EXAMINATION

- Foreign bodies like hairs and fibres.
- Colour, position, shape and size of blood stain and relationship with surrounding to be recorded.
- Examined by naked eye or simple lens.

Various shapes of stains

- Blood streak – venous blood.
- Parts of blood of irregular shape – abundant haemorrhage.
- Circular stains – drop of blood falls on horizontal surface from a small height.
- Ova stains – drop of blood hitting vertical or oblique surfaces.
- Splashes or line shaped stain are usually- projection of arterial blood
- Smears – blood wiped off.

CHEMICAL EXAMINATION

Benzidine test

- Press piece filter paper over stain.
- Add few drops of 10% benzidine sol in glacial acetic acid.
- Add few drops of fresh, acti 3% hydrogen peroxide.
- Green or blue colour positive.
- Sensibility – 1: 200,000.
- Not specific for blood.
- Most oxidising agents like permanganate, chromates, bichromates etc.
- Give strong reaction with our addition of H₂O₂.

- Positive result big pas, secretion of throat and nose by plant juices, by commercial formalin, fly excretion.

Phenolphthalein test

Solution preparation

- Phenolphthalein – 18 ml
- Sodium hydroxide- 20 pm
- Distilled water – 200ml
- Mix phenolphthalein and sodium hydroxide in distilled water and boil it under a reflux condenser.
- Add 20gm of granulated Zinc (20 to 30 mesh) while boiling.
- Boiling continued until solution become colorless

Procedure

- 1 to 2 ml of saline suspension of blood stain in a test tube.
- Boil it for 1/2 minutes to destroy any oxidases.
- Cool it and add 5 drops of phenolphthalein reagent and followed by some drops of 3% H₂O₂.
- Pink colour develop in position cases.
- Colour lasts no longer the 3mts.
- Sensitivity 1: 1,000,000.
- Not specific for blood.
- Direct oxidising compound copper, potassium ferricyanide and nickel – positive reaction.

Leucomalachite green test

Reagent

- Leucomalachite green – 18 m
- Acetic acid- 100ml
- Distilled water – 150ml
- Add one drop of the reagent on the stained filter paper
- Leave it for 10 seconds
- one drop of H₂O₂
- Green colour develops – if blood.
- More sensitivity – not commonly used.

MODULE-22: EVIDENCE PROCEDURE IN COURTS

- **Learning outcome**

- This module is focusses on the details about the court, types of court, types of magistrates, and evidence procedure in courts. In addition this module focusses on the slaughter methods, euthanasia, traditional slaughter methods, and procedure of slaughter.

INDIAN COURTS

Types of courts in India

Two types

- Civil courts: which deals with the civil matters relating to the individual claims of monetary transactions, property etc.
- Criminal courts: which deals with the crimes against animals etc.

Highest courts

- Highest judicial court at national level – Supreme court
- Highest judicial court at state level – High court

Supreme court

- Highest court in the country.
- Has powers of superintendence over all high courts.
- Law declared by Supreme Court, becomes the law of the country and is binding on all courts and tribunals.
- Has got the authority to hear appeals against the judgments of high courts of the states.

High court

- Highest judicial at state level.
- Got full control over the judicial institutions in the state.
- Got authority of hearing appeals against judgments passed by lower courts.
- Got powers of appointing and transfer of subordinates judges and judicials while the executive magistrates are appointed by State Govt.

Criminal courts

- There are 3 classes of criminal courts in India
 - The High Courts
 - The Courts of Sessions
 - The Courts of Magistrates

CLASSES OF MAGISTRATES

- Judicial magistrates of the first class present in every district as Chief Judicial Magistrates for the purpose of general control in every district. Similarly in every metropolitan area like New Delhi, Mumbai, Kolkata, Chennai, Chief Metropolitan Magistrates acts as Judicial Magistrates of first class.
- The above two magistrates are appointed by High court. The High Court may also appoint Special Judicial Magistrates and Special Metropolitan Magistrate for a term not exceeding one year. The Judicial Magistrate of first class when placed in charge of a subdivision is known as the Sub-Divisional Judicial Magistrates.
- Judicial magistrate
- Chief judicial magistrates.
- Additional judicial magistrates.
- Judicial magistrates of the 1st class or of the 2nd class.
- Sub-divisional judicial magistrates.
- Special judicial magistrates.
- Chief metropolitan magistrates.
- Additional metropolitan magistrates.
- Special metropolitan magistrates.
- Executive magistrate
- District magistrates.
- Additional District magistrates.
- Special Executive magistrates.
- **Sentence which High Courts and Sessions Judges may pass**
 - High Court may pass any sentence authorized by law.
 - Session Judges or Additional Session Judges may pass any sentence authorized by law; but any sentence of deaths passed by any such Judge shall be subject to confirmation by the High Court.
 - Assistant Session Judge may pass any sentence authorized by law except a sentence of death or imprisonment for life or of imprisonments for a term exceeding ten years.

SENTENCE WHICH MAGISTRATES MAY PASS

- Chief Judicial Magistrate may pass any sentence authorized by law except a sentence of death or imprisonment for life or imprisonment for a term not exceeding seven years.
- Magistrates of the first class may pass a sentence of imprisonment for a term not exceeding 3 years or of fine not exceeding five thousands rupees or of both.
- Magistrates of the second class may pass a sentence of imprisonment for a term not exceeding not one year or of five not exceeding one thousand rupees or of both.
- Chief metropolitan magistrates shall have the power of the court of a chief judicial magistrate.
- Metropolitan magistrate have the power of the court of a magistrate of the first class.
- Executive magistrates may award such term of imprisonment in default of payment of fine as authorized by law.

Subpoena / summons

- Is a document compelling the attendance of a witness in a court of law under a penalty. When summons is served to the witness the must do so punctually.

- Non – compliance of witness in a criminal case may render him liable to fine or imprisonment unless a reasonable excuse is forthcoming.

Oaths

- “Oaths” includes a solemn affirmation substituted by law for an oath, and any declaration required or authorized by law to be made before a public servant or to be used for the purpose of proof, whether in a court of justice or not.
- On the witness box, before your evidence is taken, you are to be “sworn”.
- i.e., you have to state an oath that the evidence you are going to give touching the matter before the court shall be the truth, the whole truth and nothing but truth. Then only your evidence is recorded by the judge.
- Perjury is willful utterance of falsehood under the oath. The witness liable to be prosecuted for Perjury under section 193 of I.P.C. if he fails to state what he known or believes to be true.

EVIDENCE

It has been defined in the Indian evidence Act (1872).

Two types of evidence

- **Oral evidence**
 - All statements which the court permits or required to be made before it by witness in relation to matters of fact under enquiry. Oral evidence in the evidence of a witness who has seen, heard or perceived an event. Oral evidence is more important than documentary.
- **Documentary Evidence**
 - All documents produced for the inspection of the court. It includes
 - Certificates,
 - Vetero – legal reports,
 - Examination of exhibits,
 - Electronic records.
- **Certificates**
 - Simple forms of evidence usually, certifying health, soundness, death etc. of the animals.
 - It should be carefully written with sense of responsibility for the opinion expressed in them.
 - It will not be accepted in the court of law unless issued by a qualified veterinarian who has registered in the State Veterinary Council Act.
 - Death certificate should contain identification of the animal and causes of death.
 - Certificates should be given promptly only after seeing the dead animal himself.
- **Vetero legal reports**
 - Documents prepared by veterinarian in obedience to demand by an authorized police officer or a magistrate referred to chiefly in criminal cases.
 - It may be passed to the higher courts and placed in the hands of pleaders.
- **Examination of Exhibits**
 - Articles weapons etc. sent for examination should be described fully to facilitate their identification in the Court later on.
 - They should be labelled and returned to the police officer or a magistrate in a sealed cover.

- Articles to be sent to the Chemical Examiner should be kept in the custody of veterinary officer.
- **Electronic records**
 - Any information contained in electronic record which is printed on a paper, stored, recorded in optical or magnetic media produced by a computer shall be deemed to be also a document.

The evidence is recorded in the following manner:

- **Examination – in - chief**
 - First examination of a witness by party who calls him. If the witness is summoned by a private party, he is at first examined by the pleader of the party.
 - Cross – Examination:
 - Held by the counsel for the accused who tries to elicit facts.
 - Leading questions are permissible.
- **Leading questions**
 - Any question suggesting the answer which the person putting it wishes or expects to receive.
 - Leading questions are not asked in an examination- in-chief or in a re-examination except with the permission of the court.
 - In cross examination, he may be asked any question which tends to test his veracity.
 - Discover who he is and what his position in life is shake his credit by injuring his character. Although the answer to such question might tend to directly or indirectly to incriminate him, or might expose or tend to directly or indirectly to expose him to a penalty or forfeiture.
- **Re-examination**
 - The counsel who conducts the examination- in- chief has the right of re-examination of the witness to explain any misunderstanding occurred during cross-examination the adverse party may further cross examine upon this matter.
- **Question put by the judge**
 - The judge may also put questions in order to discover or to obtain proper proof of relevant facts or may order the production of any document or thing.
 - The witness may refresh his memory by referring to any writing made by himself at the time of the transaction concerning.
 - The witness may also refer to any such writing made by any other person, and read by the witness within the time aforesaid.
 - A witness may refresh his memory by reference to any document, he may, with the permission of court.
 - An expert may refresh his memory by reference to professional treatise.

WITNESSES

- There are two kinds
 - Common witness
 - Expert witness
- **Common witness**
 - “Statement of facts” which an ordinary witness has actually observed.
 - “Fact” is defined by the Indian Evidence Act.
 - Anything capable of being perceived by the senses.

- Any mental condition of which any person is conscious.
- One fact is said to be relevant to another referred to as relevance of facts.
- **Expert witness**
 - Is one given by the person having professional knowledge / training who is capable of deducing opinions and inferences from the facts observed by himself or noticed by others when the court has to form an opinion upon a point of foreign law or of science or art, as to that point of fact, such as the identification of handwriting (or finger prints), the opinion upon that point of fact given by a person specially skilled.
 - A veterinary witness is an expert witness.

Hints for giving evidence

- Always prepare the facts of the case before going to Court by reference to your original notes and consider what possible questions you may be asked and be prepared with your answers.
- Speak slowly and distinctly.
- Avoid technical terms as far as possible.
- Answer questions with a 'Yes' or 'No' as far as possible.
- If you are unable to answer any question say 'I can not answer'.
- Do not lose your temper while under cross-examination by a lawyer. They may put questions of an irritating nature.
- Quotations from books should be avoided.

EVIDENCE

RULES FOR GIVING EVIDENCE

- The veterinary practitioner, when summoned to court as an expert witness, must remember that he is there to tell the truth, the whole truth and nothing but the truth, and should, therefore, give his evidence irrespective as to whether it was likely to lead to conviction or acquittal of the accused. His evidence should be relevant, reliable and clear.
- He should speak slowly, distinctly and audibly to enable the judge and counsel to hear him and to take notes of his evidence.
- He should use plain and simple language, avoiding all technicalities, as the bench and the bar are not expected to be familiar with veterinary terms. It is no use showing his erudition by using these terms; however, if he cannot help using any veterinary terms he should try to explain it in ordinary language as far as possible.
- He should avoid long discussions, especially theoretical arguments. His answers should be brief and precise, and in the form of "yes" or "no". However, by doing so, if he finds that what he means to say is not understood he can explain his answer after obtaining permission from the judge.
- If he does not know or remember any particular point, he should not hesitate to say so and must not hazard a guess in a doubtful case; also he should never assume the function of the judge or jury by giving an opinion on the merits of the case.
- He should remember that the lawyer has practically unlimited license and latitude in putting questions to the witness in cross-examination, and consequently he should never lose his temper, but should appear cool and dignified, even if questions of an irritable nature be put to him. However, he should have no complaint against lawyers. They have great regard for him and have shown the greatest amount of courtesy to him at the time of his deposition in court.

- The veterinary witness may refresh his memory from his own report already forwarded to the court, but should not do so from his private notes, unless they agree word for word with the original, made at the time of, or immediately after, the occurrence of the event, and were written by him or certified to be correct if written by his assistant; besides, he should be prepared to have them put in as exhibits if desired by the judge or counsel.
- He should not quote the opinion of other veterinarians or quote from text-books regarding the case. He is supposed to express an opinion from his own knowledge and experience.
- When counsel quotes a passage from a text-book and asks the witness whether he agrees with it, he should, before replying, take the book, note the date of its publication, read the paragraph and context, and then state whether he agrees or not; for, counsel usually reads only that portion which is favourable to his case, and the meaning may be completely altered when it is read in reference to context. In spite of this precaution he should stick to his opinion if it is still his opinion, and if he finds that it differs from the one expressed in the book. To avoid being surprised by such quotations, however, it is advisable to study all the available literature on the subject before giving evidence in court.

VOLUNTEERING OF A STATEMENT

- It is said that a witness is not supposed to volunteer a statement in court, unless called upon to do so. This may be true in the case of a lay witness, but it cannot be so in the case of a veterinary witness. Even though a veterinary witness is called by one side to give evidence in court, he must not forget his duty towards the opposite party-of honesty and fair dealing.
- He must also remember that the judge regards him not as a veterinary advocate put forward by one side to establish the case but as an officer of justice helping the court elicit the truth. It is, therefore, the duty of a veterinary witness to state fairly all the veterinary facts bearing on the case without any reservation. Hence it is proper for him to volunteer statements and suggest questions to court, especially when he finds that there is danger of justice being miscarried owing to the court having failed to elicit any important point.

PROFESSIONAL SECRETS

- Under Section 126 of the Indian Evidence Act, a lawyer can claim privilege and will not at any time be permitted to disclose in court any communication made to him in the course and for the purpose of his employment as such by his client except with his express consent, but a veterinary witness cannot claim such privilege as regards professional secrets obtained by him or communicated to him by the owner of the animal under his care in the course of examination and treatment. Nevertheless, he should, on no account, volunteer these secrets, but should divulge them under protest to show his sense of moral duty, when pressed by the court to do so. Non-compliance with the order of the court may render him liable for contempt of court proceedings.

NEGLIGENCE

Liabilities under the law of courts are:

- Civil action - for damage.
- Consumer Protection Act - Private Veterinarian's liability under the Act for rendering deficient service due to negligence. Government Veterinarians are exempted from this Act as no considerable amount is charged for rendering services.

MODULE-23: EUTHANASIA

- Euthanasia is the production of quiet, painless death in an animal for humane reasons.
- It is performed for the hopeless cases, when the treatment would be ineffective and the animal is suffering severe pain.
- Owners, request to euthanize animals, if the animals have behaviour problems. If the owners no longer wish to own the said animal.
- Veterinarians should not advocate or recommend euthanasia in any specific case. They should present the alternatives and try to give the client, the facts upon which to make a decision. The decision to euthanize an animal is the ultimate responsibility of the owner.
- In addition to client relations and ethical problems which may occur in a case terminating in euthanasia the veterinarian should consider the legal implications. Prime importance is to obtain the consent of the rightful owner or authorized agent in writing on a consent form. Failure to obtain owner and/or insurer permission invites a lawsuit.
- Veterinarian has a legal obligation to euthanize an animal.
- Owner retains all property rights to the animal's body and may either remove it or ask the veterinarian to dispose of.
- Before euthanasia is performed, it has to be ascertained that the animal involved is indeed the one to be euthanized.
- Selection of the method of euthanasia is dependant upon the
 - Species of the animal
 - Available means of control of the animal
 - Numbers of animals
 - Economic considerations
 - Wishes of the owner
- Selection-method of euthanasia points to be considered
 - Method should produce death without pain
 - Time to produce unconsciousness and death should be short
 - Method should be reliable.
 - Method should minimize undesirable psychological stress
 - Method should be compatible with its purpose
 - Should minimize the emotional effect upon observer and operators
 - Should be economically feasible
 - Should have a restricted environmental impact
 - Should be safe for the personnel involved
- Need for euthanasia:- In hopeless cases, to alleviate pain and suffering of animals.
- Pre requisite for euthanasia:- Consent of the owner, who has all the property rights of the body of the animal.
- Points to be kept in mind while selecting the method of euthanasia
- The method should be Economical Reliable and Should produce death in a very short period without causing pain to the animal.

EUTHANASIA WITH INHALANTS

- Chloroform, ether, halothane and methoxyflurane. They are useful in the euthanasia of young animals like pup and kittens when venipuncture is difficult.
- **Carbonmonoxide**: Carbonmonoxide 6% is effectively used to euthanize small animals like dogs and cats.

- *Carbondioxide*: used to euthanize small laboratory animals such as mice, rat, rabbit and poultry birds. A combination of CO₂(40%) and CO (3%) can also be used for euthanasia.

EUTHANESIA WITH NON INHALANTS OR INJECTABLES

- Most of the agents in this category can be given i/v by intra-thoracic, intra-cardiac or intra-peritoneal.
- *Barbiturates*: Sol. Phenobarbitone soln (200 mg/ml) at a dose rate of 1 ml/kg b.wt can be used to euthanize the small animals.
- *Chloral hydrate*: Used to euthanize large animals at the dose rate of 2mg/10 kg b.wt.
- *Strychnine*: Produces violent muscular contraction and produces a painful death.
- *Magnesium sulphate*: Saturated solution of Mag.sulphate is an inexpensive method of euthanasia in both large and small animals.
- *Potassium chloride*: Sat.solution of potassium chloride can be used for euthanizing the animals.

EUTHANASIA WITH PHYSICAL METHODS

- *Electrocution*: Is feasible to electrocute individual animals which is relatively humane.
- *Shooting*: Is a humane and rapid method of euthanasia but it causes excessive haemorrhage.

MODULE-24: CATTLE SLAUGHTER PREPARING LIVESTOCK FOR SLAUGHTER

- At the time of slaughter, animals should be healthy and physiologically normal. Slaughter animals should be adequately rested. They should be rested, preferably overnight, particularly if they have travelled for some times over long distances. However, pigs and poultry are usually slaughtered on arrival as time and distances travelled are relatively short and holding in pens is stressful for them. Animals should be watered during holding and can be fed, if required. The holding period allows for injured and victimised animals to be identified and for sick animals to be quarantined.
- When ready for slaughter, animals should be driven to the stunning area in a quiet and orderly manner without undue fuss and noise. Droving can be facilitated using flat canvass straps, rolled plastic or paper, and in the case of stubborn animals, prodders can be used occasionally. Animals should never be beaten nor have their tails twisted. Animals should be led in single file into the stunning area where they can be held in appropriate restraining device(s) before stunning.

RESTRAINT DEVICES

It is very important that slaughter animals should be properly restrained before stunning or bleeding. This is to ensure stability of the animal so that the stunning operation can be carried out accurately and properly. Different types of restraints are appropriate for different species:

Cattle

- A stunning box is the most common method of restraining cattle. The size of the box should be just wide enough to prevent the animal from turning around, and so be difficult to stun. The floor of the box should be non-slip. A simple neck crush used by farmers to restrain cattle for weighing is suitable for small-scale operations. Restraining tame cattle outside the stunning box by securing the head in a halter and then pulling the rope through a metal ring in a concrete floor is effective.

RELIGIOUS OR RITUAL SLAUGHTER (HALAL AND KOSHER)

Most developed and many developing countries of the world require by law an animal to be rendered unconscious before it is slaughtered. This is in order to ensure that the animal does not suffer pain during slaughter. However, exceptions are made for the Jewish (Kosher) and Muslim (Halal) slaughter of livestock. Here stunning generally is not allowed and the animal is bled directly using a sharp knife to cut the throat and sever the main blood vessels. This results in sudden and massive loss of blood with loss of consciousness and death. However, many authorities consider that religious slaughter can be very unsatisfactory and that the animal may not be rendered unconscious and suffer considerable discomfort and pain in the slaughter process.

A number of factors must be given serious consideration before this type of slaughter is acceptable:

- Animals that are slaughtered according to Kosher or Halal requirements should be securely restrained, particularly the head and neck, before cutting the throat. Movement results in a poor cut, bad bleeding, slow loss of consciousness (if at all) and pain. This has serious implications for animal welfare. The knife that is used to cut the throat and the carotid and jugular blood vessels must be razor sharp and without blemishes and damage. This is to ensure a swift, smooth cut across the throat behind the jaw and to ensure immediate and maximum gush of blood. Poor bleeding causes slow loss of consciousness and reduces meat quality.
- Animals should not be shackled and hoisted before bleeding. This causes them severe discomfort and stress. Hoisting should be done only after the animal has lost consciousness. Restraining equipment should be comfortable for the animal.
- Operator competence is of great importance in order to carry out satisfactory religious slaughter, and the authorities should license all slaughter personnel. A poor technique will result in great suffering and cruelty to the animal. Religious slaughter should be carried out paying attention to detail and ensuring the method, equipment and operators are correct. The slaughter process is slow.

The captive bolt gun is suitable for this stunning when using the mushroom shaped head of the bolt. The mushroom gun is an improvement on the plain bolt, as this bolt does not penetrate the brain and cause death. This should be more acceptable to the religious authorities, and its use would encourage more humane slaughter amongst Muslims in developing countries, thereby improving animal welfare.

Fortunately, many Muslim authorities accept some forms of pre-slaughter stunning. Many Muslim authorities permit electric stunning of cattle, sheep and poultry, whose meat is destined for Muslim communities, because the animals subjected to this stunning method would recover if no bleeding was carried out. Electric stunning is also the method of choice in meat exporting countries where stunning of slaughter animals is required by law, for export to Muslim countries. Similarly, Muslim minorities in countries with stringent animal welfare regulations are allowed to use Halal slaughter methods, but in combination with electrical stunning.

Any kind of prestunning for livestock to be slaughtered according to the Jewish Kosher method has not yet been accepted.

- It is recommended that the operator should be positioned behind protective steel bars.

MODULE-25: LIABILITY AND INSURANCE

- **Learning outcome**
- This module focusses on the insurance aspects in animal husbandry practices. The insurance acts, claims and details about insuring livestock are discussed in this module.

LIVESTOCK INSURANCE IN INDIA

Objects of Livestock Insurance

- To guarantee the farmer and livestock owner, the compensation for loss due to the death of the insured animal.
- To enable the livestock owner to repay the loan given to him for purchase of animals to the bank.
- In case of permanent or total or disability of draught animals or the permanent loss of quarters in milch cows, 75% of the insured value is paid to the animal owner.
- To enable livestock owner to get fresh loans again for purchasing similar type of animals.

Conditions for Insurance

- Livestock owners are required to fulfill the following conditions in regard to the animals intended to be insured:
- Provide suitable housing accommodation having byres with sufficient space for manger, watertroughs and suitable housing for poultry.
- Offer good and balanced feed to the animal(s).
- Animals must be free from ecto and endo-parasites and be treated periodically for parasites.
- Animals must be protected against bacterial and viral diseases like anthrax, blackquarter, haemorrhagic septicaemia and foot-and-mouth disease.

REQUIREMENTS FOR LIVESTOCK INSURANCE

- **Health Certificate:** It is obligatory on the part of the owner to produce a certificate from a qualified veterinarian to the effect that his animal is healthy and free from any disease. The owner is required to pay a fee for this certificate as per prevailing rates or as prescribed by Government Insurance Company.
- **Identification of Animals:** The identification of the insured animal is of great importance and must be done thoroughly and faithfully. Emphasis has to be given to the breed, age, sex, body colour, shape of horns, height, identification marks, (natural or acquired by branding, tattooing and tagging). Tagging must be done by a veterinarian.
- The animal owner should note the ear tag number of his animal.
- If the ear tag falls or is lost due to tearing of ears while rubbing, the fallen tag should be preserved and this must be reported to the bank or insurance company in writing.
- When an insured animal dies its ear tag is valuable proof and must be preserved carefully. While claiming compensation for the loss of a deceased insured animal a piece of its ear along with tag must be submitted to the insurance company for verification.

- **Principle of No tag-No claim:** Nowadays insurance companies have adopted the principle of 'No tag-No claim' i.e. no claim is paid if the specific ear tag is not submitted. Thus, timely retagging of the animal is very important whenever its tag is lost or falls off.

INSURANCE POLICY

- Livestock can be insured through two types of policies
 - proposal-cum policy and
 - master policy agreement.

Proposal-cum-Policy

- The owner is required to fill the proforma called "proposal-cum-policy with receipt and veterinary certificate" after obtaining the health certificate. This proforma may be filled by the authorised agent of the insurance company and by a qualified veterinarian. The proforma contains information about:
 - Owner - his name, address and occupation.
 - Period of insurance, from to
 - Particulars of animal-species, breed, sex, colour, age, identification marks, height, purpose for which used, total number of calvings, date of last calving, present market value and sum for which to be insured.
 - Location of dairy farm
 - Premium amount
 - Cause of loss and number of animals lost during past three years.
 - Special remarks if any.
- There is a declaration in this form by the owner stating that all information given by him is true to the best of his knowledge and belief There is also a certificate by a qualified veterinarian indicating that the animal mentioned is in good and sound health, free from any vice, and has been vaccinated against prevailing contagious diseases. The lower part of this form has a detachable receipt showing that the premium amount has been received by an authorised agent of the company. After payment of the premium and completion of the above formalities the animal is deemed as insured from this date.

Master policy agreement

- Since financed animals have to be insured in large numbers and as a day-to-day practice, a Master Policy Agreement is generally entered into by the DRDA or various banks with an insurance company. Under this agreement the bank or the DRDA agrees to insure all its finance cases through one particular insurance company for an agreed period. Only one master policy is issued at the inception. The various animals financed during the period of this policy agreement, which may range from three to five years, are covered under the policy through individual insurance certificates. Certificate books are provided to different banks or DRDA by the company. Whenever a case is financed, they issue a certificate of insurance giving all particulars and send it to the company along with the health certificate and premium amount. The advantage here is that owner does not have to wait for the insurance agent for insurance and also, under the agreement, the animals are covered right from the date and place of purchase.

RATE OF PREMIUM

- The rate of premium varies from species to species and from company to company. Usually the companies charge 5% of the insured amount in case of cattle.

Non-Scheme animals

- 5% of insured value.
- 1% extra for crossbred.
- 2% extra for exotic.
- 4% in case of cattle belonging to organised dairy farms and cooperative societies.
- 2.25% for livestock other than cattle financed under schemes. Generally scheme animals are insured for three years or more. In such case, there is a long term discount of 25% on the total premium.

Insured value

- The amount of insurance depends upon the prevailing market rates of animals, yield of milk, meat etc. However, certain insurance companies may fix a maximum amount for various species of animals.

Owner's duties in case of sickness/accident of insured animal

- When the insured animal gets sick for one reason or the other, the owner must inform the concerned insurance company in writing. He must arrange for proper and timely treatment by a qualified veterinarian for early recovery. The treatment of an insured animal by a quack or Veterinary Field Assistant, Stockman or Compounder will not be accepted by an insurance company and will disqualify the claim in the event of death of an insured animal. Qualified veterinarian means a person holding a B.V.Sc./B.V.Sc.&A.H. degree from a University/statutory body constituted by the State Government or as prescribed by the Indian Veterinary Council Act of 1984.
- As soon as the accident occurs the owner must report to the nearest police station and obtain a receipt for the same. This receipt is an important document to be included in claim papers. A certificate from an attending qualified veterinarian and the treatment chart indicating detailed treatment given to an insured animal must be obtained and attached with the insurance claim papers.

HOW TO PROCEED FOR INSURANCE CLAIM

- In the event of the death of the insured animal the owner must proceed systematically for making a claim. The following line of action is suggested:
 - The owner must intimate in writing to the concerned insurance company at once and get a receipt for this intimation.
 - In case purchase of the animal has been financed by a bank he must inform the bank concerned and get a receipt for the same.

- If the insured animal dies on a Sunday/Holiday, the message must be sent telegraphically/personally to the Branch Manager. The receipt of the telegram must be preserved as proof.
- The owner, after intimating the insurance company/bank, should wait for on the spot verification to be conducted by the representative of the insurance company and arrange for post-mortem.
- The carcass should be removed for post-mortem either by a Khadi Gramodyog /locally available chamar with the instruction that post mortem will be conducted by a veterinarian and to collect a piece of the ear having ear tag, which will be submitted to the insurance company.
- Sometimes insured animal dies at such a place where it is not possible to get the post-mortem conducted by a qualified veterinarian. In such circumstances some insurance companies accept a joint certificate issued by any two of the following:
 - Village sarpanch.
 - Manager of Milk Collection Centre.
 - Officer of Cooperative Central Bank/ Supervisor of DRDA or its Inspector.
- Requirement of post-mortem has been waived in cases of animals financed under government schemes and in such cases if the post-mortem examination has not been conducted, then a death certificate signed jointly by any of the following two persons will be sufficient:
 - Attending veterinary doctor.
 - Sarpanch/Pradhan.
 - Official/Chairman of Milk Cooperative Society.
 - Official of DRDA.
 - Manager of a bank other than the financing bank or other village cooperative.
- However, post-mortem examination is a must for animals other than the above.

CLAIM FORMS

- After the death of the insured animal the owner should approach the insurance company and obtain the claim forms which include the following:
 - **Cattle/other livestock claim form:** This form is to be filled by the owner. Care must be taken to fill the form and the details given should not differ from the particulars mentioned in the policy. It will be better if he consults his veterinarian and takes his help to fill up the form. In case of financed cattle, the form has to be countersigned by the bank.
 - **Veterinary Death-cum- Valuation Certificate:** This is the second form in the claim papers and is to be filled by the attending veterinarian. It incorporates the physical features, date of illness/death and value before illness.
 - **Treatment chart:** This is a date-wise account of the treatment given by a veterinarian to the sick insured animal. The treatment given should correspond to the symptoms noted and post-mortem findings. It is required only for private or other non-scheme animals.
 - **Post-mortem report:** The post-mortem report proforma is usually supplied by the insurance companies. The attending veterinarian must carefully mention the post-mortem findings in this form. It is advised that he should consult a standard text-book of Pathology or Medicine prior to writing the post-mortem report. He should mention identification marks of the animal and its tag number as also whether the tag was duly fixed in the ear or not.
 - **Photograph of deceased animal:** The photograph of a dead insured animal constitutes good evidence of its death if taken correctly showing tag, if possible, and other identification marks and salient features.

- **Confirmation of the post-mortem:** Some insurance companies require a confirmation certificate from the attending veterinarian that he has really conducted the post-mortem of the deceased insured animal. The text of the certificate is given as below:

At the request of *Sri* *S/o*

Rio Village Tehsil.

District.. ..

I have actually conducted the post-mortem of (animal) on
..... (date) at (time). The details of the deceased animal
were as follows:

Breed, sex, colour, horns, identification marks, tag no switch of tail, height, age etc.
This certificate is being issued in confirmation of my having conducted the post-mortem of the said
animal.

Date:

Signature
Designation Seal

LAIM DOCUMENTS

- The total claim documents required by insurance companies in urban and rural areas are as follows:

In urban area

- Livestock claim form
- Veterinary certificate
- Treatment chart
- Post-mortem report
- Valuation certificate
- Photographs
- Post -mortem confirmation certificate
- Police report, if any.

In remote rural area

- Livestock claim form

- Two certificates signed by either
 - Village Sarpanch
 - Chairman, Co-operative Society
 - Manager, Milk Collection Centre
 - Officer of Cooperative Central Bank/Supervisor of DRDA or Inspector. In case of animals financed under government schemes, in order to help ignorant villagers in completing the various formalities for a cattle claim, insurance companies have come up with a single page claim form which incorporates all the certificates required from the owner, bank, veterinarian and others. No other form is required to be filled.
- Panchnama
- Valuation certificate
- Police report, if any.
- The above documents, after completion, should be submitted to the insurance company for necessary action for payment. Sometimes these documents are sent through the financing bank

VALUABLE ADVICE FOR LIVESTOCK OWNERS TO GET CLAIM

- Look after the insured animal well by proper feeding and management.
- Never shift the insured animal without intimating the insurance company in writing.
- Never sell the insured animal or keep it away from the owner without informing the insurance company.
- Never hide any fact or information regarding injury or death of the insured animal.
- Never cause any injury to the insured animal nor show any disinterest or slackness in the treatment of a sick insured animal.
- The treatment and post-mortem of an insured animal must be done by a qualified veterinarian and not by para-veterinary staff, as this would disqualify the claim.
- In the event of the death of an insured animal immediately inform the insurance company.
- Keep the ear tag along with a piece of the ear in your custody and submit it to the insurance company.
- When an accident occurs never forget to report to the nearest police station.
- All the insurance claim forms must be completed in all respects and then submitted. Incomplete forms cause unnecessary delay.

Repudiation of Insurance Claims

- There are many reasons for which claims are liable to be rejected / repudiated. Some of these are:
- When there is difference in the identification of the animal as given by the owner and as mentioned by a veterinarian in various claim papers.
- If the period of insurance expires prior to the death of the insured animal.
- When fraud is suspected.
- In cases where the treatment and post-mortem are not done by a qualified veterinarian, with the exception indicated earlier in remote villages.
- When the insurance claims are not submitted in time, or the death intimation is not given timely and the carcass is disposed of under suspicious circumstances, the company entertains the claim with doubts.
- In case the insured animal dies due to contagious disease for which vaccination can be done, the companies repudiate such claims. But under special circumstances and on the certificate of a qualified veterinarian that the insured animal died in spite of vaccination, the claims are paid.
- When there is change in location of animal and owner of an insured animal the claim is rejected.

- Where more than one veterinarians are involved in the treatment and post-mortem of an insured animal and there is no synchronization between the two, the claim is likely to be rejected on technical grounds.

FEES OF THE VETERINARIAN

- The fee rate varies from State to State as per Veterinary Manuals. In most of the States fee rates have not been revised as per price index and pose a serious legal problem. Veterinarians can charge fees as per recommendations of the Veterinary Manual or as per prevailing market rates for issuing health certificates and for conducting post-mortems.

COMMON MALPRACTICES IN INSURANCE CLAIMS

- *Insurance on ill animals:* Sometimes animals in sick conditions are insured in order to submit claims for obtaining benefits. Death of an insured animal soon after its insurance may lead to suspicion. Such claims are detected and disallowed.
- *Willful killing of an insured animal:* It has been observed that sometimes after getting the desired lactation, the animals are killed by some unscrupulous methods and claim is submitted for receiving benefit from the insurance.
- *Claim for an uninsured animal:* Sometimes a livestock owner uses a tag from an insured animal to claim insurance on some other dead uninsured animal. This can be detected and claim rejected.
- *Insurance of old and unproductive animals:* Many a time with the connivance of more than one agency, old and unproductive animals are insured. These animals die within the insurance period and claims are submitted for payment. A technical probe can detect such frauds and claims of this nature are liable to be disallowed.
- *Tampering with ear tags:* At the time of issuing health certificate the veterinarian is required to certify that the tag has been fixed to the animal's ear. Therefore, a veterinarian should get the tag fixed in his presence to avoid any complications. Sometimes ineligible animals are insured by a an unscrupulous owner and this brings liability upon the veterinarian. Sometimes ear tags are fabricated by fraudulent owners who attach it to the ear of the dead animal. This can be detected by very careful examination which will indicate that:
 - Recently fixed tag will show scratches due to stone or hammer used for fixing the tag.
 - There would be some hair growth around the tag hole in the case of a recent punch but this would be usually absent in an old tag hole.
 - Sometimes the tag is new and does not correspond to the time when the animal was insured.
 - A fresh ear hole in which the tag has been put will not show the changes usually seen in an old hole. Histopathology in such cases will reveal the presence of fibrous connective tissue at the edge of the hole if the tag is put during life time but it would be absent in case the tag is put after death or just before death.

MODULE-26: CODE OF CONDUCT AND ETHICS FOR VETERINARIANS

- **Learning outcome**

- Veterinary Council of India is the authority in India for Veterinarians. It drafts rules and regulations for veterinary profession in India. In addition it develops a common curriculum for Veterinary Colleges all over India for UG and PG education. In this module we are focusing on the common regulations for veterinary practice and the code of conduct for Veterinarians.

PRINCIPLES OF VETERINARY PROFESSIONAL CODE OF CONDUCT AND PROFESSIONAL ETHICS

- To develop, improve and conserve the livestock resources.
- To prevent cruelty to animals.
- To provide protection against spread of diseases and relief to the suffering animals.
- To impart sound, technical opinions in Vetro-legal cases.
- To create proper market for sale / purchase of animals and their products.
- To render services to the society for their economic uplift.

VETERINARY COUNCIL OF INDIA

- In exercise of the powers conferred by section 66 read with section 31 of Indian Veterinary Council Act, 1984 (52 of 1984) the Veterinary Council of India, with previous approval of the Central Government, hereby makes the following regulations, namely:
- **Short title and commencement**
 - These regulations may be called, the Veterinary Council of India (Standard of Professional Conduct, Etiquette and Code of Ethics, for Veterinary Practitioners) Regulations, 1992.
- **Definitions**
 - “veterinarian” means a registered veterinary practitioner as defined in the Act;
 - “Act” means the Indian Veterinary Council Act, 1984 (52 of 1984);
 - “regulations” means the Standard of Professional Conduct, Etiquette and Code of Ethics Regulations. The definition limits its meaning to the terms used within the said regulations;
 - “patient” shall mean an animal, bird or a group of them being treated or managed or advised to be treated or managed by a veterinarian;
 - “client” means the legal owner of the patient or his representative or the person or persons directly assigned responsibility of the patient;
 - “profession” means veterinary profession covering all its branches, namely, animal health, animal production and technology, and shall include any sister
 - profession wherein veterinary service is extended or is sought or is made use of, namely, laboratory animal medicine, animal experimentation and vaccine production.
- **Declaration and oath by a veterinarian**
 - Being admitted to the profession of veterinary medicine, I solemnly swear to use my scientific knowledge and skills for the benefit of society through the protection of animal health, the relief of animal suffering, the conservation of livestock resources, the promotion of public health, and the advancement of knowledge in veterinary science.

- I shall practice my profession conscientiously, with dignity, and in keeping with the principles of veterinary medical ethics.
- I accept as a lifelong obligation the continuous improvement of my professional knowledge and competence.

GENERAL PRINCIPLES

- **Character of the veterinarian**
 - The primary object of the veterinary profession is to render service to humanity by facilitating better animal production, ensuring veterinary public health as also caring for the sick animals and alleviating their suffering; financial reward is always a subordinate consideration.
 - Whosoever chooses the profession assumes the obligation to conduct oneself in accordance with the noble ideals of serving the dumb and mute creatures. The veterinarian shall be an upright person and shall be sincere and diligent in caring for the sick and managing the healthy animals. He/She shall be modest, sober, and prompt in discharging his/her duties.
 - He/She shall conduct himself/herself with propriety in his/her profession and in all actions of his/her life.
- **Standards of character and moral**
 - The veterinary profession expects from its members the highest level of character and morals. Every veterinarian owes to the profession and the public a duty to achieve such a level.
- **Veterinarian's responsibility**
 - A veterinarian shall merit the trust and confidence of his/her clients, rendering their animals full measure of service with devotion.
- **Advertising**
 - Solicitation of clients by a veterinarian directly or indirectly through an advertisement in news papers, handbills etc. in a manner projecting himself/herself to be in a position of superiority which he/she himself/herself has not achieved, is unethical.
 - A veterinarian practitioner is permitted formal announcement in press about the following matters namely
 - starting of his/her practice;
 - change of type of practice;
 - change of address;
 - temporary absence from duty;
 - resumption of practice;
 - succeeding to another's practice.
- He/She shall further not advertise through price lists or publicity materials of manufacturing firms or traders with whom he/she may be connected in any capacity, nor shall he/she publish cases, operations or letters of thanks from patients in non-professional newspapers or journals.
- **Fee for professional service**
 - A veterinarian engaged in practice / advisory service / consultancy, shall limit the source of his/her income to the fee received for services rendered to the client. Remuneration received for such services shall be in the form and amount specifically announced to the client at the time the service is rendered.
- **Patent and copyrights**
 - A veterinarian may patent surgical instruments, appliances and medicine or copyright applications of methods and procedure.
- **Running an Open Shop:** (Dispensing of drugs and appliances by veterinarians)

- A veterinarian shall not run an open shop for sale of medicine for dispensing prescriptions by doctors other than himself.
- **Rebates, gifts and commissions**
 - A veterinarian shall not give, or receive any gift, or commission from any person for in consideration for the referring or procuring of, any client.
- **Secret Remedies**
 - The prescribing or dispensing by a veterinarian of secret medicines or other secret remedial agents, of which he/she does not know the composition or the manufacture, or promotion of their use, is unethical.
- **Evasion of Legal Restrictions**
 - A veterinarian shall observe the laws of the country in regulating the practice of veterinary medicine and will not assist others to evade such laws.

DUTIES OF VETERINARIANS TO THEIR CLIENTS AND PATIENTS

- **Obligation to the patient**
 - Though a veterinarian is not bound to treat cases of each and every one asking for his/her service, except in emergencies, he/she shall for the sake of humanity and the noble tradition of the profession, not only be ever ready to respond to the need of the sick and injured animals/birds, but shall also be mindful of the high character of his/her mission and the responsibility he/she incurs in the discharge of his/her professional duties.
- **Patient not to be neglected**
 - A veterinarian is free to choose whom he/she will serve, provided that he/she shall respond to any request for his/her assistance in an emergency or whenever temperate public opinion expects the service. Once having undertaken a case, a veterinarian shall not neglect the patient nor shall he/she withdraw from the case without giving notice to the client sufficiently in advance, to allow him secure another professional veterinarian.
- **Termination of service**
 - A veterinarian shall withdraw his/her service under following valid reasons:
 - Where he/she finds another veterinarian in attendance.
 - Where remedies/advice other than those prescribed by him/her are being used.
 - Where his/her remedies/advice and instructions are refused.
 - Where he/she feels that the illness or the need of his/her service is an imposture and that he/she is being made a party to a false pretence.
 - Where a client is resorting to the use of intoxicating drugs, means of quackery or witchcraft against medical advice; and
 - Where complete information/history concerning the fact and a client does not supply circumstances of the case.
 - The discovery that the malady is incurable / the client's problems are unsolvable, is no excuse to discontinue attendance by a veterinarian as long as his/her client desire the service.
- **Act of negligence**
 - No veterinarian shall willfully commit an act of negligence that may deprive patient of the care that is absolutely necessary.
 - A veterinarian is expected to show such diligence and skill in service as would be expected of another veterinarian of similar qualifications, experience and attainments.
 - His/Her acts of commission or omission shall not be judged *by any non-veterinary standards*, but as those expected from a veterinarian of his/her training, standing and experience.

- A veterinarian shall use any drug prepared under standard pharmacological principles and shall adopt all necessary preparations / precautions, like sterilization and verification of doses as are normally prescribed for that drug.
- **Behaviour to patients**
 - The demeanour of a veterinarian towards his/her patient shall always be kind, tender and full of patience. Every patient shall be treated with attention, consideration and concentration.
- **Visit**
 - Notwithstanding the fact that a veterinarian is not bound to visit his/her patient/client, he/she shall make it at the indicated hour, once he/she had committed to do so.
- **Prognosis**
 - A veterinarian shall neither exaggerate nor minimize the gravity of a patient's condition. He/she shall ensure that the legal owner of the patient has such knowledge of the patient's condition as will serve the best interest of the patient and its owner (client).
 - In case of dangerous manifestations, or when grave and highly communicable diseases are encountered he shall not fail to inform the client and those to whom the disease can be of potent danger.

DUTIES OF VETERINARIANS TO THE PROFESSION

- **Upholding the honour of the profession:** A veterinarian is expected to uphold the dignity and honour of his profession.
- **Membership of societies:** For the advancement of his/her profession a veterinarian may affiliate himself/herself with professional societies and contribute his/her time means and energy to their progress, so that they may represent and promote the ideals of the profession better.
- **Safeguarding the profession:** Every veterinarian shall laid in safeguarding the profession against admission to it of those who are deficient in moral character or education. He/she should not employ in connection with his/her professional practice any unqualified personnel to treat or perform operation upon patients.
- **Exposure of unethical conduct:** Exposure of unethical conduct on the part of any member of the profession shall be done without fear of favour. The incompetent corrupt, dishonest or unethical conduct on the part of any member of the profession shall be discouraged at any cost.
- **Appointment of substitute:** Whenever a veterinarian (hereinafter mentioned as absent veterinarian) requests another veterinarian to attend to his/her patient/clients during his/her temporary absence from practice professional courtesy requires the acceptance of such an appointment by the latter, if consistent with his/her other duties. The veterinarian under such appointment shall give utmost consideration to the interest and reputation of the 'absent' veterinarian. He/she shall not charge either the patient or the absent veterinarian or his/her service, except in case of special arrangement between them. All such patients shall be restored to the care of the absent veterinarian upon his/her return.
- **Professional services of veterinarians to one another**
 - There is no rule that a veterinarian shall not charge another veterinarian or a member of a sister profession for service rendered. But a veterinarian shall consider it a pleasure and privilege to render gratuitous service to his/her professional brethren, if they are in his/her vicinity, or to a veterinary student.
 - When a veterinarian is called from a distance to attend to a case of a fellow veterinarian or a member of a sister profession, reimbursement shall be made for traveling and other incidental expenses.

- A veterinarian called in any emergency to visit a patient under the care of another veterinarian, shall when the emergency is over, retire in favour of the latter; but shall be entitled to charge the client for his/her services.
- When a veterinarian is consulted at his/her own residence, it is not binding on him/her to inquire if the patient has been under the care of another veterinarian. But in the interests of the patient he/she shall, while ascertaining history, go through the treatment followed if any. However, it is unethical that his information be used to malign or instigate against, directly or indirectly, the veterinarian who happened to attend on the patient previously. When a veterinarian sees a patient at the request of another veterinarian, it shall be the duty of the first veterinarian to write a letter to the veterinarian making the request stating his/her opinion of the case with the modes, of treatment he/she thinks proper to be adopted.
- If a veterinarian is engaged to attend on a patient of dystocia or a similar distress he/she shall do so Refusal to do so under excuse of another engagement is unethical except when he/she is already engaged in similar or another serious case (s).
- When a veterinarian engaged to attend a serious case is absent and another veterinarian is sent for, the latter shall be entitled to legitimate fees; provided that he/she shall secure the client's consent to withdraw on arrival of the former at a mutually consented or logical phase.
- When it becomes the duty of a veterinarian occupying an official position to see and report on an illness or injury or any other professional problems concerning a patient, he/she shall do so, but communicate to the veterinarian in attendance, so as to give him/her an option to be present. The former shall avoid remarks on the diagnosis or treatment that has been adopted. This does not prevent him/her from discussion the matter with the later in isolation.

DUTIES OF PRACTITIONERS IN CONSULTATION

- **Consultation shall be encouraged:** In case of serious illness / doubtful or difficult conditions, a veterinarian shall always request consultation; he/she shall also do so in the case of major surgical interventions, out break of herd diseases, poisoning, endemic or undiagnosible situations.
- **Punctuality in consultations:** Punctuality shall be observed by a veterinarian in attending a consultation. If a consultant veterinarian does not arrive within reasonable time, the consulting veterinarian shall be at liberty to see the patient; provided that he/she shall leave his/her conclusion in writing in a closed envelope. The same rule is applicable if the consulting veterinarian fails to turn up in time.
- **Patient referred to another veterinarian**
 - When a patient is referred to another veterinarian, a statement of the history, observation, clinical and laboratory examination as well as the tentative treatments undertaken shall be given.
 - On receipt of the above statement the veterinarian receiving the referred case shall communicate his/her opinion to the attending veterinarian and either shall continue the treatment as per his/her opinion or return the case along with his/her opinion communicated.
- **Consultation for patient's benefit:** In every consultation, the benefit of the patient shall be of prime importance. It should not be done with an ulterior motive of evasion of responsibilities or on monetary considerations.
- **Conduct of consultation**
 - In consultation there shall be not place for insincerity, rivalry or envy. All due respect shall be shown to the veterinarian in charge of the case and no statement or remark shall be

- made which would impair the confidence reposed in him/her by the client. For this purpose no such discussion shall, therefore, be made before the client.
- All statements to the client shall be made in the presence of consulting veterinarian, except as otherwise agreed. Announcement to the client also shall be made by the consulting veterinarian to the extent possible.
 - Difference of opinion shall not be divulged, unless both feel that it would be in the interest of the patient or dignity of the profession.
 - It shall be open to the client to take further advice and the consulting veterinarian with the consent of the client may consult yet another veterinarian, the rules of ethics remaining the same.
- **Cessation of consultation:** Attendance of consulting veterinarian shall cease when consultation is concluded; unless another appointment is arranged by the attending veterinarian.
 - **Treatment after consultation**
 - No decision shall restrain the attending veterinarian from making such subsequent variation in the treatment as any unexpected change or development may demand; provided that at the next consultation, reasons for variations are stated.
 - The same privilege, with its obligations, belongs to the consultant when sent for in an emergency during the absence of attending veterinarian. The attending veterinarian may prescribe at any time; but the consultant only in case of emergency or instant need. All treatments by both shall be stated to each other.
 - **Consultant not to take charge of the case:** When a consultant veterinarian is called upon to be consulted, none but the rarest and most exceptional circumstances shall justify the consultant taking charge of the case. He must not do so merely on solicitation of the client or his representative.
 - **Bar on consulting non-registered veterinary practitioners:** No veterinarian shall have consultation

DUTIES OF VETERINARIANS TO THE PUBLIC

- **Veterinarian as citizen**
 - Veterinarians as good citizens possessed of special training, shall advise concerning the health and husbandry of the animals, in the localities wherein they dwell. They shall play their part in enforcing the laws of the community (not favouring any communal overtones or prejudices) and in substantiating the institutions that advance the interests of humanity. They shall cooperate with the authorities in the humanity.
 - They shall cooperate with the authorities in the observance and enforcement of sanitary laws and the laws relating to drugs poisons and pharmacy, made for the protection of health.
- **Public health**
 - A veterinarian engaged in public health work shall enlighten the public concerning quarantine regulations and measures for the prevention of epizootic, zoonotic and food borne diseases and intoxications and communicable diseases. At all times, the veterinarians shall notify the respective authorities of every case of communicable diseases under their care in accordance with the laws, rules and regulations of the authorities.
 - When an epidemic prevails, the veterinarian shall continue his/her labour without regard to the risk to his/her own health. He/She shall endeavour to ensure hygienic handling of animal products and wastes thereof and educate the public of food-borne diseases and intoxications.

- **Livestock production and technology**
 - A veterinarian shall strive for the betterment of animal production through timely advice / propaganda on scientific management involving economically viable and hygienic housing, adoption of scientific breeding schedule, disease prevention of routines, reproductive health monitoring, hygienic and systematic care before, at and following parturition, care of new born, hygienic collection of farm products and proper disposal of animal products and wastes thereof.
 - He/She should strive to educate the public and Para veterinary staff personnel regarding timely insemination and aseptic handling during A.I. and discourage the unscientific unhygienic inseminations.
- with a veterinary practitioner who is not registered.

DISCIPLINARY ACTION

- **The Veterinary Council of India brings to the notice of the registered veterinary practitioners the following**
 - In matters regarding offences and form of professional misconduct, which may be brought before the appropriate Veterinary Council for disciplinary action, in view of the authority conferred by section 31 of the Act, the appropriate Veterinary Councils may award such punishment's as deemed necessary or may direct the removal altogether, or for a specific period, from the register, the name of any registered veterinary practitioner, who has been convicted of any such offence. The appropriate Veterinary Council may also direct that any name so removed shall be restored.
 - It must be clearly understood that the instances of offences and of professional misconduct, which are given, do not constitute a complete list of the infamous acts which may be punished. Circumstances may and do arise from time to time, in relation to which may occur questions of professional misconduct, which do not come within any of these categories. Every care should be taken that the code is not violation in letter or spirit. In such instance, as in all others the Veterinary Council of India and/or State Veterinary Council(s) has / have to consider and decide upon the facts brought before the Veterinary Council of India and/or State Veterinary Council(s).
- **Professional Misconduct:** The following actions shall constitute professional misconduct
 - Improper conduct with a patient or maintaining an improper association with a client.
 - Conviction by a Court of law of offences involving moral turpitude.
 - Tampering with or falsely providing professional certificates, reports and other documents. The registered veterinary practitioners are in certain cases bound by law to give or may from time to time be called upon or requested to give signed certificates, reports and other documents of kindred character by courts, administrative offices, banks as part of veterinary service. (See section 30 of the Act).
 - Notwithstanding the existence of other similar and/or kindred documents such documents shall include:
 - A vaccination certificate for licensing of pets.
 - A vaccination certificate for transport across border from one state to another.
 - A vaccination certificate for export of animals.
 - A soundness certificate for a horse or health certificate for cattle for the purpose of insurance, registration, etc.
 - A certificate indicating results of test (diagnostic) like tuberculin test.

- Post-mortem reports in vetero-legal cases, for purpose of insurance claims etc.
- Trauma certificate indicating the extent and nature of damage and/or deterioration, in vetero-legal cases involving accidents, cruelty or vindication.
- Birth certificates required at times for purposes of insurance, registrations, loans etc.
- A certificate indicating, reproductive defects, diseases and other conditions rendering an animal uneconomic (fit for culling).
- A certificate for euthanasia where treatment is unlikely to be rewarding and; a certificate of performance of euthanasia.
- Quarantine clearance certificate for having cleared quarantine at places of disembarkation of animals/birds/livestock products.
- Any registered veterinary practitioner who shall be proved to have signed or given under his/her name and authority any such certificate, report or document of a kindred character which is untrue, misleading or improperly relating to the several matters, some of which are specified above, is liable to have his/her name erased from the Indian Veterinary Practitioners' Register.
- A registered veterinary practitioner shall maintain a Register of Veterinary Medical Certificates giving full details of certificates issued.
- When issuing such a certificate the identification mark of the patient, name or number, if any, age and other details including owner's name and keep a copy of the Certificate for record with him. The client's signature and/or thumb impression and a declaration that he is the genuine owner of the patient or is his authorized representative shall invariably be obtained.
- Contravention of the provisions of law relating to drugs and regulations made thereunder.
- Selling a drug or poison regulated by law to the public or his/her clients, save as provided by that law.
- Performing, or enabling an unqualified person to perform, any illegal operation for which there is no medical, surgical or psychological indication.
- Affixing a signboard on a chemist's shop or in places where the veterinarian does not reside or work. This should not be construed to mean announcement, boards or direction indicators.
- While displaying a board or a name plate, to write anything other than his/her name, qualifications obtained from a University or a statutory body, titles and names of his/her specialty if any (such name being the content of his/her prescription papers), and the registration number.
- Before euthanasia or before performing an operation not obtaining in writing the consent of the client.
- Using touts or agents for procuring patients.
- **Disciplinary Committees:** There shall be a Disciplinary Committee charged with the duty of considering and determining:-
 - (a) Any case of indiscipline referred to them by State Veterinary Council or appeal by aggrieved registered veterinary practitioners.
 - (b) Any other case of which the Disciplinary Committee has cognizance.
- The State Veterinary Council shall investigate, through Committee (s) / Sub Committee (s) (by fact – finding and receiving complaints), regarding:
 - (a) Registered veterinary practitioners being convicted of any criminal offence which in the opinion of the committee renders unfit to practice veterinary profession.

- (b) Any such veterinarian detected / judged to have been guilty of disgraceful conduct in any professional respect.
 - (c) Any person whose name is fraudulently entered in the register.
- The State Veterinary Council (s) shall on receipt of complaints regarding any unregistered individual and individuals who while rendering minor veterinary services described under sub-section (b) of section 30 of the Act, undertake unsupervised treatment/veterinary service assuming himself/herself to be a veterinarian, shall make necessary investigations and advise the concerned State government to take appropriate action or shall themselves initiate action under the law

MODULE-27: DETECTION OF FRAUDS - DOPING, ALTERNATION OF DESCRIPTION AND BISHOPING

- **Learning outcome**

- This module focusses on the malpractices on and with animals and detection of frauds like doping, alternation of description and bishoping.

DOPING

- Introduced by American trainers in 1903 Administration of drugs to act as a stimulant before racing dates.
- Doping was defined as the administration to a horse a short time before race, of certain substances capable of exciting the physical and nervous impulses of the animal, before and during the race, by an artificial action produced on the nervo muscular system.
- Doping originated from an old Dutch word, doop indicating any thick liquid used as an lubricant or excitant
- Doping is defined as the administration to an animal of any substance, other than a normal nutrient, for the purpose of affecting its speed, stamina, courage or conduct in a race.
- Stimulants compounds used in horse racing an
 - Caffeine Adrenaline Sympathine
 - Strychnine Leptazol Methamphetamine groups
 - Picrotoxin Amphetamine
 - Cocaine
- Drugs used for impairing an animal's performance Sedatives
 - Chloral hydras Paraldehyde Strontium
 - Barbiturates Bromides
 - Morphine Camphor

VETERO-LEGAL ASPECTS OF DOPING

- For determining whether or not a horse has given drugs should include
 - Clinical examination before the race
 - Demonstration of the chemical cleanliness of utensils used in sampling.
 - Recovery of adequate volumes of samples fluids.
 - Allowing the owner split samples of the fluids to be analysed or right to have an approved analyst present at the official analysis.
 - Use of tests, which distinguish drugs from alkaloids occurring naturally in the body.
 - Specific identification and precise determination of the amount of any organic substances detected.

ALTERATION OF DESCRIPTION

- The description of the animal is altered in case it has been stolen or there are signs which are unsound. Alteration of description is done in the following ways.
 - Castration
 - Clipping of mane and tail

- Docking
- Colouring of white patches by hair dye
- Provision of signed and witnessed laboratory record protocols by the analyst.

BISHOPING

- Bishoping is the act of changing the appearance of equine teeth to make the horse look younger. This process can involve such processes as filing down the Galvayne's Groove or changing the shape or length of the teeth, or using silver nitrate to artificially create cups. This dishonest practice does not reverse other signs of age shown in enamel rings, the shape of the holes in the smooth-tabled incisors to make new cups which were then blackened with wearing surface or the angle of the incisors from the side.

MODULE-28: WOUNDS

- **Learning outcome**

- This module focusses on the wounds in animals. This module is related to vetero-legal approach of wounds in animals and its types, depth, shape, how it's made, diagnosing the causative agents, firearms and its applications in detail.

DEFINITION

- A wound is defined as a forcible breach in the continuity of the soft tissues of the body including the skin, mucous membrane and cornea. Wounds from the vetero-legal aspect may be classified as:
 - Incised wounds
 - Punctured wounds
 - Lacerated wounds and
 - Firearm wounds

INCISED WOUNDS

- An incised wound is produced by a sharp cutting object such as a knife, razor, sword, gandas (chopper), axe, or any object which has a sharp, cutting edge. An incised wound is always broader than the edge of the weapon causing it owing to the retraction of the divided tissues. It is somewhat spindle-shaped and gaping, its superficial extent being greater than its depth.
- This gaping is greater in deep wounds when the muscle fibres have been cut transversely or obliquely. Its edges are smooth, even, clean-cut, well defined and usually everted. The edges may be inverted if a thin layer of muscular fibre is closely united to the skin, as in the scrotum. They may be irregular in cases where the skin is loose or the cutting edge of the weapon is blunt or heavy. Length of the incised wound has no relation to the length of the cutting edge of the weapon. A curved weapon, such as a scythe or sickle, first produces a stab or puncture and then an incised wound; and sometimes the intervening skin may be left intact.
- While describing an incised wound it is always necessary to note its direction. At commencement the wound is deep, and it gradually becomes shallower and tails off towards the end, but no direction is noticeable if the weapon has not been drawn while inflicting the wound.
- Haemorrhage in the case of incised wounds is usually much more than in other wounds, as a clean-cut blood vessel bleeds considerably more, and bleeding may be so severe as to cause death, especially if a main artery has been cut.

PUNCTURED WOUNDS (Stabs)

- These are termed 'penetrating' wounds when passing through the tissues. They enter a cavity of the body, such as the thorax or abdomen. Such wounds are produced by a piercing or stabbing object, such as a pin, needle, knife, scissors, pickaxe, arrow, etc. A punctured wound caused by a sharp-pointed and cutting object has clean-cut edges which are almost parallel but slightly curved towards each other and have sharp angles at the two extremities. An object having one cutting and one blunt edge will show a certain amount of bruising and raggedness at one end of the wound.

- The wound is generally wedge-shaped, if it is produced by an object with a thick, broad back and only one cutting edge and circular or slit-like opening if produced by sharp-pointed and cylindrical or conical object. The aperture of a punctured wound in the skin is usually smaller in length than the breadth of the weapon used. It is sometimes larger as the weapon enlarges the wound, if it is withdrawn with lateral movement.
- The depth of a punctured wound is much larger than its length or width, and may be equal to or less than the length of the blade of the object causing it. In some cases the depth may even be greater than the length of the blade owing to the fact that the force of the blow may depress the tissues of the part struck, allowing the point to reach the deeper tissues such as in the abdomen.
- There may be very little external haemorrhage and yet profuse haemorrhage may take place internally owing to some vital organ having been penetrated.
- In the case of a perforating wound, there are two wounds, one, the wound of entry and the other, the wound of exit. The wound of entry is usually larger with inverted edges and the wound of exit is smaller and has everted edges. The edges of the entrance wound may be found everted, when the weapon used is rough and rusty.
- In some cases two or more punctures may be found in the soft parts with only one external orifice. This shows that the object had been partially withdrawn after it pierced the tissues, and thrust again in a new direction.
- Sometimes it is argued that a punctured wound may have been caused by a fall on a sharp-pointed piece of an earthenware pot or broken glass. In that case the edges of the wound are irregular and more or less bruised, and fragments of such article may be found embedded in the soft tissues.

LACERATED WOUNDS

- These are tears or splits produced by blows with blunt objects and missiles, violent falls on sharp and hard projecting surfaces, by machinery and railway accidents, by the wheels of a vehicle, by the claws, teeth or horns of animals and by projecting nails. These wounds do not generally correspond in shape or size to the object producing them. Their edges are torn, jagged, irregular and swollen or contused. The tissues are torn and the skin around the seat of injury is ecchymosed and the underlying bones are likely to be fractured, while the internal organs may be injured.
- Foreign bodies such as earth, grease, machine oil, cinder, hair, fibres etc. are frequently found in the wounds which are, therefore, predisposed to infection.
- When produced by a blunt weapon such as a club (lathi), crowbar, stone, etc., a lacerated wound is usually accompanied by a considerable amount of bruising of the surrounding and underlying tissues, and has everted and irregular edges. When a heavy weight like a wheel of heavy or a truck passes over any of the extremities it tears the skin and crushes the muscles and soft parts beneath it, releasing considerable blood fat in them. Crush syndrome or fat embolism may occasionally follow.
- The direction of shelving of the margins of a lacerated wound indicates the direction of the blow applied to cause the wound.
- Hemorrhage in lacerated wounds is, as a rule, not extensive owing to the - that the arteries are not cut evenly but are torn across irregularly so as facilitate clotting of the blood. In lacerated wounds of the head, the oral arteries often spurt as freely and forcefully as when cut cleanly.

FIREARM WOUNDS

- **Types of firearms**
 - *Shotguns* - which fire a number of shots together, with barrels varying in size from 22 to 30 bore are smooth inside and are commonly used to kill birds and small animals. An average shotgun cartridge varies in size from 2 to 3 inches. Ordinarily shotguns are effective within a range of thirty to forty yards.
 - *Rifled firearms* - These are firearms which fire a single bullet and have one barrel only. These have firing range of upto 1000 yards and can cause serious or fatal injury to man or animals.
- Firearms generally produce two wounds or apertures, viz., one of entrance and the other of exit of the projectile. The wound of entrance is usually smaller than the projectile because of the elasticity of the skin, and is round when the projectile strikes the body at right angles and oval when it strikes obliquely. When a wound of entrance is present, but not the wound of exit, it means that a bullet is lodged in the body, except in those cases where the bullet has been coughed up after entering the respiratory passages or lost in the stool after entering the intestinal tract and also when a hard bullet coming in contact with a bone is so deflected as to pass out through the same orifice by which it entered. If a bullet is lodged in the body it must be taken out if death has occurred, and forwarded to the Superintendent of Police in a sealed envelope containing its description in the Veterinary Officer's handwriting as it forms evidence of the greatest value. While searching for a bullet it must be borne in mind that it may take a very erratic and circuitous course while passing through the body. In a case where death has not occurred, the bullet should be located by means of X-ray, if possible.
- The flame and the forceful expansion of the gases of explosion in the skin and subcutaneous tissues usually cause a large entry wound, the edges of which are ragged and everted. Wadding or debris may be found lodged in the wound with the skin surrounding it being scorched and tattooed with particles of unconsumed gunpowder. The entry wound of a revolver fired very near or in contact with the skin is stellate or cruciform in shape instead of being circular. When it is fired from beyond a distance of 12 inches, there are no gunpowder marks around the wound. If the revolver is fired close to the skin but held at an angle, the smudging and tattooing is limited only to one side of the bullet hole. The wound of exit is often larger than the wound of entrance, and its edges are irregular and everted, but free from scorching and tattooing. The edges of both the wounds of entrance and exit may be everted in fat animals due to protrusion of fat into the wounds, and in decomposed bodies because of the expansible action of the gases of putrefaction. The edges of the wound of exit may be very ragged and torn, if the projectile was discharged at close quarters, had passed through the bone or was deformed by striking elsewhere first (wound by recoil). These characters of the wound are due to the wobble of the projectile, its deformed condition, laceration of the skin by fragments of bone expelled from the body along with the projectile or by the splintered pieces of the projectile itself.
- Large bullets cause greater damage to the internal organs than small ones. Round bullets produce larger wounds than conical ones. They cause extensive laceration of the tissues and comminuted fracture of the bones if they strike the body at a different angle and sometimes their course is arrested by coming in contact with chains or other hard articles. Conical bullets produce much less laceration than round ones, and the rounds caused by them are punctured in appearance. Conical bullets rarely split in the tissues, though round ones often do.
- Modern, steel-jacketed bullets used in army weapons have the shape of an elongated cone and owing to their great velocity usually pass straight and directly through the body without any deflection or deviation, and without causing much damage. The wounds of entry and exit are almost circular and similar in appearance, without any bruising or laceration of the surrounding

parts. Such wounds heal very rapidly. Even wounds caused by such bullets in the brain, lungs, or intestines often run a perfectly normal course, and heal without any difficulty. Expanding, grooved, dum-dum bullets are very destructive and produce extensive wounds with ragged margins.

- Fragments of shell are also destructive and cause extensive wounds. Irregular missiles, such as pieces of stone, iron, kankar, beads of brass or nickel anklets or wristlets, seeds etc. used in muzzle-loading guns produce several irregular, lacerated wounds, and the exit wounds are larger than the entrance wounds. It is possible for a single pellet or shot to cause death. Wadding or

VELOCITY OF PROJECTILE

- A bullet travelling at high velocity produces a clean, circular, punched-out aperture or slit, as in a stabbing wound, and usually perforates the body.
- It is not deflected from its path by striking a bone, but may cause its comminution or splintering. On the other hand, a bullet of low velocity causes contusion and laceration of the margins of the wound of entrance. It is easily deflected and deformed by striking some hard object, and often lodges in the body.
- The track made by a bullet widens as it goes deeper. This is the reverse of a punctured wound. If a bullet grazes a bone, it may produce a gutter, with or without fracturing it, and may or may not give the direction or deflection of the bullet.
- gunpowder may cause laceration and may produce death by penetrating the internal organs of the body, even if a blank cartridge is discharged close to the body.

DISTANCE OF FIREARM

- If a firearm is discharged very close to the body or in actual contact with it, subcutaneous tissues over an area of two or three inches round the wound of entrance are lacerated and the surrounding skin is usually scorched and blackened by smoke and tattooed with unburnt grains of gunpowder.
- If the gunpowder is smokeless there will be no blackening of the skin, but there may be a greyish or white deposit on the skin around the wound. No blackening or scorching is found, if the firearm is discharged from a distance of more than four feet.
- Moreover, these signs may be absent even when the weapon is pressed tightly against the body, as the gases of the explosion and the flame, smoke and particles of gunpowder will all follow the track of the bullet in the body. The effects produced by small shot fired from a shotgun vary according to the distance of the weapon from the body, and the choking device.
- A charge of small shot fired very close to, or within a few inches of, the body enters in one mass like a single bullet, making a large irregular wound with scorched and contused edges, and is followed by the discharge gases which greatly lacerate and rupture the deeper tissues.
- Particles of unburnt powder expelled from the weapon behind the missile are driven to some distance through the wound, and some of them are found embedded in the wound and the surrounding skin which is also singed and blackened by the flame and smoke of combustion.
- The exit wound of a close range shows greater damage to tissues than the entrance wound, the margins everted, but there is no evidence of blackening or singeing.
- At a distance of one to three feet small shot make a single aperture with irregular and erated edges corresponding in size to the bore of the muzzle of the gun, as the shots enter as one mass, but are scattered after entering the wound and cause great damage to the internal tissues.
- The skin surrounding the wound is blackened, scorched and tattooed with unconsumed grains of powder. On the other hand, at a distance of six feet the central aperture surrounded in an area of

about two inches in diameter by separate openings made by a few pellets of the shot which spread out before reaching mark.

- The skin surrounding the aperture is not blackened or scorched, but is tattooed to some extent.
- At a distance of twelve feet the charge of shot spreads widely and enters the body as individual pellets producing separate openings in an area of five to eight inches in diameter, but without causing blackening, scorching or tattooing of the surrounding skin. As the distance increases, the damage caused by a single pellet diminishes, until at about 30 feet, it is only capable of penetrating the skin of an animal.

TIME OF FIRING

- After recent discharge a black deposit of potassium sulphide mixed with carbon is found in the barrel of the firearm, if black gunpowder was used. Up to five or six hours this deposit forms a strongly alkaline solution when dissolved in distilled water and emits an offensive odour of sulphuretted hydrogen. If the solution is filtered, and the filtrate is treated with a solution of lead acetate, a black precipitate of lead sulphide is formed. After exposure to air and moisture for a few days the potassium sulphide becomes converted into thiosulphate, thiocyanate and finally potassium sulphate, that gives a neutral solution with distilled water that gives a white precipitate with lead acetate. Later, oxides of iron (iron rust) with traces of iron sulphate are formed in the barrel.
- Smokeless nitro-powders leave a dark grey deposit in the barrel of a recently discharged firearm. It does not change with the lapse of time, gives a neutral solution with distilled water, and contains nitrites and nitrates, but not sulphides. If chromate or bichromate powder is used, the residue in the barrel is usually of a greenish tint.
- It should be borne in mind that the composition of the deposit will vary considerably if the firearm was dirty at the time of its discharge, and a veterinarian has no means to know its condition prior to discharge. Again, a deposit would not be found if the weapon had been thoroughly cleansed after discharge.

VETERO-LEGAL ASPECTS OF WOUNDS : EXAMINATION OF THE INJURED ANIMAL

- The Veterinary Officer may be supplied by the Police Superintendent or The Magistrate the following form which he is required to fill in after examining the injured animal.

1	2	3	4	5	6	7
Nature of injury ether cut, wound, bruise, burn, fracture or dislocation	Size of each injury in inches, that is length, breadth and depth	On what part of the body inflicted	Simple, grievous or dangerous	By what weapon	Whether the weapon was dangerous or not	Remarks

- The Veterinary Officer should be very careful in filling in this form. First of he should write at the left hand top corner of the form identification of animal and the name and number of the police constable accompanying it and should note the mark or marks of identification to enable him to

recognize the injured animal. He should then note the exact time of the examination, viz. hour, date, month and year, and proceed with the examination proper as below:

Nature of Injury

- While describing the injuries in columns 1, 2 and 3 of the form, the Veterinary officer should carefully note their nature and number, the character of their edges, their size as regards length, breadth and depth, the line of direction and their situation. Presence of any extraneous material in the wound should be carefully noted. If necessary, he should use a magnifying lens. All the injuries should be measured with a tape-measure, and the exact measurements in inches must be given. While mentioning the exact situations a reference to some bony prominences or anatomical contours should be made.

Simple, Grievous or Dangerous Injury

- In column NO.4 it must be mentioned whether the injury is simple, grievous or dangerous to life. A simple or slight injury is one which is neither extensive nor serious, and which heals rapidly without leaving any permanent deformity or disfiguration.
- Grievous injuries as described in Section 320, I.P.C., are as follows
 - Emasculation.
 - Permanent privation of the sight of either eye.
 - Permanent privation of the hearing of either ear.
 - Privation of any membrane or joint.
 - Destruction or permanent impairing of the powers of any membrane or joint.
 - Permanent disfiguration of the head or face.
 - Fracture or dislocation of a bone or tooth.
 - Any hurt which endangers life or which causes the sufferer to be, during a period of twenty

KIND OF WEAPON & AGE OF INJURY

- In the fifth column the kind of weapon by which the injury was inflicted should be mentioned. In the remarks column the age of injury should be noted.

Age of Injury

- The age of a bruise may be ascertained from the colour changes which its ecchymosis undergoes. These changes start from eighteen to twenty- four hours after its infliction. On histological examination at necropsy the polymorphonuclear leucocytes in a haematoma begin to disintegrate after 3 to 5 hours, mostly they are fragmented within 21 hours and by 30 hours the basophilic nuclear fragments have either undergone autolysis or have been engulfed by the phagocytic monocytes.
- The age of a wound may be ascertained by observing stages of its healing process.
- A wound, which is not thoroughly aseptic and is, gaping owing to loss of tissue, heals by the formation of granulation tissue.
- The vascular endothelium shows distinct proliferative changes, and vascular buds are given off from the minute vessels at the periphery during twenty-four hours.
- A complete network of new capillary vessels is formed in thirty-six to forty hours.

- Spindle-shaped cells, which run at right angles to the vessels in the deeper parts of the wound, are visible in forty-eight to fifty-six hours.
 - Definite fibrils are seen running parallel to the long fibroblasts in three to six days.
 - The cellular structure and vessels are obliterated gradually, and are replaced by a dense fibrous scar tissue in three to four weeks.
 - Pus may appear in a septic wound in about thirty-six to forty-eight hours.
 - In bone fracture the reparative process indicates its approximate time.
 - Signs of inflammation and exudation of blood in the soft parts and around the fractured ends - first to the third day.
 - Soft provisional callus- fourth to the fifteenth day.
 - Ossification of callus- fifteenth day to the fifth week. Complete absorption of callus- six to eight weeks.
 - Bone formation does not occur in comminuted fractures.
 - In the case of dislocation of a joint, the time can be judged from the colour changes of the bruise.
 - When a tooth has been hit and has dropped out, bleeding from its socket may continue for a few hours. The cavity of the socket usually fills up in seven to ten days, and the alveolar process becomes quite smooth after fourteen days.
- days, in severe bodily pain, or unable to follow ordinary pursuits.

CAUSES OF DEATH FROM WOUNDS

- The causes of death from wounds may be immediate or direct and remote or indirect.

Immediate or direct

- **Haemorrhage**
 - It may be external or internal. External haemorrhage may produce marked fall in blood pressure and consequent shock, causing death either rapidly, if a large blood vessel, such as the carotid or femoral artery, has been wounded, or slowly, if a number of small vessels have been injured. Sudden loss of blood is more dangerous than the same quantity lost slowly. The loss of one-third of the blood in the body is almost enough to cause death. Internal haemorrhage may occur in penetrating and gunshot wounds. It need not be profuse for a fatal result; for a small quantity of haemorrhage in the brain or pericardium may prove rapidly fatal by disturbing the functions of the brain or heart owing to mechanical pressure on them. Blood flowing into the windpipe may cause death mechanically by asphyxia. Rupture of internal organs like spleen, liver, lungs and heart usually causes fatal internal haemorrhage, only the symptoms are often noticed after a few hours.
- **Injury to a vital organ**
 - Severe injury to a vital organ, such as crushing of the brain, heart, lungs, etc., is, as a rule, rapidly fatal.
- **Shock**
 - Death may occur from shock without any visible injury due to paralysis of the heart by a hit in the cardiac region. Death from shock occurs easily in animals that are severely ill and weak, old and young.

Remote or indirect

- Inflammation of the internal organs, such as meningitis, cerebritis, pleurisy, pneumonia, peritonitis, etc.
- Septic infection of a wound causing septicaemia, pyaemia or exhaustion from prolonged suppuration.
- Gangrene or necrosis resulting from severe crushing of parts and tearing of the blood vessels or crush syndrome.
- Thrombosis in veins and embolism.
- Infective diseases, such as erysipelas and tetanus, which may develop through the entrance of the causal organism through a wound.
- Fat Embolism occurring after fracture of long bones and injury to fatty tissue.
- In injuries to the jugular, cephalic and femoral veins, air may be sucked in because of the negative pressure during inspiration and intravenous injection may cause death.
- Neglect of Injured Animal
- Result of Surgical Operation - If death follows a surgical operation performed for the treatment of an injury caused by the offender, he shall be responsible for the result, if it is proved that the death was inevitable even without the operation and that the operation was thought necessary and was performed by a competent surgeon with reasonable care and skill.

DIFFERENCE BETWEEN WOUNDS INFLICTED-DURING LIFE AND AFTER DEATH

- There is more or less copious haemorrhage in all wounds. The effused blood is forced into the tissue interspaces in the vicinity of the wounds, and is found infiltrated into the cellular and muscular tissues.
- There is consequent staining of the edges of the wounds and the neighbouring tissues, which cannot be removed by washing, but the staining caused by the blood effused from post-mortem wounds is easily removed by washings.
- Inhaled blood will be found in the lungs and bronchi, if the area of haemorrhage communicates with the bronchial tree. There will be clots of effused blood in the wounds and tissues, and in the neighbourhood of the body.
- Clotting of the blood normally occurs in about four to ten minutes. There will also be signs of spouting of arterial blood on the body, or in its vicinity. In a contusion ecchymosis, absorption, changes in the colour and a swelling of the neighbouring tissue will be present.
- On dissection coagulated blood will be found in the subcutaneous tissues. Retraction of Edges of the Wound will be present if caused in life. Likewise Signs of Inflammation and Reparative Processes will be present if made during life.

MODULE-29: TRANSPORT OF ANIMALS RULES, 1978

- **Learning outcome**
- Transport of animals in India is still a crude practice. This module focusses on the transport of animals by road, train, air and by ship. In addition we are discussing about the healthcare while

DEFINITION

- **Rule 1: Short title**
 - These rules may be called the Transport of Animals Rules; 1978.
- **Rule 2: Definitions**
 - In these rules unless the context otherwise requires
 - Qualified veterinary surgeon means one who holds a diploma or a degree of a recognised veterinary college;
 - "Schedule" means a schedule appended to these rules.
- transport. The laws and rules for transport of animals were already discussed.

TRANSPORT OF DOGS AND CATS

- **Rule 3**
 - Rules 4 to 14 shall apply to the transport of dogs and cats of all breeds whether by rail, road, inland waterway, sea or air.
- **Rule 4**
 - A valid health certificate by a qualified veterinary surgeon to the effect that the dogs and cats are in a fit condition to travel by a rail, road, inland waterway, sea or air and are not showing any sign of infectious or contagious disease including rabies, shall accompany each consignment and the certificate shall be in the form specified in Schedule-A.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for transport.
- **Rule 5:** No dog or cat in an advanced stage of pregnancy shall be transported.
- **Rule 6:** Dogs or cats to be transported in the same container shall be of the same species and breed. Unweaned puppies or kittens shall not be transported with adult dogs or cats other than their dams. No female dog or cat in season (oestrus) shall be transported with any male.
- **Rule 7:** Any dog or cat reported to be vicious or exhibiting a vicious disposition shall be transported individually in a cage, muzzled and labelled to give warning to the handlers. In extreme cases, the dogs or cats shall be administered with sedative drugs by a qualified veterinary surgeon.
- **Rule 8**
 - When dogs or cats are to be transported for long distances
 - They shall be fed and given water at least two hours prior to their transport and shall not be packed for transport if they are hungry or thirsty
 - They should be exercised as late as possible before despatch;
 - They shall be given adequate water for drinking every four hours in summer or every six hours during winter;

- They shall be fed once in twelve hours in the case of adult dogs or cats and they shall be fed once in four hours in the case of puppies and kittens in accordance with the instructions of the consignors if any;
 - Adequate arrangements shall be made for their care and management during the journey.
 - When the dogs or cats are to be transported by rail involving a journey of more than six hours, an attendant shall accompany the dogs or cats to supply them with food and water on the way and the attendant shall have access to the dogs or cats for this purpose at all stations and no dog or cat shall be exposed to the direct blast of air during such journey.
- **Rule 9**
 - Where dogs or cats are to be transported for short distance by road in a public vehicle, the following precautions are to be taken namely
 - they shall be put in a cage and the cage containing the dogs or cats shall not be put on the roof of the vehicle but shall be put inside the vehicle preferably near the end of the vehicle;
 - the vehicle transporting the dogs or cats shall as far as possible maintain constant speed, avoiding sudden stops and reducing effects of shocks and jolts to the minimum;
 - at least one attendant shall be present at all times during transit who shall ensure that proper transit conditions are observed and shall also replenish food and water whenever necessary.
- **Rule 10**
 - Where dogs or cats are to be transported by air
 - The cages shall be properly cleaned and disinfected before the dogs or cats are put in the cages.
 - Sufficient dry straw or saw dust or paper cuttings shall be provided for cats in the cages as resting material.
 - For international transport, the dogs or cats shall be kept in a pressurized compartment with regulated temperature.
- **Rule 11:** The size and type of crates for transport of dogs and cats shall conform as clearly as may be to the size and type specified in Schedule-B and Schedule-C respectively.
- **Rule 12:** All containers of dogs or cats shall be clearly labelled showing the names, address and telephone number (if any) of the consignor.
- **Rule 13:** The consignee shall be informed about the train or transport arrival or flight number and its time or arrival in advance.
- **Rule 14:** Consignment of dogs or cats to be transported by rail or road shall be booked by the next passenger or main train or bus and should not be detained after accepting the consignment for booking.

TRANSPORT OF MONKEYS

- **Rule 15:** Rules 16 to 23 shall apply to the transport of all types of monkeys from the trapping area to the nearest rail-head.
- **Rule 16**
 - A valid health certificate by a qualified veterinary surgeon to the effect that the monkeys are in a fit condition to travel from the trapping area to the nearest unit-head and are not showing any sign of infections or contagious disease shall accompany each consignment.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for transport.

- The certificate shall be in a form specified in Schedule-D.
- **Rule 17**
 - Monkeys from one trapping area shall not be allowed to mix with monkeys from any other trapping area for preventing the dangers or cross-infection.
 - The time in transit from trapping area to the nearest rail-head shall be as short as possible and factors causing stress to monkeys shall be reduced to the minimum.
 - If the travel time is longer than six hours provision shall be made to feed and to give water to the monkeys en route.
 - During transit, precautions shall be taken to protect the monkeys from extreme weather conditions and monkeys that die en route shall be removed at the earliest available opportunity.
- **Rule 18:** Monkeys that are not completely weaned, that is, under 1.8 kilogram in weight, shall not be transported except when specifically permitted by the Central Government.
- **Rule 19**
 - Pregnant and nursing monkeys shall not be transported except when specifically permitted by the Central Government.
 - Pregnant and nursing monkeys as well as monkeys weighing more than 5 kilograms shall be transported in compartmented cages.
- **Rule 20:** All monkeys in the same cage shall be of the same species and of approximately the same weight and size.
- **Rule 21:** Monkeys captured within their natural habitat shall be placed in new, sterilized or thoroughly cleaned cages and subsequent transfer, if any, shall also be in new, disinfected or thoroughly cleaned cages.
- **Rule 22:** Monkeys shall be transported from trapping area to the nearest rail-head by the fastest means of transport available and the monkeys shall not be left unattended at any time during the journey.
- **Rule 23**
 - Monkeys shall be transported in suitable wooden or bamboo cages, so constructed as not to allow the escape of the monkeys but permit sufficient passage of air ventilation.
 - No nails, metallic projections or sharp edges shall be exposed on the exterior or in the interior of the cages.
 - Each cage shall be equipped with appropriate water and feed receptacles which are leak proof and capable of being cleaned and refilled during transit.
 - The floor of the cages shall be made of bamboo reapers and the space between each reaper shall range between 20 mm and 30 mm.
 - To facilitate carriage of these cages, provision may be made for rope loops at the four top ends.
 - The weight of any loaded cage shall not exceed 45 kilograms.
 - The following two sizes of cages shall be used.
 - 910 x 760 x 510 mm - to contain not more than twelve monkeys, weighing between 1.8 and 3.00 kilograms each or ten monkeys weighing between 3.1 and 5.0 kilograms each.
 - 710 x 710 x 510 mm - to contain not more than ten monkeys weighing between 1.8 and 3.00 kilograms each or eight monkeys weighing between 3.1 and 5.00 kilograms each.
 - Provided that wooden cages as specified in Schedule-F to these rules may also be used for carrying monkeys from the trapping area to the nearest rail-head.

- The construction details of two types of cages shall be as given in Schedule-E.
- **Rule 24:** Rules 25 to 32 shall apply to transport of monkeys from a rail-head to another rail-head or from a rail-head to nearest airport.
- **Rule 25**
 - Loading and unloading shall be carried out quickly and efficiently.
 - Cages shall be stored in such a manner that ventilation is adequate and the monkeys are not exposed to draught and direct heat or cold.
 - Monkeys found dead shall be removed as quickly as possible for suitable disposal.
- **Rule 26:** The transport cages shall be in accordance with specifications given in Rule 28.
- **Rule 27**
 - Due provision shall be made by the sender for a sufficient supply of food and water for the journey.
 - In case the journey is over six hours an attendant shall accompany the monkeys to supply them food, water, and such other things, on route and he shall have access to, the monkeys for feeding, giving water and attention at all stations en route.
 - The food and water containers shall be checked at least every six hours and refilled, if necessary.
 - Monkeys shall not be disturbed during the night hours.
- **Rule 28:** Not more than one cage shall be placed over the other and gunny packing shall be placed between two cages, when one is placed over the other.
- **Rule 29:** Monkeys shall be brought to the airport sufficiently early.
- **Rule 30:** Monkeys shall be provided with food and water immediately before loading on the aircraft.
- **Rule 31**
 - The cages shall be clearly labelled showing the name, address and telephone number (if any) of the consignor and the consignee in bold red letters.
 - The consignee shall be informed about the train in which the consignment of monkeys is being sent and its arrival time in advance.
 - The consignment of monkeys to be transported shall be booked by the next passenger or main train and should not be detained after the consignment is accepted for booking.
- **Rule 32**
 - A valid health certificate by a qualified veterinary surgeon to the effect that the monkeys are in a fit condition to travel from the nearest rail-head to another rail-head or from a rail-head to the nearest airport and are not showing any signs of infectious or contagious disease shall accompany each consignment.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for transport.
 - The certificate shall be in a form specified in Schedule-D.
- **Rule 33:** Rules 34 to 45 shall apply in relation to the transport of monkeys by air.
- **Rule 34:** The time in transit shall be as short as possible and factors causing stress to monkeys shall be reduced to the minimum.
- **Rule 35:** Monkeys that are not completely weaned, that is, under 1.8 kilograms in weight, shall not be transported except when specifically permitted by the Central Government.
- **Rule 36:** Pregnant and nursing monkeys shall not be transported except when specifically permitted by the Central Government. Pregnant and nursing monkeys and monkeys weighing over 5 kilograms shall be transported in specially designed individual cages.

- **Rule 37:** All monkeys in the same shall be of the same species and of approximately the same weight and size.
- **Rule 38**
 - In view of the dangers of infection, only monkeys of the same species shall be transported in the same cabin or compartment of the aircraft.
 - Apparently sick or disabled monkeys exhibiting external injuries or infested with parasites shall not be transported.
 - Transport of other species of animals, birds, fish, food stuff or poisonous materials, such as pesticides and insecticides, in the same cabin or compartment shall not be permitted.
- **Rule 39**
 - At no time during transit shall the monkeys be left unattended when carried in a freighter aircraft.
 - At least one attendant shall be present at all times when the aircraft is on the ground.
- **Rule 40**
 - Monkeys shall be transported in suitable wooden cages, so constructed as not to allow the escape of the monkeys and shall allow sufficient passage of air for ventilation; no nails, metallic projections or sharp edges shall be exposed in the interior or on the exterior of such cages. Each cage shall be equipped with water and food receptacles which shall be leak-proof and be capable of being cleaned and refilled during transit. A suitable absorbent material such as saw dust shall be kept in the dropping trays.
 - The weight of anyone loaded cage shall not exceed 45 kilograms many case.
 - The following two sizes of cages shall be used:
 - 460 x 460 x 460 mm - to contain not more than ten monkeys weighing from 1. 8 to 3.0 kilograms each or four monkeys weighing from 3.1 to 5.0 kilograms each; and
 - 760 x 530 x 460 mm - to contain not more than ten monkeys weighing from 1. 8 to 3.0 kilograms each or eight monkeys weighing from 3.1 to 5.0 kilograms each.
 - The construction details of the two types of cages shall be as given in Schedule-F.
 - The construction details of the two types of cages used for the transport of pregnant and nursing monkeys shall be as given in Schedule-G.
- **Rule 41**
 - The cages shall be clearly labelled showing the name, address and telephone number (if any) of the consignor and the consignee in bold red letters.
 - The consignee shall be informed in advance about the flight number of the freighter aircraft in which the consignment of monkeys is being sent and its arrival time.
 - The consignment of monkeys to be transported shall be booked by the next flight of the freighter aircraft and should not be detained after the consignment is accepted for booking.
- **Rule 42**
 - A valid health certificate by a qualified veterinary surgeon to the effect that the monkeys are fit to travel by air and are not showing any signs of infectious or contagious disease shall accompany each consignment of monkeys.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for shipment.
 - The form of a certificate under sub-rule (1) be as given in Schedule-D.
- **Rule 43**
 - The air shall be changed not less than twelve times per hour and draughts shall be avoided and there shall be no dead pockets of air.
 - Except when the monkeys are being fed and given water; they shall travel in semi, darkness to make them quieter and less inclined to fight and thus give them better opportunities of resting.

- **Rule 44:** The food and water containers shall be checked at every stop and refilled; if necessary, and a sufficient stock of food shall be available on the aircraft and at likely stopping places.
 - Note: About 85 grams of food per monkey is required daily. Suitable foods are dry cereal grains or gram. It is recommended that whole gram made into biscuits or wheat meal bread should be fed. A minimum of 140 ml of water shall be allowed for each monkey per day.
- **Rule 45:** An empty cage of the usual dimensions with its sides covered except 50 mm at the top to allow for ventilation shall be provided in the freighter aircraft for housing the monkeys which fall sick or are injured during the journey.

TRANSPORT OF CATTLE

- **Rule 46:** Rules 47 to 56 shall apply to the transport by rail of cows, bulls, bullocks, buffaloes, yaks and calves (hereinafter in these rules referred to as cattle).
- **Rule 47**
 - A valid certificate by a qualified veterinary surgeon to the effect that the cattle are in a fit condition to travel by rail or road and are not suffering from any infectious or contagious or parasitic diseases and that they have been vaccinated against rinderpest and any other infectious or contagious or parasitic diseases shall accompany each consignment.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for transport.
 - The certificate shall be in the form specified in Schedule-E.
- **Rule 48:** Veterinary first -aid equipment shall accompany all batches of cattle.
- **Rule 49**
 - Each consignment shall bear a label showing in bold red letters the name, address and telephone number (if any) of the consignor and consignee the number and types of cattle being transported and quantity of rations and food provided.
 - The consignee shall be informed about the train or vehicle in which the consignment of cattle is being sent and its arrival time in advance.
 - The consignment of cattle shall be booked by the next train or vehicle and shall not be detained after the consignment is accepted for booking.
- **Rule 50:** The average space provided per cattle in Railway wagon or vehicle shall not be less than two square metres.
- **Rule 51**
 - Suitable rope and platforms should be used for loading cattle from vehicles.
 - In case of railway wagon the dropped door of the wagon may be used as a ramp when loading or unloading is done to the platform.
- **Rule 52:** Cattle shall be loaded after they are properly fed and given water.
- **Rule 53:** Cattle in advanced stage of pregnancy shall not be mixed with young cattle in order to avoid stampede during transportation.
- **Rule 54:**
 - Watering arrangements on route shall be made and sufficient quantities of water shall be carried for emergency.
 - Sufficient feed and fodder with adequate reserve shall be carried to last during the journey.
 - Adequate ventilation shall be ensured.
- **Rule 55:** When cattle is to be transported by rail:
 - An ordinary goods wagon shall carry not more than ten adult cattle or fifteen calves on broad gauge, not more than six adult cattle or ten calves on metre gauge, or not more than four adult cattle or six calves on narrow gauge.
 - Every wagon carrying cattle shall have at least one attendant.

- Cattle shall be loaded parallel to the rails, facing each other.
- Rations for padding, such as straw, shall be placed on the floor to avoid injury if a cattle lies down and this shall not be less than 6cms thick.
- Rations for the journey shall be carried in the middle of the wagon.
- To provide adequate ventilation, upper door of one side of the wagon shall be kept open properly fixed and the upper door of the wagon shall have wire gauge closely welded mesh arrangements to prevent burning cinders from the engines entering the wagon and leading to fire outbreak.
- Cattle wagons should be attached in the middle of the train.
- Cooking shall not be allowed in the wagons nor hurricane lamps without chimneys.
- Two breast bars shall be provided on each side of the wagon, one at height of 60 to 80 cm and the other at 100 to 110 cm.
- Cattle-in-milk shall be milked at least twice a day and the calves shall be given sufficient quantity of milk to drink.
- As far as possible, cattle may be moved during the nights only.
- During daytime, if possible, they should be unloaded, fed, given water and rested and if in milk, milking shall be carried out.
- **Rule 56:** When cattle are to be transported by goods vehicle the following precautions are to be taken namely:
 - Specially fitted goods vehicles with a special type of tail board and padding around the sides should be used;
 - Ordinary goods vehicles shall be provided with anti-slipping material, such as coir matting or wooden board on the floor and the superstructure, if low, should be raised;
 - No goods vehicle shall carry more than six cattle;
 - Each goods vehicle shall be provided with one attendant.
 - While transporting the cattle, the goods vehicle shall not be loaded with any other merchandise; and
 - To prevent cattle being frightened or injured, they should preferably face the engine.

TRANSPORT OF EQUINES

- **Rule 57:** Rules 57 to 63 shall apply to the transport by rail, road, and sea of horses, mules and donkeys (hereinafter in these rules referred to as 'equines').
- **Rule 58**
 - A valid certificate by a qualified veterinary surgeon to the effect that the equines are in a fit condition to travel by rail; road or sea and are not suffering from any infectious or contagious disease or diseases shall accompany each consignment.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for transport.
 - The certificates shall be in a form specified in Schedule-I.
- **Rule 59**
 - Each consignment shall bear a label showing in bold red letters the name, address and telephone number (if any) of the consignor and consignee the number and type of equines being transported and quantity of rations and food provided.
 - The consignee shall be informed in advance about the train or vehicle or ship in which the consignment of equines is being sent and its arrival time.
 - The consignment of equines to be transported shall be booked by the next train or vehicle or ship and shall not be detained after the consignment is accepted for booking.
- **Rule 60**

- Pregnant and young equines shall not be mixed with other animals.
- Different species of equines shall be kept separately.
- Equines shall be loaded after being fed and given water adequately, watering arrangements shall be made en route and sufficient food carried to last during the journey.
- Veterinary first-aid equipment shall accompany all batches of equines.
- Adequate ventilation shall be ensured.
- Suitable ramps and platforms, improvised where not available, shall be used for loading and unloading equines.
- **Rule 61:** For the transport of equines by rail, the following precautions shall be taken
 - Equines shall be transported by passenger or mixed trains only;
 - Ordinary goods wagon when used for transportation shall carry
 - Not more than eight to ten horses or ten mules or ten donkeys on broad gauge and not more than six horses or eight donkeys on metre-gauge;
 - In extreme hot water weather, shall be sprinkled over the wagons containing equines by the railway authorities to bring down temperature. Ice slabs in specially made containers may be placed inside the wagon, if recommended by a qualified veterinary surgeon;
 - Every wagon shall have two attendants if the equines are more than two in number;
 - Equines shall be loaded parallel to the rails, facing each other;
 - Material for padding, such as paddy straw, shall be placed on the floor to avoid injury if an animal lies down and this shall not be less than 6 cm thick;
 - To provide adequate ventilation, upper door of the side of the wagon shall be kept open and properly fixed and the upper door of the wagon shall have wire gauge closely welded mesh arrangements to prevent burning cinders from the engines entering the wagon and leading to fire breakout;
 - Two breast bars shall be provided on each side of the wagon, one at a height of 50 to 80 cm and the other at 110 cm.
- **Rule 62:** For the transport of equines by goods vehicles, the following precautions shall be taken, namely:
 - Specially fitted vehicles with a special type of tail-board and padding around the sides shall be used;
 - Ordinary goods vehicles shall be provided with anti-slipping material on the floor and the super structure, if low, should be raised;
 - Bamboo poles of at least 8 cm diameter between each animal and two stout batons at the back shall be provided to prevent the animal from falling;
 - To prevent horses from being frightened or injured their heads
 - Should face left away from the passing traffic;
 - Each vehicle shall not carry more than four to six equines; (f) each vehicle shall be provided with one attendant;
 - The vehicle shall be driven at a speed not more than 35 kilometres per hour.
- **Rule 63:** For the transport of equines by sea the following precautions shall be taken, namely:
 - Horses may normally be accommodated in single stalls and mules in pens, each pen holding four to five mules;
 - Ample ventilation shall be ensured by keeping portholes and providing permanent air trunks or electric blowers on all decks, and exhaust fans shall be installed to blowout foul air;
 - All standings shall be athwart the ship with heads facing inwards;
 - To avoid distress specially during hot weather, the ship may go under way immediately after embarking and disembarking shall be done as early as possible after anchoring;
 - Colts and fillies shall be kept on the exposed decks;
 - A pharmacy and spare stalls for five per cent of equines shall be available;

- Passage between two rows of pens shall not be less than 1.5 metres.

TRANSPORT OF SHEEP AND GOATS

- **Rule 64:** Rules 65 to 75 shall apply to the transport of sheep and goats by rail or road involving journeys of more than six hours.
- **Rule 65**
 - A valid health certificate by qualified veterinary surgeon to the effect that the sheep and goats are in a fit condition to travel by rail or road and are not suffering from infectious or contagious or parasitic disease shall accompany each consignment.
 - In the absence of such a certificate, the carrier shall refuse to accept the consignment for transport.
 - The certificate shall be in a form specified in Schedule-J.
- **Rule 66**
 - Each consignment shall bear a label showing in bold red letters the name, address and telephone number (if any) of the consignor and consignee the number and type of sheep or goats being transported and quantity of rations and food provided.
 - The consignee shall be informed in advance about the train or vehicle in which the consignment of sheep or goats are being sent and its arrival time.
 - The consignment of sheep or goats shall be booked by the next train or vehicle and shall not be detained after the consignment is accepted for booking.
- **Rule 67**
 - First-aid equipment shall accompany the sheep or goats in transit.
 - Suitable ramps shall be provided for loading and unloading the sheep or goats.
 - In the case of a railway wagon, when the loading or unloading is done on the platform, the dropped door of the wagon shall be used as a ramp.
- **Rule 68:** Sheep and goats shall be transported separately; but if the lots are small special partition shall be provided to separate them.
- **Rule 69:** Rams and male young stock shall not be mixed with female stock in the same compartment.
- **Rule 70:** Sufficient food and fodder shall be carried to last during the journey and watering facility shall be provided at regular intervals.
- **Rule 71:** Material for padding, such as straw, shall be placed on the floor to avoid injury if an animal lies down, and this shall be not less than 5 cm. thick.
- **Rule 72:** The animals shall not be fettered unless there is a risk of their jumping out and their legs shall not be tied down.
- **Rule 73:** The space required for a goat shall be the same as that for a woolled sheep and the approximate space required for a sheep in a goods vehicle or a railway wagon shall be as under:-

Approximate weight of animal in KG	Space required in Sq mts.	
	Woolled	Horned
Not more than 20	0.18	0.16
More than 20 but not more than 25	0.20	0.18
More than 25 but not more than 30	0.23	0.22

More than 30	0.28	0.26
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Broad gauge		Metre gauge		Narrow gauge
(1)	(2)	(3)	(4)	(5)
Area of wagon	Area of wagon	Area of wagon	Area of wagon	
Less than	21.1	Less than	12.5	
21.1	Square Metres and Above	12.5	Square Metres and Above	
70	100	50	60	25

- **Rule 74**

- No railway wagon shall accommodate more than the following number of the sheep or goats
- Adequate ventilation shall be provided in every wagon. Upper door of one side of wagon shall be kept open and properly fixed and closely welded mesh to prevent burning cinders from the engine.

- **Rule 75**

- Goods vehicle of capacity of 5 or 4.5 tons shall carry not more than forty sheep or goats.
- Partitions at every 2 or 3 mts across width in large wagons.
- Separate panels for young ones.

TRANSPORT OF ANIMALS (AMENDMENT) RULES, 2001 NOTIFICATION

- S. O. 269 (E) whereas certain draft rules further to amend the Transport of Animals Rules, 1978 were published as required by sub-section (1) of Section 38 of the Prevention of Cruelty to Animals Act, 1960 (59 of 1960) under the notification of the Government of India in the Ministry of Social Justice & Empowerment number S.O. 1164 (E) dated 26th December 2000 in the Gazette of India. Extraordinary, Part II, Section 3, sub Section (ii) dated the 27th December, 2000 inviting objections and suggestions from all persons likely to be affected thereby, before the expiry of the period of sixty days from the date on which copies of the Gazette containing the said notification are made available to the public.
- And, whereas copies of the said Gazette were made available to the public on the 1st January 2001.
- And, whereas no objection or suggestion has been received from the public in respect of the said draft rules by the Central Government.
- Now, therefore in exercise of the powers conferred by sub section (i) and (ii) of Section 38 of the Prevention of Cruelty to Animals Act 1960 (59 of 1960), the Central Government hereby makes the following rules further to amend the Transport of Animals Rules 1978.

TRANSPORT OF POULTRY BY RAIL, ROAD AND AIR

- **Rule 76: Definition** - In this Chapter unless the context otherwise requires, "Poultry" includes day old chicks and turkey poults, chickens, quails, guinea fowls, ducks, geese and turkeys.
- **Rule 77: General requirement in transport of poultry by rail, road or air.**
 - These rules may be called the Transport of Animals (Amendment) Rules, 2001.
 - They shall come into force on the date of their publication in the Official Gazette .
 - In the Transport of Animal Rules 1978 (hereinafter referred to as the said rules), after chapter VI., the following chapters shall be inserted, namely
 - the container shall be properly cleaned and sterilised before the poultry is placed in them.
 - poultry shall not be exposed to the sunlight, rain and direct blast of air during transport.
 - poultry shall not be transported when the temperature exceeds 25 degree Celsius or when the temperature falls below 15 degree Celsius.
- **Rule 78: Day-old chicks and turkey poults**
 - In transport of day old chicks and poultry by rail, road and air
 - chicks and poults shall be packed and dispatched immediately after hatching and shall not be stored in boxes for any length of time, before dispatch.
 - Note: In the said transport endeavor shall be made by the consignor or his agent so that consignments shall arrive at destination within the shortest possible time after being taken out of the incubator. Seventy two hours shall normally be regarded as the maximum period to be taken from incubator to brooder in winter and 48 hours in summer
 - chicks or poults shall not be fed or watered before and during transportation.
 - every effort shall be made to ensure that chicks and poults arrive as quickly as possible at the dispatching site
 - personal attention shall be given by the consignor or the forwarding agent to ensure that all consignments are kept out of direct sunlight, rain and heat;
 - care shall be taken to carry the boxes in a level position so that chicks are not in danger of falling over on to their backs. The putting up of other merchandise over and around chick boxes shall be avoided.
- **Rule 79: Poultry other than day-old chicks and turkey poults**
 - In transport of poultry other than day old chicks and turkey poult by rail, road or air.
 - the poultry to be transported shall be healthy and in good condition and shall be examined and certified by a veterinary doctor for freedom from infectious diseases and fitness to undertake the journey.
 - poultry transported in the same container shall be of the same species and of the same age group.
 - poultry shall be properly fed and watered before it is placed in containers for transportation and extra food and water shall be provided in suitable troughs fixed in the containers.
 - arrangements shall be made for watering and feeding during transportation and during hot weather; watering shall be ensured every six hours;
 - male stock shall not be transported with female stock in the same container.
- **Rule 80: Road Travel**
 - In transport of poultry by road the container shall not be placed one on the top of the other and shall be covered properly in order to provide light, ventilation and to protect from rain, heat and cold air.
- **Rule 81: Rail Travel**
 - In transport of poultry by rail
 - in case the journey is for more than twelve hours, an attendant shall accompany the consignment;
 - poultry shall not be exposed to rain or direct blast of air;

- as far as possible poultry shall be transported in wagons having adequate facilities for ventilation and no other merchandise which may result in mortality of birds shall be loaded in the same wagon;
- **Rule 82:Air travel**
 - In transport of poultry by air or for international transport the containers carrying poultry shall be kept in pressurized compartments with regulated temperature and the container shall preferably be kept near the door and shall be unloaded immediately on arrival.
- **Rule 83:Containers for transportation**
 - In transport of poultry by rail, road or air
 - containers used to transport poultry shall be made of such material which shall not collapse or crumble and they shall be well ventilated and designed to protect the health of poultry by giving it adequate space and safety.
 - the containers shall be so designed as to render impossible for birds to crowd into the corners during transportation, and to avoid the danger of boxes being stocked so close together as to interfere with ventilation.
 - all the containers shall be clearly labelled showing the name, address and telephone number of the consignor and the consignee.
 - the minimum floor space per bird and the dimension of the containers for transporting poultry shall be specified in the Table below, namely:

S.No.	Kind of Poultry	Minimum	Demension			Number in a container
		Floor space cm ²	Length cm	Width cm	Height cm	
i.	Month old chickens	75	60	30	18	24
ii.	Three month old chickens	230	55	50	35	12
iii.	Adult sock (excluding geese and turkeys)	480	115	50	45	12
iv.	Geese and turkeys	900	120	75	75	10 youngs
		1300	75	35	75	2 growing
		1900	55	35	75	1 grown up
v.	Chicks	-	60	45	12	80
vi.	Poult	-	60	45	12	60

- **Rule 84**
 - Special requirement of containers for chicks and poultry In transport of poultry by road, rail or air
 - wire mesh or a net of any material shall not be used as bottom for the containers.
 - the containers shall be properly secured to avoid pilferage
 - the following instruction shall be printed on a label and fixed to the lid or printed directly on sides, namely "Care in Transit".

- the consignee shall be informed about the train, transport or flight number and its time of arrival well in advance.
- poultry shall not be transported continuously for more than 6 hours and whole batch shall be inspected at every 6 hours interval
- the transportation shall not remain stationary for more than 30 min and during this period, it shall be parked in shade and arrangement shall be made for feeding and watering.
- all precautions against fire shall be taken and provision of fire extinguishers in transport shall be provided.

TRANSPORT OF PIGS BY RAIL ON ROAD

In this chapter, unless context otherwise requires, "pigs" includes piglets, hogs, hoglets and animals of pig family.

- **Rule 86: Duration of travel**
 - Rules 87 to 95 shall apply to the transport of pigs by rail or road involving journeys of more than six hours.
- **Rule 87: Health Certificate**
 - A valid health certificate by a veterinary doctor *to the* effect that the pigs are in a fit condition to travel by rail or road and are not suffering from infectious or contagious or parasitic disease shall accompany each consignment in the transport of pigs by rail or road.
 - In the absence of a certificate under sub-rule (i), the carrier shall refuse to accept the consignment for transport.
 - The certificate under sub rule (I) shall be in a foI11'J specified in Schedule K.
- **Rule 88: Identification of consignor and consignee for the purpose of this Chapter.**
 - each consignment shall bear a label showing in bold red letters the name, address, and telephone number (if any) of the consignor and consignee, the number and type of pigs being transported and quantity of
 - rations and food provided to them. .
 - the consignee shall be informed in advance about the train or vehicle in which the consignment of pigs is being sent and its arrival time.
 - the consignment of pigs shall be booked by the next train or vehicle and shall not be detained after the consignment is accepted for booking.
- **Rule 89: First aid**
 - In transport of pigs by rail or road.
 - first aid equipment shall accompany the pigs;
 - suitable ramps shall be provided for loading and unloading the pigs;
 - in the case of a railway wagons, when the loading or unloading is done on the platform the dropped door of the wagon shall be used as a ramp.
- **Rule 90: Group of pigs.**
 - In transport of pigs by rail or road, male young stock shall not be mixed with female stock in the same compartment.
- **Rule 91: Facility of food and water.**
 - In transport of pigs by rail or road, sufficient food and fodder shall be carried to last during the journey and watering facility shall be provided at regular intervals.

- **Rule 92: *Padding of floor during travel***
 - In transport of pigs by rail or road, material for padding, such as straw, shall be placed on the floor to avoid injury if an animals lies down, and this shall be not less than 5 cm thick.
- **Rule 93: *Ban on fettering.***
 - In transport of pigs by rail or road, the animals shall not be fettered unless there is a risk of their jumping out and their legs shall not be tied down.
- **Rule 94: *Space requirement during rail travel***
 - In transport of pigs by rail
 - no railway wagon shall accommodate more than the number of pigs as specified in the Table below:

Broad gauge		Metre gauge		Narrow gauge
(1)		(2)		(3)
Area of Wagon	Area of Wagon	Area of Wagon	Area of Wagon	
Less than 21.1	More than 21.1	Less than 12.5	More than 12.5	
squares Metres	Square Metres	Square Meter	Square Meter	
Number of Pigs	Number of Pigs	Number of Pigs	Number of Pigs	Not allowed
35	50	25	30	

- adequate ventilation shall be provided in every wagon and the upper door of one side of wagon shall be kept open and properly fixed and the upper door of the wagon shall have wire gauge closely welded mesh arrangements to prevent burning cinders from the engines entering the wagon and leading to fire breakout.
- **Rule 95: *Space requirement during road travel***
- In transport of pigs by road
 - goods vehicles of capacity of 5 or 4.5 tons, which are generally used for transportation of animals, shall carry not more than twenty pigs.
 - in the case of large goods vehicles and containers, partition shall be provided at every two or three metres, across the width to prevent the crowding and trapping of pigs.
 - in the case of pigs under six weeks of age, separate panels shall be provided.

MISCELLANEOUS

- **Rule 96: *Issues of certificate before transportation***
 - A valid certificate issued by an officer or any person or Animal Welfare Organisation duly recognised and authorised for this purpose by the Animal Welfare Board of India or the Central Government, shall be procured by any person making transport of any animal before transportation of such animal verifying that all the relevant Central and State Acts rules and orders pertaining to the said animal including the rules relating to transport of such animals have been duly complied with and that the animals are not being transported for any purpose contrary to the provision of any law.

- In the absence of such certificate, the carrier shall refuse to accept the consignment for transport.
- **Rule 97: Cancellation of permit or authorisation for transport**
 - In the event of contravention or non-compliance of any of the rules contained in these rule for transport of animals, if it is pointed out in writing by any officer or person or Animal Welfare Organisations authorised for the purpose by the Animals Welfare Board of India or the Central Government, then, any permit or authorisation issued for such transport shall be immediately cancelled by the concerned authority and It shall be the duty of the police to stop further transport even from the intermediate station and proceed against the said offenders and deal With the animals in accordance with law;
 - The custody of the animals immediately after unloading from the rail wagons, truck or any other vehicle shall be given to the authorised Animal Welfare Organisation if available, till the competent authority or the magistrate having jurisdiction decides about their care and upkeep.
- **Rule 98: General conditions of transport**
 - Animals to be transported shall be healthy and in good condition and such animals shall be examined by the veterinary doctor for freedom from infectious diseases and their fitness to undertake the journey; provided that the nature and duration of the proposed journey shall be taken into account while deciding upon fitness.
 - An animal which is unfit for transport shall not be transported and the animals who are new born, diseased, blind, emaciated, lame, fatigued or having given birth during the preceding seventy two hours or likely to give birth during transport shall not be transported.
 - Very young animals shall not be mixed with other animals during transport.
 - Different classes of animals shall be kept separately during transport.
 - Diseased animals, whenever transported for treatment, shall not be mixed with other animals.
 - Troublesome animals shall be given tranquilisers before loading for transport.
 - Animals shall be transported in their on-farm social groups (established at least one week prior to journey).

TRANSPORT OF ANIMALS ON FOOT RULES, 2001

(S.O. 268 (E) (26th March, 2001)

In exercise of the powers conferred by sub-sections (1) and (2) of Section 38 of the Prevention of Cruelty to Animals Act, 1960 the Central Government hereby makes the following rules, namely:

- **Short title and commencement:** These rules may be called the Prevention of Cruelty to Animals (Transport of Animals on Foot). Rules 2001.
- **Definitions:** In these rules unless the context otherwise requires:(a) "animal" means livestock and includes the following animals namely:
 - cattle including cow bulls and bullock, buffalo bulls and bullock, cows, buffaloes, mithuns, yaks and calves;
 - equines including horses, ponies, mules and donkeys
 - horse including stallions, geldings, brood mare, colts and fillies
 - goat including adult goat, male or female of two years age and above
 - buck including male goat

- kid-young goat below one year of age
- nanny-female goat
- sheep including adult sheep, male or female of two years age and above
- ewe-female sheep
- Lamb-young sheep below one year of age; (xi) ram-male sheep
- wether includes male Lamb that has been castrated before reaching sexual maturity
- pig includes adult pig, male or female of one year of age or above
- piglet includes young pig below one year of age
- veterinary doctor" means a person registered with the Veterinary Council of India established under the Indian Veterinary Council Act, 1984(52 of 1984):
- "Schedule" means a schedule appended to these rules
- ***Application of the rules:*** These rules shall apply to transport of animals on foot when the distance from the boundary of Village or town or city of the origin of such transport to the last destination is 5 km or more than 5 km.
- ***Condition of health of animals transported on foot***
 - Every animal to be transported on foot shall be healthy and in good condition for such transport.
 - A certificate of a veterinary doctor in respect of each animal to be transported to the effect that such animal is in a fit condition for such transportation and is not suffering from any infectious, contagious or parasitic diseases and that it has been vaccinated against any infectious, contagious or parasitic diseases shall accompany such animal.
 - The certificate under sub rule (1) shall be in the form as specified in the First Schedule.
- ***Certain animals do not transport on foot:*** New born animals of which the navel has not completely healed, diseased, blind, emaciated, lame, fatigued, or having given birth during the preceding seventy two hours or likely to give birth during transport shall not be transported on foot.
- ***Transport in on-farm social group:*** Animal shall be transported in their on farm social groups (established at least one week prior to journey)
- ***First aid equipment to accompany animals transported on foot:*** The owner of the animals shall provide veterinary first aid equipment to be accompanied with such animals while transported on foot.
- ***Certificate to be carried during transportation:*** In case the person transporting the animals on foot is not the owner of the animals then such person shall carry a certificate as specified in the Second Schedule during such transportation. .
- ***Watering arrangement during transportation of animals:*** The owner of the animals shall make watering arrangement en route during transport of such animals on foot.
- ***Feed and fodder arrangements during transportation of animals:*** Sufficient feed and fodder with adequate reserve of such feed and fodder for the animals shall be made available by their owner during their transport on foot.
- ***Prohibition of the use of whip, etc. during transportation of animals on foot***
 - No person shall use a whip or a stick in order to make the animals to walk or to hasten the pace nor such person shall apply chillies or any such substance to any part of body of the animals for this purpose during their transportation on foot.
 - If any animal needs to be tied during transport on foot, it shall be tied by a rope covered with suitable cushioning such as cloth around its leg and such animal shall not be tied by its nose, all legs or any other part of the body except by its neck.
 - If more than one animal is to be tied adjacent to one another by a single rope during their transport on foot, the space between any two of such animals shall be minimum two feet and animals so tied shall be of similar physical condition and strength and no more than two such animals shall be tied adjacent to each other by a single rope.

- ***Certain prohibition on transport of animals on foot***
 - No person shall transport on foot an animal before sunrise or after sunset.
 - No animal shall be transported on foot beyond the distance, time, rest interval and temperature specified for such animal in the following Table:
 - No animal shall be made to walk under conditions of heavy rain, thunderstorms or extremely dry or sultry conditions during its transport on foot.
- ***Transportation of animals in certain cases not permitted without shoes:*** Animals whose hooves are not provided with shoes (as in the case of pack or draught animal) shall not be transported on foot on hard cement, bitumen-coated or metalled roads, steep gradients or hilly and rocky terrain, irrespective of weather conditions (summer or winter).
- ***Power of Police to require the owner to take animal to nearest Magistrate***
 - If any police officer or any other person authorized on this behalf by the Government or by the Animal Welfare Board of India has reason to believe that an offence has been or is being committed in respect of an animal in contravention of these rules he may require the owner or other person in charge of such animal to take the animal to the nearest magistrate.
 - If the owner or the person in charge of the animals referred to in sub rule (I) refuses to comply with the demands of the police officer under that sub rule, it shall be lawful for such police officer or such other persons to take the animal to the nearest magistrate.

Species (Animal)	Maximum distance covered/day/hour	Maximum no. of walking/day of hours Travelling)	Period of rest (interval)	Temperature range Max. Min
Cattle (Cows)	30 km/day 4 km/hr	8 hours	At every 2 hours for drinking and at every 4 hrs for feeding	12 deg. C to 30 deg.C
Buffaloes	25 km/day 3 km/hr	8 hours	At every 2 hours for drinking and at every 4 hrs for feeding	12 deg. C to 30 deg. C
Cows and Buffaloes Calves	16 km/day 2.5 km/hr	6 hours	At every 1½ hrs for drinking and at every 3 hrs. for feeding	15 deg. C to 25 deg.C
Horses, Ponies, Mules,	45 km/day 6 km/hr	8 hours	At every 3 hrs for drinking and at every 6	12 deg. C to 30 deg. C

Donkeys			hrs. for feeding	
Young ones (Foal)	25 km/day	6 hours	At every 2hrs. for drinking and at every 4 hrs. for feeding	15 deg. C to 25 deg. C
	4 km/hr			
Goats and Sheep	30 km/day 4 km/hr	8 hours	At every 2 hrs. for drinking and at every 4 hrs. for feeding	12 deg. C to 30 deg. C
Kids and Lambs	16 km/day 2.5km/hr	6 hours	At every 1 1/2 hrs. for drinking and at every 3 hrs. for feeding	15 deg. C to 25 deg. C
Pigs	15 km/day 2 km/hr	8 hours	At every 1 Y2 hrs. for drinking and at every 3 hrs. for feeding	12 deg. C to 25deg. C
Piglets	10 krn/day 1.5 km/hr	6 hours	At every 1 Y2hrs. for drinking and at every 3 hrs. for feeding	15 deg. C to 25 deg. C

Note: After being provided with water every animal shall be given a break of 20 minutes before the commencement of the transport of the animal on foot and in case of feeding the break shall be given for one hour before the commencement of the transport of the animal on foot.

MODULE-30: COMMON FRAUDS IN THE SALE OF LIVESTOCK AND LIVESTOCK PRODUCTS

Learning outcome

Sale of meat, milk and other products of animals to the public makes a big commercial market. This makes the businessmen to go for adulteration of the products. This module focusses on common frauds in the sale of livestock and livestock products in detail.

FRAUDS IN THE SALE OF LIVESTOCK

- **Alteration of description**
 - The description of the animal is altered in case it has been stolen or there are signs which are unsound. Alteration of description is done in the following ways.
 - Castration of entire
 - Clipping of mane and tail
 - Docking
 - Colouring of white patches by hair dye
- **Bishoping**
 - By this method animals are made to appear younger. It is done by making small holes in the table of incisors and filling them with some black substance such as marking nut. Air is also blown under the skin of supra orbital fossa of the animals. The chronic ulcers and fistulae are often covered with coloured mud. Frauds in the sale of livestock are punishable under Section 420 I.P.C.
 - I.P.C. Section 420 whoever cheats and thereby dishonestly induces the person deceived to deliver any property to any person, or to make, alter or destroy the whole or any part of a valuable security, or anything which is signed or sealed, and which is capable of being converted into a valuable security, shall be punished with imprisonment of either description for a term which may extend to seven years and shall also be liable to fine.
 - Comment:- Simple cheating is punishable under section 417. But where there is delivery or destruction of any property or alteration or destruction of any valuable security resulting from the act of the person deceiving, this section comes into operation.

LIVESTOCK PRODUCTS

FRAUDS IN THE SALE OF MILK

- The adulteration of milk is very common in India. There are various chemical tests to detect these adulterations. (eg) Addition of water to milk – detected by
 - Determination of specific gravity of milk by lactometer.
 - Test for nitrate in milk: Normal milk does not contain nitrates but ordinary water does. The presence of nitrates in the milk is a sure test of adulteration of milk with water.

Routine methods in the adulteration of milk

Adulteration	Method
Reduction of fat	- By adding water - By skimming of milk - By skimming and watering
Addition of thickening agents	- This is done to increase the specific gravity and consistency of the milk. - It is done by adding starch gelatin and cane sugar
Addition of colouring agent	- Sometimes colours are added to make milk more attractive - The colours generally used are annatto and coal tar dyes.
Addition of preservatives	- Preservatives like formalin, boric acid, hydrogen peroxide, sodium bicarbonate, sodium carenate are added to retard the decomposition with an objective to - Delay the curdling of milk - Destroy the pathogenic bacteria
Accidental adulteration	- Some adulterants are mixed accidentally in the milk (eg) urine or animal, dung, vegetable cells, dirty water. - These agents give a foul smell.

FRAUDS IN THE SALE OF GHEE

Some of the methods of adulteration of ghee are by adding vegetable oils, hydrogenated oils (sesame oil) or animal fat (Mutton or beef fat).

- **By adding vegetable oils**
 - Ghee is adulterated with vegetable oils or hydrogenated oils. Vegetable fat contains phytosteryl which is not present in ghee. Phytosteryl acetate test is carried out to prove a sample of ghee has been adulterated with vegetable oil or hydrogenated oils.
- **By adding hydrogenated oils**
 - It is compulsory to add a definite quantity of sesame oil to all vegetable ghee in India.
 - Presence of sesame oil in supposedly pure ghee is a sure indication of adulteration with vegetable ghee.
 - Boudouin test is used to detect presence of sesame oil in ghee.
- **By adding animal fats**
 - Ghee is adulterated with mutton or beef fat.
 - Adulteration of milk and milk products is punishable under I.P.C Section 272
 - Selling of unfit food or drink is punishable under Section 273 I.P.C Section 272 whoever adulterates any article or food drink, so as to make such article noxious as food or drink intending to sell such article as food or drink or knowing it to be likely that the same will be sold as food or drink, shall be punished with imprisonment of either description for a term which may extend to six months, or with fine which may extend to one thousand rupees, or with both.

FRAUDS IN THE SALE OF MEAT

- Adulteration of meat involves substitution of costly or superior quality meat with cheaper and inferior quality meat. The substitutions of meat generally practiced are
 - Beef for buffalo meat
 - Mutton for goat meat
 - Rabbit for chicken
- Since, meat adulteration is a fraudulent practice, which is objectionable on the grounds of health, religion and economics, so identification of meat is very important.

IDENTIFICATION OF MEATS

- The common methods used for the identification of meats of various species of animals are as follows:
- To avoid sale of meat unfit for human consumption, proper ante mortem and post mortem inspection should be carried out.

PHYSICAL EXAMINATION

- The muscles, fat and bones of different animals have different characteristics. Hence, it is essential to know the physical character of different types of meats to detection of fraud.

- Rigor Mortis: The onset time of rigor mortis varies in different species. So by looking at the rigor mortis we can identify the type of the meat.
- Meat of dead animals: Animals which have died of natural causes show fullness of blood vessels and the meat is dark red due to lack of proper bleeding. Such carcasses undergo decomposition very quickly. So such meat should not be allowed for human consumption.

Mutton

- Meat is dark red in colour
- Ammonical odour
- Muscle fibres are firm and dense in consistency
- Rich fat deposition in between groups of muscles
- Fat is not intermixed with muscles
- Fat is white in colour, hard and firm
- Bone marrow is slightly red

Goat meat

- Paler than mutton
- Muscles contain very little fat
- Goat meat odour resembles odour of buck

Pork

- Whitish grey in colour and in advanced age it becomes grey-red in colour
- Consistency is soft
- Fat is intermixed with muscles
- Fat is white and granulated
- Bone marrow is pink-red in colour
- Urine-like odour

Dog meat

- Fat is slightly intermixed with muscles
- Dark red in colour
- Fat is white in colour

Beef

- Saturated Red in colour with a brown tinge
- Fat is intermixed with muscles
- After chilling fat acquires a firm consistency

Milch cow

- Lighter in colour

- Coarse fibres
- Fat is yellow and less firm

Old cattle

- Yellow fat
- Bone marrow is pure white to reddish yellow

Veal

- Fine fibres
- Fat is not inter mixed with muscles
- Bone marrow is pink-red in colour

Horse meat

- Meat is dark red in colour
- When meat is exposed to air to becomes blackish in colour
- Fat is golden to dark yellow in colour
- Bone marrow is greasy

Poultry meat

- Muscular tissues are firm with fine fibres
- Fat is intermixed with muscles

Fish meat

- Meat is white in colour
- Entire muscular mass of each side comprises a single muscle plate
- Fat is finely distributed in the muscles
- Meat contains large quantity of water
- Apart from the physical examination of meat, examination of bones is also helpful in determining adulteration.

CHEMICAL EXAMINATION

Following tests are used

Glycogen test

- Meat of horse contains more of glycogen than the meat of other animals.
- Pig's liver contains a high percentage of glycogen. Glycogen test should be done whenever a piece of liver is examined.

- Horse flesh is supposed to contain maximum glycogen 2.28 where as in normal mammalian muscles it varies from 0.5 to 1%.

Tests based on fats of the animals

- Sometimes, the fat of horse is mixed with the fat of cattle and sheep. Fat of horse is identified by the presence of 1 to 2 percent linolenic acid.
- In fat of other animals, the percentage of linolenic acid is not more than 0.1 per cent.

Iodine value

- Iodine value is the amount of iodine absorbed by unsaturated fatty acids present in the fat.

Species	Iodine value
Horse	71 to 86
Ox	38 to 46
Sheep	35 to 46
Pig	50 to 70

Refractive Index

- The fat of different animals have different refractive index. For this purpose, the fat of the animal is converted into liquid by heat and refractive index is estimated.
 - Refractive index of horse fat is 35.5
 - Refractive index of ox fat is not above 40
 - Refractive index of pig fat is not above 51.9
- Some other chemical methods are now increasingly being applied in meat inspection. These methods are specific and often the only means available for detection of foreign proteins in the meat or meat products.

Serological tests employed for differentiation of flesh

- Immuno – precipitation test
- Electrophoretic methods
- Isoelectric focusing

Rigor Mortis time Refractive Index and Iodine value – various species

Species	Rigor mortis Time (Min)	Refractive index	Iodine value
Sheep	80	-- --	35 to 46
Lamb	60	40	--
Ox Pig	163	51.9	38 to 46
Horse	50	53.5	50 to 70
	238		71 to 86

ANTEMORTEM INSPECTION OF MEAT

- Antemortem inspection should be carried out / made on the premises of the slaughter house on the day of slaughter.
- Following categories of animals should be condemned
 - Any animal suffering from high temperature ($> 106^{\circ}\text{F}$).
 - All animals showing symptoms of rabies, scrapie, tetanus or anthrax.
 - All animals showing symptoms of some disease condition that would cause condemnation of carcass on postmortem inspection.
 - Any animal found to be dead or in dying condition.
 - Any animal used for research investigations.
 - Any animal vaccinated against Anthrax in last 6 weeks.
 - Hogs showing symptoms of hog cholera or swine erysipelas.
 - Animals treated with some medicines whose residues are expected.
- If any animal on ante-mortem inspection is suspected of being affected with any disease or condition that would cause condemnation of only part of the carcass then on postmortem examination should be marked “suspect”.
- Following categories of animals should marked suspect
 - Any animal seriously crippled.
 - Animals which have reacted to tuberculin test.

POST MORTEM EXAMINATION OF MEAT

- A careful postmortem inspection of the carcass and the parts thereof should be made at the time of slaughter.
- Whole carcass and its parts found to be healthy, wholesome and fit for human consumption should be marked “Inspected” and passed.
- Each carcass or parts, which is found to be unhealthy and unfit for human food should be marked “Inspected and condemned”.
- Carcasses of animals in a well nourished condition showing uncomplicated localized lesions of actinomycosis, or actinobacillosis may be passed after the infected organ or parts have been removed and condemned.
- As per the detailed list of the conditions on the basis of which the carcass is completely condemned (Anthrax generalized T.B).

- Carcasses in following conditions should be condemned.
 - Carcasses showing generalized pigmentary deposits.
 - Carcasses of animals affected with mange, scab in advanced stages.
 - Carcasses showing any degree of icterus.
 - All carcasses of animals so affected that consumption of the products thereof may give rise to food poisoning (eg) Generalised inflammatory conditions septicemia or gangrenous conditions -Abscess.
 - Carcasses which give off pronounced odour of urine (or) of any medicinal or chemical substance.
 - Carcasses of all unborn and still born animal.
 - Carcasses of those animals, the meat of which give appearance of water soaked, flabby and tears easily.
 - Carcasses infested with parasites transmissible to man.
 - Carcasses of the animals affected with or showing lesions of any of the following diseases should be condemned except when recovery has occurred to the extent that only localized.

PHYSICAL CHARACTERISTICS OF MEAT OF VARIOUS SPECIES

Meat	Colour	Consistency	Fat
Sheep	Dark red	Firm & Dense Ammonical odour	White fat deposition in between groups of muscles
Buffalo	Dark red	Firm	Pale white
Horse Flesh	Dark red	Soft	Golden to dark yellow
Dog meat	Dark red	--	White fat is slightly intermixed with muscles
Beef	Red	Fairly firm	Yellowish white fat is intermixed with muscles
Camel	Red	Fairly firm	Pale yellow
Goat	Light red	very firm Similar to Buck	Pure white, muscles contain very little fat
Pork	Light red	very soft Urine like odour	White and granulated fat is intermixed with muscles
Poultry	White	Firm	Fat intermixed with muscles
Fish meat	White	Entire muscular mass of each side comprises a single muscle plate	Fat is finely distributed in muscles

ADULTERATION OF FOOD OR DRINK

- Adulteration of food or drink is punishable under Section 272 I.P.C

Section 272 I.P.C.

- Whoever adulterates any article of food or drink so as to make such article noxious as food or drink, intending to sell such article as food or drink or knowing it to be likely that the same will be sold as food or drink, shall be punished with imprisonment of either description for a term which may extend to six months or with fine which may extend to one thousand rupees or with both.

Selling of unfit food or drink is punishable under Section 273 I.P.C

- Whoever sells or offers or expose for sale, as food or drink, any article which has been rendered or has become noxious, or is in a state unfit for food or drink, knowing or having reason to believe that the same is noxious as food or drink, shall be punished with imprisonment of either description for a term which may extend to six months, or with fine which may extend to one thousand rupees or with both.

Important terms related to Animal welfare, ethics and jurisprudence

A

Acute Pain

:Results from a traumatic, surgical or infectious event that is abrupt in onset, relatively short in duration, and generally alleviated by analgesics. Associated distress may be responsive to tranquilizers.

Analgesia

:Absence of the sense of pain without loss of consciousness.

Anesthesia

:Loss of bodily sensation with or without loss of consciousness.

Animal

:Any living, non-human, vertebrate animal (mammal, bird, reptile, amphibian, or fish), unless otherwise specified.

Animal Welfare Assurance

:Key document in defining the relationship between the institution and the Public Health Service (PHS), signed by the Institutional Official, sets forth the responsibilities and procedures of the institution regarding the care and use of animals, and requires the Institutional Official to file an annual report with the PHS. An "Assurance" can remain in place up to four years and is renewable.

Animals as Human Food Source

:Agricultural production of animals or animal products intended to enter the food distribution system.

AV

:Attending Veterinarian. A veterinarian with training or experience in the care and management of the species attended, and who has direct or delegated authority for activities involving animals at Cornell University.

B

Biohazardous Agents

:Infectious/pathogenic agents classified as Class 2, 3, and 4 bacterial, fungal, parasitic, viral, rickettsial or

chlamydial agents, as well as other agents that have the potential for causing disease in healthy individuals, animals or plants (including infectious agents, regulated pathogens & pests, cell cultures, tissues, human-derived materials, biological toxins, recombinant DNA molecules)

Biomedical Studies

:Studies in which animals (including agricultural or farm animals) are used in research or teaching for which the direct or indirect objectives are to improve human health.

Breeding Protocol

:An animal protocol in which animals are used for planned propagation. USDA Category B applies only to animals used solely for breeding purposes. If parents or offspring are subjected to any other procedures (including sampling of tissue for genotyping, or euthanasia) they must be assigned to USDA Category C.

C

Cervical Dislocation

:Euthanasia accomplished by "breaking the neck" or "snapping the spine" by a method intended to be quick and painless.

Chronic Pain

:Results from a longstanding physical disorder or emotional distress that is usually slow in onset, has a long duration, and is generally not totally alleviated by analgesics, but frequently responds to tranquilizers combined with environmental manipulation and behavioral conditioning.

Client Owned Animals

:Vertebrate animals not owned by Cornell and normally kept in non-Cornell owned housing or facilities. Owners may bring these animals to Cornell for veterinary care, or for demonstration purposes. Conversely, Cornell researchers may visit the animals and carry out procedures at the client location. Animals on non-Cornell farms are client owned.

Conspecifics

:Individuals of the same species. This term is usually used to refer to interactions within an animal enclosure or proximate area.

D

Distress

:Undesirable physical or mental stress resulting from pain, anxiety, or fear. Its acute form may be relieved by tranquilizers, whereas sustained distress requires environmental change and behavioral conditioning, and

does not respond to drug therapy.

E

Euthanasia

:The humane destruction of an animal accomplished by a method that produces rapid unconsciousness and subsequent death without evidence of pain or distress, or a method that utilizes anesthesia produced by an agent that causes painless loss of consciousness and subsequent death.

Exemption

:An exception approved by the IACUC to the standards of animal care specified in the Guide or the Ag Guide. Exemptions from Standards of Care must be approved by the IACUC every six months.

Experimental Animal Endpoints

:The time at which live animals (or groups of animals) are no longer undergoing experimental treatment or procedures. In many studies the experimental endpoint is euthanasia.

F

Field Studies

:Studies conducted with wild animals in their natural habitat. Wild animals may be captured or held in the field for limited periods of time.

G

General Anesthesia

:State of unconsciousness characterized by absence of pain and some muscle relaxation.

Generalized Discomfort

:Any procedure that would be expected to cause more than momentary discomfort in a human and for which a human would receive pain-reducing medications including aspirin, ibuprofen, acetaminophen, lidocaine, or prescription medications.

H

Hypnosis

:Anesthesia produced by placing the patient in a trance-like state.

Hypothermia

:Lowering of body temperature, either locally or generally, to a level at which anesthesia occurs.

L

Local anesthesia

:Loss of sensation in a limited area of the body.

M

Major surgery

:Surgery which penetrates and exposes any body cavity, including the cranium and the perineum (except castration), involves orthopedic surgery, or produces significant impairment of anatomical and/or physiologic function.

Minor surgery

:Surgery which does not expose a body cavity and causes little or no physical impairment.

N

Narcosis

:A reversible condition defined as a drug-produced state of sleep accompanied by analgesia.

Neuroleptanalgesia

:A state of sedation and analgesia produced by the combined use of a tranquilizer (neuroleptic) and a narcotic analgesic.

Non-surgical procedure

:Any procedure performed on a live animal that does not meet the definitions of surgery, restraint or euthanasia.

Non-survival surgery

:Surgery in which the animal is euthanized before recovery from anesthesia.

P

Pain

:Awareness of acute or chronic discomfort occurring in varying degrees of severity resulting from injury, disease or emotional distress and evidenced by biological or behavioral changes or both.

Perfusion

:The passage of a fluid through a specific organ or an area of the body, often used as a therapeutic measure whereby a drug intended for an isolated part of the body is introduced via the bloodstream.

Protocol

:A description of the proposed procedures involving any use of animals as required on Cornell University's protocol for the use of live vertebrate animals. See esirius.research.cornell.edu/esirius/.

R

Regional anesthesia (nerve block)

:Loss of sensation to a large, though limited area of the body.

Restraint

:Physical (manual or mechanical) or chemical (anesthetic or tranquilizer not used in conjunction with surgery) means to limit some or all of an animal's normal movement or behavior.

S

Sedation

:A mild degree of CNS depression in which the animal is awake but calm and free of nervousness.

Surgery

:A procedure involving (major) incisions to remove, repair, or replace a part of a body.

Surgical anesthesia

:Unconsciousness, accompanied by muscular relaxation to such a degree that surgery can be performed painlessly and without struggling on the part of the animal.

Survival surgery

:Surgery in which the animal awakes from surgical anesthesia.

T

Tranquilization

:State of behavioral change in which the animal is relaxed and unconcerned by its surroundings; the

animal is often indifferent to minor pain.